Final Review Report

Large-Scale Aerodynamic Wake Survey System

by

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Executive Overview

LAWSS Objectives

The Large-Scale Aerodynamic Wake Survey System (LAWSS) aims to survey the wakes of large-scale bodies such as buildings, wind turbines, and ships in order to provide real world data for aerodynamic analysis of large scale bodies, which have historically relied on computationally expensive simulations and small-scale wind tunnel experiments. To address this, the LAWSS system is designed to determine the aerodynamic forces within 5% of the true value and qualitatively identify the major flow characteristics behind real large-scale bodies. The aim of this report is to provide a baseline for the design by performing market, functional, risk, and sustainability analyses to inform the formulation of VALID stakeholder, mission, and system requirements and the formulation of a pruned design option tree for subsequent trade-off.

Market Analysis

Mission Architecture

Functional Analysis

Functional Flow Diagram

Functional Breakdown Structure

Requirements

Technical Resource Management

Risk Analysis

Sustainability Analysis

Design Option Tree and Straw-man Elimination

1 Introduction

2 Market Analysis

The market analysis serves two purposes. First, it establishes who needs the system, what they are willing to pay and how large the opportunity is. Second, it translates that into a target cost and a set of requirements the design must meet to be viable. The Baseline Report produced a first version of this analysis. This is the updated version and the main change is that the system now has a defined product and a bottom-up cost so the analysis rest on actual specifications rather than on the general concept.

Before starting, the project constrained us to assume a €10,000 pay per acquisition day and a two year payback period. These parameters are fixed and will be indicated as optimistic or constraining along the chapter. The objective is not to present the system favourably but to test whether it holds up against the market and to state the result plainly.

2.1. Stakeholder Analysis

A stakeholder is any party that affects the system or is affected by it. Stakeholders matter here because they drive requirements, and because their conflicting interests must be resolved in the design.

2.1.1. Identification

The paying parties are the customers: wind energy companies (Eneco, Vattenfall and similar), turbine manufacturers and certification bodies, architecture and facade firms, wind-assist propulsion suppliers and research institutes. The funding parties are the investors. The regulating parties are EASA and the Dutch ILT for operating a drone swarm in shared airspace, together with EASA and FAA for wider operation. Government bodies (Dutch, local, EU) and the public sit alongside the regulators, concerned mainly with safety. The building and operating parties are the engineers and operators. Finally there are the suppliers, the drone manufacturers, wind tunnel centres and CFD firms which act as partners in some cases and as competitors in others.

2.1.2. Power and interest mapping

Each stakeholder is placed on a power versus interest grid seen in Figure 2.1. Regulators and government sit high on power, because they can prevent the system from flying, but lower on interest, since it is one of many areas they oversee. Investors sit high on both. Customers sit high on interest and lower on power since they shape requirements through willingness to pay but do not control the design. Among customers, energy companies hold the most spending power, followed by shipbuilding then architecture, urban planning and research. Research institutes hold the highest interest of all because full-scale wake data is precisely what they cannot generate cost-effectively in-house. The public sits low on both axes, its main concern being safety.

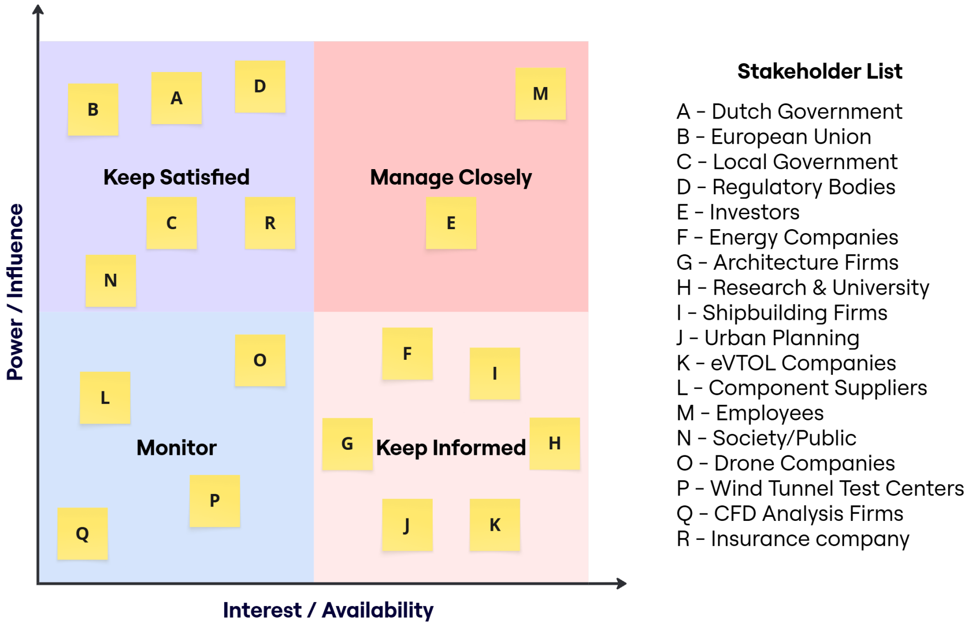

Figure 2.1: Stakeholder Map

2.1.3. Conflicting interests

The most common conflict is between investors and engineers. Investors favour a cheaper faster build to protect margin, while engineers prefer the budget and time to optimise the product. The €10k per day ceiling and the two year payback are where this conflict is resolved and both bear directly on unit cost.

The second conflict is between the company and its competitors since the system aims to take a share of work currently performed by wind tunnel centres, CFD firms and LiDAR service providers. Several of those firms could also act as partners which complicates the relationship.

The third is between customers and regulators. Customers want the most capable and least restricted system, while regulators require that it never endangers anyone. Regulation takes precedence, so the design absorbs the cost of compliance even where this reduces efficiency.

The fourth is between investors and customers over price. Investors favour a high day-rate and customers a low one and demand falls sharply if the price exceeds the value of the work. This constraint keeps the day-rate close to €10k.

2.1.4. Shared interests

Safety aligns government, the public, regulators and the insurance company which strengthens every safety requirement. Customers and the company both want an efficient, accurate product which is reflected in the technical requirements. Engineers and investors both want the system to be profitable, which depends largely on reducing unit cost. Academic institutions and the energy companies both want high-resolution flow data although for different reasons: the academics to improve its flow models, the energy companies to lay out wind farms that lose less power to wakes

2.2. Product definition

Before the market is analysed, the system is stated in concrete terms because the design has progressed far enough that the analysis should rest on actual specifications rather than on the general concepts. LAWSS is a swarm of 80 measurement drones operated from a single ground station by a two-person crew. Each drone carries a TriSonica Mini three-component sonic anemometer sampling at 10 Hz, an RTK-GNSS receiver giving position to within a few centimetres relative to a local base station and a WiFi HaLow radio that links the drone into a mesh back to the ground station. The drones measure a plane of about 100 meter by 100 meter and a Gaussian process regression (GPR) reconstruction turns the point measurements into a continuous three-component velocity field with an uncertainty estimate at every point. Collision safety is handled onboard by ORCA deconfliction and flight by a PX4 controller with an onboard state estimator. The ground station is a single electric vehicle carrying the drones, a generator, a charging station and the RTK base. Set-up to first measurement is under 90 minutes.

This definition matters for the market analysis because each of these choices sets a boundary on what the system can sell. The drone count sets the spatial resolution and therefore the largest structure that can be usefully resolved. The sensor sets both the velocity accuracy and most of the unit cost. The RTK base and the HaLow mesh set how far from the ground station the swarm can fly, which caps the area covered in a single set-up and is the main reason offshore is a further pathway rather than a current segment. The single vehicle and two-person crew set the mobilisation cost, which sets the shortest campaign that is still economic. The rest of this chapter refers back to these figures rather than to the concept.

2.3. Client and customer

The Baseline report used the word client for two distinct roles. They are separated here, because they want different things.

The first is the commissioning client, the party that set the requirements: 10,000€ per acquisition day, two year payback, a two-person crew and deployment in under 90 minutes from a standard vehicle. This client is concerned with the business case closing rather than with which segment is served. Its requirements are the hard constraints the rest of the analysis must respect.

The second is the paying customer, the parties that book measurement campaigns. There are four groups and each is matched here to what an 80-point plane can actually resolve.

Wind energy is the primary group. Developers, turbine manufactures, test laboratories and certification bodies need wake data to validate power-performance test under IEC 61400-50-3 to verify wake-steering controllers and to plan re-powering. A modern onshore turbine wake at the 100 meter scale is well matched to the system's plane and resolution, which is the technical reason this is the lead segment and not an aspiration. Willingness to pay is high because the stakes are large. One percentage point of wake loss on a 500 MW farm is worth roughly a million euros a year, so a measurement that de-risks a much larger decision is inexpensive by comparison.

Built environment is the second group. Architects and facade engineers commission wind studies for tall buildings, stadiums and bridges often because a code requires it. In the Netherlands this is NEN 8100 for pedestrian wind comfort with comparable rules across Europe. Building and stadium wakes sit at a similar spatial scale to the turbine case, so the same plane and resolution apply. Each job is smaller than a wind job but there are more of them.

Maritime decarbonisation is small but high value. Suppliers of wind-assist propulsion need full-scale thrust and side-force validation to support commercial deployment under tightening IMO rules. The jobs are few but large. This segment is reachable on land where the rotor is tested ashore or dockside, but full at-sea validation runs into the offshore limits.

research and academia is the fourth group. Institutes need validation data they can not produce in-house. willingness to pay is lower but academic deployments build the track record and published results that wind energy customers will expect before they place trust in the system.

The unifying need across all four is the same: spatially resolved, three-component, full-scale wake data that simulation and scaled testing cannot reliably provide.

Serving all four groups at once risks serving none of them well, so the first target is wind energy validation, specifically the IEC 61400-50-3 power-performance use case. The clearest entry point is an independent validation house such as DNV or Deutsche WindGuard since these firms already sell wake-measurement campaign, already hold the customer trust the project lacks and operate under a standard that requires the three-component data which a single LiDAR delivers only with difficulty. The complication which should be stated directly is that these same firms are competitors. The realistic first move is therefore to supply them as a measurement subcontractor or partner rather than to displace them and to use that work to build the field record needed to approach operators such as Eneco or Vattenfall directly.

2.4. Market need and existing solutions

Across wind, construction and shipping, the aerodynamic behaviour of large structures is determined mainly from two indirect sources: numerical simulation and sub-scale wind tunnel testing. Both carry well-documented uncertainty that the parties relying on them cannot afford. A one-point wake-loss error on a 500 MW farm represents roughly a million euros a year. An underestimated wind load on a high-rise leads to costly retrofitting. An overstated thrust figure for a Flettner rotor results in a failed commercial deployment. In each case the missing element is the same: direct, full-scale, spatially resolved wake data against which to validate the models before major investments decisions are made.

Six categories of existing solution address parts of this need, and the system itself is added as a seventh row so the comparison is made on the same axes (Table 2.1)

Table 2.1: Existing solutions and LAWSS, compared on the axes that decide a wake-measurement purchase.[**Baseline**]

Solution	Velocity output	Coverage and resolution	Weather ceiling	Cost to client	Key limitation
Met mast (60 to 175 m)	3-comp, direct	A few fixed points on one line	Full	€18k to 500k installed	Sparse; cannot map a plane; slow to permit
Scanning LiDAR	1-comp along beam; 3-comp only reconstructed	Long range, line of sight	Degrades in rain and fog	€100k to 400k unit; €5k to 15k/day	One direct component; needs homogeneity assumption that fails in wakes
Single-drone anemometry	3-comp, direct	One moving point	Below ~Bft 6	€5k to 50k unit	No simultaneous area coverage
Tethered balloon / kytoon	3-comp, direct	Vertical line	Up to ~25 $\mathrm{m\,s^{-1}}$	€20k to 100k	Cannot map a plane; weather-fragile
CFD / LES services	3-comp, modelled	Virtual, any resolution	N/A	€20k to 200k/study	Not a measurement; needs validating
Sub-scale wind tunnel	3-comp, direct	Full plane, scaled	N/A	€50k to 500k/campaign	Reynolds and scale mismatch
LAWSS	3-comp, direct (sonic)	~100 by 100 m plane; 80 simultaneous points	Rain (IPX4) and up to 6 Bft	€10k/day, multi-day campaigns	Fixed 80 drones limits resolution on very large rotors; coverage capped by mesh and RTK base; high unit cost; no track record yet

Reading the table it can be concluded that the system occupies a position no single competitor holds. It is the only one that delivers three direct velocity components at 10 Hz rate across a full plane in though weather conditions without a permanent installation. Scanning LiDAR, the closest functional rival fails two of those axes at once. It only measures the component along its beam so a three-component field requires either several synchronised units or an assumption that the flow is uniform. That assumption breaks down in inhomogeneous wake regions that are the point of the measurement. t also loses data in rain and fog, which conflicts directly with the IPX4 and 6 Beaufort envelope.

Since the product design is more detailed the limits can also be more refined. Three of them follow directly from the design. First, resolution is fixed by the drone count. Eighty points across a 100 by 100 meter plane results comply with the error requirement whereas the offshore wind turbines span up to 250 meter which will lead to larger errors if the same amount of drones is used. Second, coverage in a single set-up is capped by the HaLow mesh range and the RTK base radio link. The HaLow mesh is limited to operate the drone swarm to a distance of 1.5 kilometres. This is the concrete reason offshore deployment is a future adaptation: the base would have to sit on a moving vessel and the mesh would have to reach across open water, neither of which the current ground station does. Third, the system competes on capability, not price. A single LiDAR unit is cheaper to buy than an 80-drone swarm, so LAWSS only wins where the three-component, all-weather, full-plane combination is actually needed and it losses any job a single LiDAR can already serve well enough.

2.4.1. Cost of getting the same dataset using LiDAR

The most useful way to express the value of the system is to ask what it would cost a customer to obtain the same dataset, at three-component velocity field over a 100 by 100 meter plane using the most capable alternative. That alternative is scanning LiDAR. On the five technical axes of Table 2.1, a single scanning LiDAR gives one direct velocity component. About 1 Hz effective rate per volume, long line-of-sight range but only reconstructed planar coverage and degraded performance in rain. To approach a three-component field it needs at least three synchronised units since three non-coplanar lines of sight are the minimum for a full velocity vector. Even then the result is a time-averaged rather than the 10 Hz field and still loses data in the rain and high wind. So no LiDAR setup truly matches the dataset. The estimate below is for the closest achievable equivalent, which means the resulting cost saving is a lower bound on the real value gap.

Table 2.2 compares the five-year total cost of ownership of LAWSS against a three-unit scanning LiDAR system sized to deliver that closest equivalent. The LAWSS figures are taken from the financial model, the LiDAR figures are rough and every soft input is flagged. The comparison excludes one-off development cost on both sides since a LiDAR buyer does not carry the LAWSS development either.

Table 2.2: Capability-matched five-year cost of ownership, LAWSS against a three-unit scanning LiDAR system delivering the closest equivalent dataset. LiDAR inputs are rough estimates.

Cost element (5-year)	LAWSS	3-unit scanning LiDAR
Hardware (capex)	€1.1M	€0.75M to 1.2M
Personnel	€0.92M	€1.4M (3 FTE incl. reconstruction)
Other operating	€1.08M	€0.9M
Total 5-year TCO	**€3.11M**	≈**€3.2M**
Equivalent datasets (5 yr)	≈490	≈ 15 to 20
Cost per dataset	≈**€6,300**	≈**€160k to 210k**

Two things stand out. First, the hardware cost and the total five-year cost are close, both near three million euros, so the difference is not the equipment or the running cost. It is what each system delivers for that cost. LAWSS turns the same annual spend into about 98 planar datasets a year, while the LiDAR turns it into three or four large site campaigns. Per delivered dataset that is the difference between roughly €6k and roughly €180k. The gap is real but the LiDAR provides a much larger and longer record of one site, whereas LAWSS provides a fast, repeatable snapshot. For a customer who needs to characterise many structures or who needs an answer in days rather than a season, the LAWSS data set is the better one for a fraction of LiDAR prices.

Second, the LiDAR dataset is not just larger, it is also weaker in the case that matters here. Dual and triple-Doppler reconstruction assumes a horizontally homogeneous flow, and that assumption fails in exactly the inhomogeneous wake regions the measurement is meant to capture. So in a wake the extra LiDAR data is partly biased, while the LAWSS area is measured directly. The comparison is therefore not only cheaper data, but valid data where the alternative struggles to provide it.

2.5. Market sizing

The Baseline Report claimed a 1 to 3% share of the European wake market but never put a number on what that market is worth. This section sizes the market from top down, narrows it to what the system can actually reach and then puts a bottom-up revenue figure on the achievable portion from the financial model.

To assess the total addressable market (TAM) the closest published proxy for what the system sells is studied. This is the global wind LiDAR market which is valued at about € 145 million in 2025 (approx. USD 157 million) and growing at 7.7% per year toward the low-300s by 2035 . That figure only counts cite hardware sales, which means it understates the actual measurement spend as most wind measurement is bought as a service. The upper bound is the wider wind resource assessment services market which runs to low billions of dollars as of 2024 but the bulk of that is routine resource assessment the system cite is not designed for. The addressable market sits somewhere between the two, closer to the LiDAR proxy.

Three filters narrow the TAM down to the market LAWSS can realistically compete for. The first is geography: Europe is the launch region and Europe accounts for roughly a third of the global wind LiDAR market [**<empty citation>**]. The second is scope: wake measurement specifically, rather than the broader wind profiling and resource assessment market the LiDAR proxy covers. The third is capability: the adverse-weather, three-component, land-deployable niche where scanning LiDAR is structurally weak means that the sub-segment where a customer cannot get what they need from the LiDAR technology is an open niche.

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Applying the European share and a wake fraction to the proxy puts the serviceable market in the low tens of millions of euros per year on a hardware basis. That figure is a floor rather than a ceiling, because LAWSS is sold as a service rather than as equipment. In service terms it competes for the same customer budget lines as LiDAR campaigns at €5k to 15k a day, wind tunnel campaigns at €50k to 500k each, and CFD studies at €20k to 200k each, so the relevant spending pool is larger than a hardware-sales comparison suggests. The European wind turbine services market alone is projected to grow by close to € 2.9 billion between 2023 and 2028 at around 8.5% per year [**euWindServices**] , confirming that the underlying spend on turbine-related measurement is both large and rising.

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The Serviceable obtainable market (SOM) is the best-grounded of the three figures because it comes directly from the project's own model. Each system bills 98 days per year at steady state and earns around one million euros. A fleet of 3 to 8 systems therefore brings in roughly €3 to 8 million of revenue and 294 to 784 acquisition days per year. That is consistent with the 300 to 800 days assumed in the Baseline report. Against the broad European wind and wake measurement services which runs to several hundred million euros the fleet is a few precent. The Baseline Report's 1 to 3% figure is in that ballpark, but that is not the market the system competes in. Against the specific niche it is actually built for (the adverse-weather, three-component, full-plane measurements that scanning LiDAR cannot deliver), the same 3 to 8 million euro is in the order of 15 to 30%. The higher bound is more relevant, because it reflects a near-unique capability rather than one product among several. The practical limit on share is capacity, not demand: each system needs two operators and a ground station, so obtainable share grows as the fleet grows. Therefore, the 3 to 8 system range is an early-phase number and not a ceiling.

percent of what? SOM?

lost me here

Overall, a TAM of around €145 million growing at roughly 8% a year in a European serviceable niche in the low tens of millions and an obtainable €3 to 8 million from an early fleet of systems. White this is a small slice of the broad market, it is a substantial niche that the system is specifically designed for.

share of what?

2.6. Future market development

The trends over the next five to ten years generally work in the system's favour, though that tailwind is not the same as a durable competitive advantage.

Wind capacity is on track to roughly double by 2035, which lifts demand for wake and resource measurement across the board. Turbines keep getting bigger which pushes wakes into spatial scales where met masts stop being practical and a 100 by 100 meter measurement plane becomes useful rather than just novel. That same trend is double-edged for LAWSS. An 400-point plane resolves a 100 meter rotor well, but the largest offshore rotors may need a bigger swarm, or deal with a coarser field. As such, moving into the offshore and very large rotor segment is a product decision rather than a natural extension of the current design. Wake steering is moving out of research and into normal operating practice. With that sub-market growing at double digit annual rates and every turbine operator adopting it, would require the exact kind of validation data that LAWSS produces. Regulation is also pulling in the same direction: IEC 61400-50-3 is building remote sensing into power-performance verification. Additionally, pedestrian wind comfort codes like NEN 8100 are spreading across Europe. This turns measurement from something optional into something required, which raises the floor of the addressable market. Adjacent segments are opening up as well, for example, eVTOL operators need low-altitude wind models for certification, generally in urban environments.

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genuinely shows up wayyy to often

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2.7. Competitive positioning

There are four groups of competitors, separated by how they make money.

Hardware vendors sell instruments: Vaisala/Leosphere, ZX LiDARs, Mitsubishi Electric and Halo Photonics. They own the scanning LiDAR segment, set the price expectations and have built up the IEC compliance record that any new entrant will work against.

Service providers sell campaigns rather than equipment: DNV, TÜV SÜD, UL Renewables, and Deutsche WindGuard. These firms are both competitors and the most likely first customers, which is the central tension in the client strategy in the client section.

Drone-based startups, such as Black Swift Technologies and Meteomatics, use flying platforms but target weather forecasting rather than wake mapping. However, these companies serve as useful evidence that the drone measurement approach is commercially viable rather than being direct threats to LAWSS.

Wind tunnels and CFD serve as substitutes, they are different products competing for the same budget line. However, wind tunnel experiments to replicate conditions of large engineering structures are unscaleable, or become extremely expensive. While specialized wind tunnels exist in Europe to accomplish this, such as the Onera F1 low-speed high Reynolds wind tunnel in Mauzac, France [31] or the Cryogenic Wind Tunnel (KKK) in Cologne, Germany [18], their Reynolds numbers would be insufficient for extremely large building models, they may only operate a handful of times per day and their operational costs run into The correctness of results derived from CFD must always be justified by real-world data, so the need for validation remains.

LAWSS's competitive position is unique, as it is the only deployable system that delivers a full 100 by 100 meter three-component velocity field at 10 Hz in adverse weather without a permanent installation.

The position is not secure as LiDAR vendors are developing wake mapping capability through dual-LiDAR configurations which undermines the three-component advantage from the remote sensing side. They will not match the adverse-weather operation easily but they do not need to beat LAWSS everywhere. In case of LiDAR, it is sufficient to present enough advantages in the cases customers actually care about. On top of that, LAWSS enters with no bank-ability track record against an installed base backed by more than a decade of IEC-classified field data. Bank-ability is earned slowly, one campaign at a time. Per-job hardware cost is also higher than a single LiDAR unit, so the system has to win on capability rather than price. The commercial case depends entirely on the cases where LiDAR physically cannot do the job. If that set of cases gets smaller, so does the commercial opportunity for LAWSS.

2.8. Business model: supplier or operator

The LAWSS product can be situated in two different business cases namely one where it is sold as a product or another where it is sold as a service. The project description implies that it will be sold as a service but the supplier option will still be explored in this section.

In the supplier model, systems are sold to parties already in the measurement space with recurring revenue from maintenance and support contracts. The advantages are speed and reach. Selling into firms that already have customer trust and bankability omits the biggest commercial weakness because the buyer brings the track record the system does not yet have. It also scales after and ties up less capital. The downsides are margin and control: revenue per system is lower than operating it and the company hands the customer relationship and the reputation to the buyer.

In the operator model, the company owns and runs the fleet and sells the data as a service. this captures the full value per campaign, builds the brand and the field record directly. Furthermore it keeps control of the data quality while bankability is still being established. The cost is that it is slower and capital intensive. Market share is capacity-limited rather than demand-limited and customer trust has to be built from scratch.

The two models are not excluding one another and the most interesting path the a phased path. Operating the fleet early is the only practical way to build the clean campaign record that eventually becomes bankability. Once the system is proven and a field record exist, a supplier or licensing track can be added to scale further than what a operator-only company could manage on its own.

2.9. SWOT

The SWOT connects the market picture to the design and risk work rather than floating as a stand-alone exercise. Internal factors are strengths and weaknesses; external factors are opportunities and threats. The matrix is shown in Figure 2.2

Strengths *internal, positive*

- Direct 3-component velocity, 100×100 m plane at 10 Hz, reconstruction
- Operates in rain (IPX4) and up to 6 Bft
- Two-person van deployment under 1.5 h
- Only system combining planar coverage, 3D velocity and all-weather use

Weaknesses *internal, negative*

- High unit cost (~€1.1M); 2-year payback only marginal
- Resolution capped by 80-drone density; largest offshore rotors out of reach
- Limited endurance; no bankability yet
- Lower TRL than LiDAR; swarm airspace overhead (EASA Specific Category)

Opportunities *external, positive*

- IEC 61400-50-3 makes wake measurement a requirement
- Wake-steering rollout needs validation data
- NEN 8100 urban comfort and maritime decarbonisation widen the market
- Partner with certification bodies (DNV, Deutsche WindGuard) to scale

Threats *external, negative*

- Dual-LiDAR vendors closing the 3-component gap
- Tightening EU drone rules; public resistance to flights near people
- CFD substitution as compute matures

Figure 2.2: SWOT analysis of the LAWSS wake-measurement system.

2.10. Target cost and financial viability

This section derives the cost ceiling the market and the payback rule together allow, then checks the actual design against it. The main change since the Baseline Report is that the design now has a real bottom-up cost, so this is a direct comparison rather than an estimate against a placeholder. All figures come from the current financial model.

The ceiling follows from two constraints in the project description: up to €10,000 per acquisition day and recovery of the non-recurring cost within two years of nominal operation. At steady state the system bills 98 days per year, calculated as 250 working days reduced by 30% weather downtime, 20% transit and setup and a 70% booking rate. At 10,000 per day that is about €1.02 million of revenue per system per year. Subtract the operating costs that do not scale with hardware and treat maintenance and insurance as 14.5% of hardware cost per year. Then the two-year payback rule allows a maximum hardware cost of €1,220,781 per system.

The bottom-up build comes to € 1,103,982, which is € 116,799 or 9.6 % below that ceiling. On a hardware basis the design meets the cost requirement with a margin. Within the total the drone fleet (80 drones, € 506,632) and the ground station (€ 335,352) account for most of the cost. Per drone, the TriSonica Mini at € 2,212 is the single largest line item, so unit cost is driven by the sensor choice more than the airframe. That is the most direct lever on cost whenever a larger drone count is considered

The picture gets less comfortable once development cost and the utilisation ramp are included and this is where the Baseline Report was very optimistic. The €1220781 ceiling excludes the one-off development cost and assumes full utilisation from day one. Development cost is half of the annual operating profit (non-steady) which is €579,631 and is added as a non recurring cost. If this development cost is added to one system, the ceiling will not be respected and LAWSS will not be profitable within 2 years. It would take about 2.8 years for it to be net profitable.

The requirement is about two years of nominal, steady-state operation at which point the system generates about €619,000 of operating profit per year. A single unit misses the target because it carries the entire €579,631 development cost alone. Spread that cost across a 3 to 8 system fleet and per-unit payback comes within the limit, as shown in Table 2.3.

Table 2.3: Steady-state payback as a function of fleet size, with development cost amortised across the fleet.

Fleet size	Development per unit (€)	Total investment per unit (€)	Steady-state payback
1	579 631	1 683 613	2.8 yr (misses)
3	193 210	1 297 192	2.1 yr (misses)
5	115 926	1 219 908	2.0 yr (meets)
8	72 454	1 176 436	1.9 yr (meets)

The cost requirement therefore closes only if five systems are built. This is the link back to the market sizing and it is a sound one because the fleet count comes from the market rather than being chosen to make the numbers work. The requirement is met from five units upward, the single and three-unit case is the only configuration where it is at risk.

3 Requirements and Assumptions

You can find the compliance matrix through this link: Compliance Matrix

we really need to define all terms clearly here including measurement periods, t/o and landing procedure times, crew operation assumptions, assumption about weather, etc

4 System Overview

The demand for the project is now clearly understood and necessary requirements from the system have been collected. In this chapter, the design process is summarized based on the provided requirements to showcase the priorities undertaken in the project. The final configuration is presented in this chapter as an overview of the work in the report.

4.1. System Configuration Overview

4.2. System Design Process

Figure 4.1 shows the N2 chart for the design process undertaken and in the section below, the design history is outlined. The design began with an initial guess based on preliminary drone sizing, totalling 80 drones. At the heart of the entire system lies the numerical method for flow reconstruction and subsequent aerodynamic force calculation. The most important outcome of the system is its ability to compute the aerodynamic forces within 5%, so the system must adapt to this. Therefore, Gaussian Process Regression and Adaptive Sampling Analysis is completed in 3 dimensions (setup description in Chapter 5, examples, configuration optimization and case study results in Chapter 7). A sensitivity study for the reconstruction method, with a cylindrical setup for the swarm of drones, provided the necessary number of measurement nodes of 400, and initially required measurement configuration for adaptive sampling of 80 points in 5 phases to reach the force error requirement.

Theoretical estimates for the necessary amount and distribution of measurement nodes for accurate force reconstruction are now apparent in perfect conditions. However, measurements in real world are far from perfect and must now be considered, which is done in Measurement Convergence Analysis (Chapter 6, in Section 6.1 and Section 6.2). Here, measurement uncertainties due to non-uniform inflow conditions and turbulence are considered together with potential sensor uncertainties. Understanding the Atmospheric Boundary Layer (ABL) provided an understanding of the required measurement time to capture atmospheric features well, without influence of high-period fluctuations. This resulted in a 24-minute measurement interval every hour, to offer results of atmospheric fluctuation effects throughout the day on the target structure. Given the drone measurement configuration and the required measurement time, the necessary convergence time can be decided, computed to be 6 minutes in total. Knowing this and the 5x80 sampling configuration allowed to simulate the necessary number of drones to cover all required measurement nodes in time. This configuration did not fulfil the convergence requirements for an individual drone. A recommendation was made to reduce the number of phases to 4, thereby sweeping 100 nodes per phase, which reached the necessary convergence time per drone when 15 drones sample 30 points in the inlet and 70 drones sample 70 points in the wake. Therefore, this analysis directly informs the hardware design in terms of required hover time and drone count. Additionally, system control and communication requirements are laid out (Section 6.3, Section 6.4) to provide subsystem requirements for the hardware design downstream, where the payload measurement sensors, communication and control hardware was decided.

Finally, the drone hardware can be finally sized given driving requirements from the software section (Chapter 8). Hardware has a strong feedback to the software design since it must handle the requirements set out by force accuracy requirements, in terms of feasible hover time. Simultaneously, the system must remain realistic in its operation, such that it fulfils space and time constraints (container footprint and 90 minute deployment by 2 operators), as well as operational condition constraints (weather requirements, capability to operate both onshore and offshore). The design process was organized in parallel with software analysis and was finalized once accuracy requirements were finalized. The battery is sized directly based on the necessary hover time. Given an understanding of the necessary control, positioning and communication layout, preliminary mass estimations can include these components. Propulsion is sized given requirements on gust rejection and manoeuvrability. The frame takes into account the propulsion system together with compactness requirements, as well as sensor placement requirements based on downwash effects from the propellers. The ground station is then designed together with operations and logistics (Chapter 9) around the total number of drones to fulfil the link budget, recharging requirements and footprint. The requirements of the system also included a detailed section regarding sustainable design strategy, which includes design goals for the system to minimize environmental, economic and social impacts. Therefore, this strategy served as an overarching negotiable at every phase of hardware design.

The most essential design parameter for the swarm was the number of drones required for its operation. This parameter was driven by contrasting requirements. On one hand, the container footprint, operational constraints, cost and the need to minimize the amount of rechargeable batteries for sustainability, meant that the total number of drones used for reconstruction had to be minimized. However, this back-propagated to software design in order to preserve acceptable measurement error in flow reconstruction and consequently the force accuracy. As such, the design loop emerged accordingly. Derived from this need came a complimentary requirement for drone hover time, which had to conform to necessary convergence time for the measurements taken by drones, but constrained by the battery life of the drone.

Thorough analysis of the design provided feedback on the feasibility of the requirements initially set. In case the design space became over-constrained, requirements were negotiated and altered. This occurred with regulations-related requirements on public space closure as well as safety requirements with regards to operation in crowded areas. Additionally, the need for real-time measurement feedback was reviewed, since customers do not require on-the-spot force values. However, this data is still valuable for operators to understand the state of the measurements and their quality, in order to implement corrections. The requirements on the number of drones and hover time were negotiated within the group between the hardware and software departments.

Requirements	Desired force prediction accuracy	Sensor accuracy	Operational distance requirements, pointing accuracy requirements, sampling rate requirements	Sustainable development strategy	Sustainable development strategy, weather condition requirements	Sustainable development strategy, weather sealing requirements	Container & closure footprint requirements	Sustainable development strategy, operator requirements, deployment time requirements
Requirements feasibility	Gaussian Process Regression + adaptive sampling analysis	Necessary distribtution and number of measurement points	Necessary pointing accuracy				Necessary distribtution and number of measurement points, Computational needs	
Requirements feasibility	Nr of total possible measurements, Sensor characteristics	Measurement Convergence Analysis	Necessary measurement time	Required operating time			Drone count	
Requirements feasibility			Control and Communications Analysis	Payload masses		Payload masses, dimensions, integration requirements	Control and communications performance specifications	
Requirements feasibility				Battery sizing	Drone mass	Battery mass	Battery capacity	Battery replacement time
Requirements feasibility	Necessary drone spacing to avoid prop-wash interference	Available operating time		Power usage	Propulsion sizing	Propeller diameter, Sensor boom length		
Requirements feasibility				Drag force	Drone mass	Frame & mounts sizing	Storage footprint	Ground footprint
Requirements feasibility	Maximum number of drones physically transportable			Available battery storage space		Container size	Ground station design	Battery charging time
Requirements feasibility		Total available measurement time		Power grid availability	Maximum available ground footprint		Setup/dismantling time	Operations

Legend:	Stakeholders	Software Section	Hardware Section

Figure 4.1: N2 chart of the design process

4.3. Drone System Subsystem integration

Having converged on the design using the design loop described previously, the top-down N2 chart for the entire system configuration can be generated. This chart informed the system interfaces and therefore highlighted further relationships to consider in the design during development, identifying modularity and critical subsystems in the mission.

The drone N2 chart is presented in Figure 4.2. For the LAWSS project, the system is split into three essential modules: the ground component, communication interface and the drone swarm component. As such, the three are coloured accordingly. Arrows in the chart are an example of the flow of the N2 Chart.

Since the ground and drone swarm components are interrelated through communication interface only (because human presence is only available on the ground), the communication subsystem was critical in the design and required extra attention in terms of redundancy and robustness. The ground component and the drone swarm component were designed in parallel to each other, while the communication subsystem designers concentrated on standardizing interfaces such that the two components function in unison.

4.3.1. Ground Component

The ground component includes the ground station, which is the human interface for the rest of the system. All other subsystems either receive commands or report back to this system during operation. Control commands by operators originate here, and final flow reconstruction/forces report back here as well. Aside from the ground station, all other subsystems here are software-based.

The flight swarm path planner will coordinate the drone swarm, position it initially based on operators' commands and later receive desired changes in the swarms' configuration from the adaptive sampling planner to improve the force prediction.

The flow field reconstruction subsystem will collect measurements from the drone swarm passed through the ground station and use GPR to reconstruct the wake around the building, and then using the momentum integral technique the flow can be reconstructed.

4.3.2. Communications Component

The communications component consists of its respective subsystem. It receives all commands and positioning information coming from the ground component and transmits it to the swarm. Simultaneously, drone swarm status, measurements and error states are transmitted from the drone swarm to the ground station and inter drone communication for collision avoidance is included.

In terms of system design, the communications subsystem will therefore also consist of a ground component and component per drone. This includes several interfaces: one for drone swarm control (commands and state reports), one for the RTK system position errors broadcast from the RTK base to all of the drones, another one for data transmission (velocity, pressure, time, positions in space), and one inter frone for collision avoidance.

4.3.3. Drone Swarm Component

The drone swarm component includes subsystems per drone in the swarm. This includes control, power, propulsion, structures and payload. The drone receives communications for position control and sensor actuation. The control subsystem must receive data from the propulsion subsystem for closed-loop feedback. The payload block includes the velocity and pressure measurement sensors that are fed back through the communications link to the ground station for data processing.

Ground station (human to system interface)	Initiate flow field reconstruction, velocity/pressure data	Initiate momentum integral	Control inputs for swarm, interrupt commands								
Pass reconstructed flow field, uncertainties	Flow field reconstruction subsystem	Pass reconstructed flow field, uncertainties		Pass high variance regions (to change the measurement grid of downstream drones)							
Calculated forces output		Momentum integral subsystem									
Report current drone positions			Flight swarm path planner		Control inputs for the entire swarm, flight plan for initial conditions, define global coordinates, no-fly zones, synchronise time, interrupt commands						
			Desired positions	Adaptive sampling planner							
Return data from swarm (u,v,w,p,t drone positions, prop&power health, initial conditions measurement, error states, flight time remaining, etc.)	Time, position of each drone, u, v, w, p		Time, position of each drone	Initial conditions measurement (inc. flow direction to inform downstream drones)	Communication subsystem	Positioning commands, define global coordinates, no-fly zones, interrupt commands				Interrupt commands	Mounting of antennas, signal converters, etc.
					Report time, position of each drone downstream	Control/positioning subsystem	Actuation commands	Position ICMA to measure incoming flow, pass interrupt commands	Position DCMA to measure downstream flow, pass interrupt commands	Power mode commands	Mounting of flight computers, flight sensors, positioning system, etc.
					Report Prop. health	Propulsion state	Propulsion subsystem			Current power demand	Mounting of motors, propellers, etc.
					Initial conditions measurements (u,v,w,p)			Initial Conditions Measurement Array (ICMA) (velocity&pressure sensors, memory buffer)		Current power demand	Mounting of ICMA payload
					Downstream measurements (u,v,w,p)				Downstream Conditions Measurement Array (DCMA) (velocity&pressure sensors, memory buffer)	Current power demand	Mounting of DCMA payload
					Deliver Power, Report Power Health (Flight Time Remaining)	Deliver Power	Deliver Power	Deliver Power	Deliver Power	Power subsystem	Mounting of power components, cables
											Structures subsystem

Legend:	Ground Component	Communication Interface	Drone Swarm Component

Figure 4.2: N2 chart of the system configuration

5 Gaussian Process Regression Software

The central function of LAWSS, converting a set of drone velocity measurements into an estimate of the aerodynamic force, is done by the numerical algorithm outlined in this chapter. Using a set of point samples of the mean velocity field $\overline{\mathbf{u}}$ around the building and a potential flow as input, it outputs a continuous reconstruction of the 3D velocity field over the survey domain, and the time-averaged aerodynamic force on the object using a control-volume momentum balance on the reconstructed field.

Reconstructing a continuous field from a finite set of measurement points is an ill posed problem [30]. The chosen method, Gaussian Process Regression (GPR) [37], addresses this by returning a posterior distribution rather than a single field. Hence, every posterior quantity carries a determined uncertainty which can be later exploited to determine where additional measurements are more valuable. Also, the incompressibility of the flow ($\nabla \cdot \mathbf{u} = 0$ can be ensured by construction, so that every point of the posterior is guaranteed to be divergence-free. This removes an entire set of unphysical reconstructions and is what makes the method data-efficient enough to be conducted by a small swarm.

The chapter is ordered so that the physical structure of the problem is imposed before any data is fitted. Section 5.1 introduces GPR in its general form. Section 5.2 builds a divergence-free kernel in 3D using helmholtz construction [2], separating the field into an irrotational and rotational part, modelling the the vector potential as a Gaussian Process, and derives the velocity covariance that follows from taking the curl of the underlying potential. Section 5.3 and Section 5.4 explain the remaining modelling choices, namely the base kernel and its hyperparameters [37]. Later, Section 5.5 sets out how an uncertainty field is created using the posterior variance.

With the velocity field and the uncertainty in hand, Section 5.6 calculates the time-averaged force from a control-volume momentum integral. It makes the time-averaging assumptions clear, identifies the systematic bias introduced by neglecting the Reynolds stresses. Section 5.7 and Section 5.8 define how measurement locations are chosen: an initial space-filling distribution, followed by an adaptive scheme that places further nodes where the posterior is most uncertain and flow most rotational. Finally, Section 5.9 introduces two CFD test cases: a high-rise building and an offshore wind turbine on which the pipeline is used.

Throughout this chapter the measurement process us treated as ideal: each measurement point is assumed to return the true mean velocity at its location instantaneously. This isolates the performance of GPR and the momentum integral itself, and yields the central result for the following chapter: the number of sample points required to bring the force estimate within the 5% accuracy target. The constraints of real measurement, namely the hover time needed to converge a mean within the atmospheric boundary layer and the resulting number of drones, are deferred to Chapter 6.

5.1. Principle of Gaussian Process Regression

Gaussian Process Regression was selected following a preliminary two-dimensional feasibility study which compared its performance against Ensemble Transform Kalman Filtering (ETKF), Physics Informed Neural Networks (PINNs), and direct control volume surface momentum integral evaluation. It was found in the subsequent trade-off that the Gaussian Process Regression is most suitable for LAWSS as it combines strong performance on sparse data, rapid on-site results, native uncertainty quantification, and low implementation risk in 3D.

Gaussian Process Regression is a Bayesian regression method which predicts field values at a set of desired test points (defined as coordinates at which the regressor evaluates) by evaluating their similarity in the input space to the training points (defined as where the data is measured) via a kernel, then computing the posterior distribution mean and variance at every test point. Formally, Rasmussen et al. define a Gaussian Process as a collection of random variables, with any subsample exhibiting a joint Gaussian distribution [37, p. 13]. Thus, a Gaussian Process can fully be defined by a prior mean function ($m(\vec{x})$) and a covariance function or kernel ($k(\vec{x}, \vec{x}'))$ [37, p. 13] and yields a normal distribution of admissible functions, the mean and variance of which can be known. The details of the implementation are documented in Subsection 5.2.1. Furthermore, the design of the custom divergence-free kernel in 3D is outlined in Subsection 5.2.2.

5.2. Divergence-free Velocity GP in 3D

Building upon the theoretical principles described in Section 5.1, this section details the mathematical architecture and practical implementation of the Gaussian Process regressors used. While predicting scalar fields like static pressure relies on standard regression techniques, predicting the 3D velocity field requires a more rigorous approach. To ensure the predicted flow strictly adheres to physical mass conservation laws, the software employs a Helmholtz decomposition to seperate the vector field into distinct physical components.

5.2.1. General GPR construction principles

The aim of this section is to describe the basic working principles of implementing a scalar Gaussian Process Regressor. Equivalently, the section explains how this study constructs the static pressure predictor GP based on sparse point measurements of static pressure. It also serves as the basis for the construction of the divergence-free velocity GP, further described in Subsection 5.2.2.

As explained in Section 5.1, a Gaussian Process is fully defined by its prior mean function and kernel function; this is shown in Equation 5.1 [37].

$$u(\mathbf{x}^*) = m(\mathbf{x}^*) + \mathbf{k}^{*^{\mathrm{T}}} \left(K + \sigma_n^2\, I\right)^{-1} \left(\mu(\tilde{\mathbf{x}}) - m(\tilde{\mathbf{x}})\right) \tag{5.1}$$

In Equation 5.1, m($\mathbf{x}$) represents the prior mean function (which can be thought of as an initial guess of the GP's output before factoring in the influence of the training points), with the superscript star (*) signifying evaluation at testing points, and tilde (˜) signifying evaluation at training points.

The kernel function is taken into account in the covariance matrix term (Equation 5.2, expressing the covariances between each pair of training points) and in the $\mathbf{k}^*$ matrix (Equation 5.3, expressing the covariances between each testing points and all training points). Both of these use the scalar kernel (covariance) function $k(\mathbf{x}, \mathbf{x}')$ which expresses the covariance between point $\mathbf{x}$ and $\mathbf{x}'$ using only their locations in the input space and choice of hyperparameters (discussed in Section 5.4). The scalar base kernel chosen for both the velocity and pressure GPs is the Matérn 5/2 kernel (elaborated upon in Section 5.3). The measurement noise is included with the addition of its standard deviation σ_n.

$$K + \sigma_n^2\, I = \begin{bmatrix} k(\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_1) + \sigma_n^2 & k(\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2) & \cdots & k(\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_n) \\ k(\tilde{\mathbf{x}}_2, \tilde{\mathbf{x}}_1) & k(\tilde{\mathbf{x}}_2, \tilde{\mathbf{x}}_2) + \sigma_n^2 & & \vdots \\ \vdots & & \ddots & \vdots \\ k(\tilde{\mathbf{x}}_n, \tilde{\mathbf{x}}_1) & \cdots & \cdots & k(\tilde{\mathbf{x}}_n, \tilde{\mathbf{x}}_n) + \sigma_n^2 \end{bmatrix} \tag{5.2}$$

$$k^* = \begin{bmatrix} k(\vec{x}_1^*, \tilde{\mathbf{x}}_1) & k(\vec{x}_1^*, \tilde{\mathbf{x}}_2) & \cdots & k(\vec{x}_1^*, \tilde{\mathbf{x}}_n) \\ k(\vec{x}_2^*, \tilde{\mathbf{x}}_1) & k(\vec{x}_2^*, \tilde{\mathbf{x}}_2) & \cdots & k(\vec{x}_2^*, \tilde{\mathbf{x}}_n) \\ \vdots & \vdots & \ddots & \vdots \\ k(\vec{x}_m^*, \tilde{\mathbf{x}}_1) & k(\vec{x}_m^*, \tilde{\mathbf{x}}_2) & \cdots & k(\vec{x}_m^*, \tilde{\mathbf{x}}_n) \end{bmatrix} \tag{5.3}$$

To synthesize the above, Equation 5.1 shows that the posterior mean of a GP at a test point is the prior mean at the test point corrected by a factor derived from the covariances between the test point and all sampling points ($\vec{k}^*$), scaled by the deviation of the measurement from the mean at the training points ($u(\tilde{\mathbf{x}}) - m(\tilde{\mathbf{x}})$) normalized to mitigate the influence of clustered measurements [$(K + \sigma_n^2\, I)^{-1}$].

Pressure GP implementation

For pressure prediction, a scalar GP based on the anisotropic Matérn 5/2 kernel (rationale outlined in **??**) is trained on scalar static pressure measurements at training points, then the posterior mean is evaluated at relevant test points. The hyperparameters of the anisotropic kernel were optimized as described in Section 5.4. The prior function $\mathrm{m}(\mathbf{x})$ was set to 0 on the entire domain, corresponding to free-stream (gauge) pressure. The coupling of pressure and velocity to generate a better pressure prior was considered but discontinued because of the lack of implementation time and sufficiency of the zero prior for force estimation.

5.2.2. The harmonic component and domain topology

The most direct way to regress the velocity field is to fit one independent scalar GPR to each velocity component, as suggested . This method treats u, v and w as unrelated fields, freeing it to violate mass find source conservation $\nabla \cdot \mathbf{u} = 0$. This would produce sources and sinks where flow is created or destroyed.

A divergence-free reconstruction therefore requires the components to be coupled so that incompressibility is enforces by construction rather than learned from data. Helmholtz decomposition achieves this, by splitting the velocity field into distinct physical components. It places a GPR on the vector potential, recovering the velocity as its curl. The Helmholtz decomposition based architecture was implemented analogously to prior demonstrations by Berlinghieri et al. [2], Shalashilin [39], and Padilla-Segarra et al. [33].

The Helmholtz-Hodge decomposition splits a vector field into three parts, each distinguished by how the divergence and curl operators act on it:

$$\mathbf{u} = \underbrace{\nabla\phi}_{\text{irrotational}} + \underbrace{\nabla \times \mathbf{A}}_{\text{solenoidal}} + \underbrace{\mathbf{h}}_{\text{harmonic}} . \tag{5.4}$$

The irrotational part $\nabla\phi$ is curl-free, $\nabla \times (\nabla\phi) = \mathbf{0}$, and is the only term able to carry the divergence of the field. The solenoidal part $\nabla \times \mathbf{A}$ is divergence free, $\nabla \cdot (\nabla \times \mathbf{A}) = \mathbf{0}$, and carries all the possible vorticity. The harmonic part $\mathbf{h}$ is the remainder that is simultaneously curl free and divergence-free.

Two questions determine whether this third term is needed: can $\nabla \times \mathbf{A}$ represent $\mathbf{h}$ and can $\nabla\phi$ represent it?

The first is answered independently of the domain. The harmonic component $\mathbf{h}$ cannot carry any vorticity by definition since the curl is zero. Since $\nabla \times \mathbf{A}$ is the solenoidal part of the flow, it will not capture $\mathbf{h}$. The question of whether the third term is required reduces entirely to whether $\mathbf{h}$ can instead be written as a gradient and absorbed in $\nabla\phi$.

The harmonic component $\mathbf{h}$ can be absorbed in $\nabla\phi$ if it can be written as a gradient of a scalar field ψ itself, e.g. $\mathbf{h} = \nabla\psi$. This means that the question becomes, is $\mathbf{h}$ a conservative field or not. If a field has zero circulation around every closed loop, it is a conservative one:

$$\oint_C \mathbf{h} \cdot d\boldsymbol{\ell} = 0 \quad \text{for every closed loop } C. \tag{5.5}$$

Deciding whether $\mathbf{h}$ is a gradient therefore amounts to checking whether all of its closed-loop integrals vanish. This is where the topology of the domain enters[4].

In a simply connected domain every closed loop C can be continuously shrunk to a point without leaving the domain, which guarantees that a surface S exist inside the domain having C as its boundary. Stokes' theorem then applies and relates the loop integral to the curl of that surface:

$$\oint_C \mathbf{h} \cdot d\boldsymbol{\ell} = \iint_S (\nabla \times \mathbf{h}) \cdot d\mathbf{S} \tag{5.6}$$

The integrand is the curl of $\mathbf{h}$, which is zero everywhere in the domain, so the surface integral vanishes, making the loop integral zero for every C. Hence, $\mathbf{h}$ is a conservative field and can be written as a gradient $\nabla\psi$. It is them absorbed into the irrotational term by redefining the potential, $\phi \leftarrow \phi + \psi$. The harmonic component vanishes and the decomposition reduces to two terms.

In a multiple connected domain, such as a 2D region surrounding an obstacle, there exist loops encircling the obstacle that cannot be shrunk to a point. Any surface bounded by such a loop must pass through the obstacle, outside of the domain, where $\mathbf{h}$ is not defined, making its curl not constrained to be zero. Stokes' theorem therefore cannot be applied to conclude that the loop integral vanishes, and the circulation may be nonzero, making $\mathbf{h}$ nonconservative and cannot be absorbed into $\nabla\phi$. The harmonic term must be retained, and the full three-term decomposition is required.

In the Midterm review , the domain was multiple connected, requiring the $\mathbf{h}$ term. The 3D domain considered here, a cuboid enclosing a solid building or wind turbine, is simply connected. The harmonic component therefore vanishes and the decomposition reduces to $\mathbf{u} = \nabla\phi + \nabla \times \mathbf{A}$, which will be modelled as a Gaussian Process.

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5.2.3. Modelling the rotational component as a Gaussian process

Within this framework the irrotational component, $\nabla\phi$ is known a priori: it is supplied by a incompressible potential flow source panel solver and returns a potential flow field that solves the equation $\nabla^2\phi = 0$. This field is treated as a deterministic prior mean and carries no uncertainty. The rotational component, $\nabla \times \mathbf{A}$, is the quantity to be inferred. Rather that placing the Gaussian Process directly on the velocity, it is placed on the vector potential $\mathbf{A}$ and the velocity is recovered as its curl. This guarantees that every output is divergence free, since $\nabla \cdot (\nabla \times \mathbf{A}) = 0$. This results in the incompressibility constraint $\nabla \cdot \mathbf{u} = \nabla \cdot \nabla\phi + \nabla \cdot (\nabla \times \mathbf{A}) = 0$ is enforced by construction rather than learned from data.

The vector potential, $\mathbf{A} = [A_x, A_y, A_z]$, is assigned a zero-mean Gaussian Process prior with covariance described by a kernel:

$$\mathbf{A} \sim GP\left(0, K\left(\mathbf{x}, \mathbf{x}'\right)\right). \tag{5.7}$$

A zero prior mean is assigned because, prior to any measurement, no direction or magnitude of the rotational correction is preferred. The informative trend of the field is carried entirely by the deterministic mean $\nabla\phi$.

In 3D, the covariance matrix relates every component of $\mathbf{A}$ at one point to every component at another. For a grid of two points, the covariance matrix has the following structure:

$$K = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix}, \tag{5.8}$$

where each block collects the component-wise covariances between point i and point j:

$$K_{ij} = \begin{bmatrix} \operatorname{cov}\left(A_x^{(i)}, A_x^{(j)}\right) & \operatorname{cov}\left(A_x^{(i)}, A_y^{(j)}\right) & \operatorname{cov}\left(A_x^{(i)}, A_z^{(j)}\right) \\ \operatorname{cov}\left(A_y^{(i)}, A_x^{(j)}\right) & \operatorname{cov}\left(A_y^{(i)}, A_y^{(j)}\right) & \operatorname{cov}\left(A_y^{(i)}, A_z^{(j)}\right) \\ \operatorname{cov}\left(A_z^{(i)}, A_x^{(j)}\right) & \operatorname{cov}\left(A_z^{(i)}, A_y^{(j)}\right) & \operatorname{cov}\left(A_z^{(i)}, A_z^{(j)}\right) \end{bmatrix}, \tag{5.9}$$

with $A_a^{(i)} = A_a(\mathbf{x}^{(i)})$. The components of $\mathbf{A}$ are assumed mutually independent, making the off-diagonal entries vanish:

$$\forall a, b \in \mathbb{R} : a \neq b \Rightarrow \operatorname{cov}\left(A_a^{(i)}, A_b^{(j)}\right) = 0. \tag{5.10}$$

Each block therefore reduces to a single scalar kernel on the diagonal, with no cross-component coupling,

$$K_{ij} = k\left(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}\right) I_3. \tag{5.11}$$

5.2.4. Covariance of the curl

The rotational velocity is the curl of **A**, so its statistics follow directly from those of **A**. The goal of this subsection is to turn the scalar kernel k assigned to each component of **A** into the full 3×3 covariance block of the rotational velocity. The route is fixed by a single fact: a Gaussian process is closed under linear operations [37, p. 191], and the curl is linear. Applying the curl to the mean of **A** yields the mean of the rotational velocity field, applying it to each argument of the kernel yields the rotational velocity covariance. The mean is treated first and the covariance second. For the covariance, a single entry of the block is worked out in full to expose the mechanism. and the remaining eight are assembled fron the same three-step pattern. The block that results is the curl of curl operator acting on k, and its off diagonal coupling is exactly what enforces $\nabla \cdot \mathbf{u} = 0$.

The curl is

$$\nabla \times \mathbf{A}(\mathbf{x}) = \left[\, \partial_y A_z - \partial_z A_y, \;\; \partial_z A_x - \partial_x A_z, \;\; \partial_x A_y - \partial_y A_x \,\right]^T . \tag{5.12}$$

Mean. The mean of the trnasformed process can be derived by recognizing that the expectation operator commutes with linear operators, such as the curl. Formally, by expanding the expectation into its integral definition and applying Leibnitz's rule to interchange the order of differentiation and integration. The zero prior mean for the vector potential $\mathbf{A}(\mathbf{x})$ results in a zero mean for the transformed process:

$$\mathbb{E}\left[\nabla \times \mathbf{A}(\mathbf{x})\right] = \int_\Omega (\nabla \times \mathbf{A}(\mathbf{x}, \omega))\, p(\omega) d\omega = \nabla \times \int_\Omega (\mathbf{A}(\mathbf{x}, \omega))\, p(\omega) d\omega = \nabla \times \mathbb{E}\left[\mathbf{A}(\mathbf{x})\right] = \nabla \times \mathbf{0} = \mathbf{0} \tag{5.13}$$

Where picking an ω picks out one possible realisation of the field. All possible ω are grouped in Ω, and $p(\cdot)$ is the probability density over the sample space, describing how likely each realisation is. The physical meaning of this zero mean becomes explicit when it is propagated through the Gaussian process posterior. For a test point $\mathbf{x}^*$ and a set of training measurements $\mathbf{x}$, the posterior mean is

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$$\mathbf{u}(\mathbf{x}^*) = \nabla\phi(\mathbf{x}^*) + k^* \left(K + \sigma^2 I\right)^{-1} \left(\mathbf{u}(\mathbf{x}) - \nabla\phi(\mathbf{x})\right) \tag{5.14}$$

Rearranging isolates the rotational part on both sides:

$$\underbrace{\mathbf{u}(\mathbf{x}^*) - \nabla\phi(\mathbf{x}^*)}_{\nabla\times\mathbf{A}(\mathbf{x}^*)} = \mathbf{k}^* \left(K + \sigma^2 I\right)^{-1} \underbrace{\left(\mathbf{u}(\mathbf{x}) - \nabla\phi(\mathbf{x})\right)}_{\nabla\times\mathbf{A}(\mathbf{x})} \tag{5.15}$$

Both the left-hand side and the bracketed term on the right are purely rotational, the process regresses only the curl part. It expresses the estimated rotational field at $\mathbf{x}^*$ as a weighted combination of the rotational fields observed at the training points. The weights $\mathbf{k}^*(K + \sigma^2 I)^{-1}$ decay with the distance through the kernel, so for a test point far from any measurement the cross-covariance vanishes, $\mathbf{k}^* \to \mathbf{0}$ and with it the entire correction term. In that limit, the posterior reduces to $\mathbf{u}(\mathbf{x}^*) = \nabla\phi(\mathbf{x}^*)$: in the absence of evidence the rotational estimate returns to zero and the velocity reverts exactly to the potential flow. This is the concrete consequence of the zero prior mean. The rotational field is introduced only where measurements support it, while the irrotational displacement around the obstacle is described everywhere by the deterministic prior.

Covariance. Before differentiating anything, note that the deterministic prior mean $\nabla\phi$ carries no uncertainty and therefore contributes nothing to any covariance. The covariance of the velocity is exactly the covariance of the curl, so building the velocity covariance matrix amounts to transforming k by the curl of both its arguments.

Take the top-left entry of the block: the covariance of the first velocity component with itself, evaluated at two points $\mathbf{x}$ an $\mathbf{x}'$. Writing the first component of $\nabla \times \mathbf{A}$ as $u_x = \partial_y A_z - \partial_z A_y$, this entry is

$$K_{xx} = \operatorname{cov}\left(\underbrace{\partial_y A_z(\mathbf{x}) - \partial_z A_y(\mathbf{x})}_{u_x(\mathbf{x})},\ \underbrace{\partial_{y'} A_z(\mathbf{x}') - \partial_{z'} A_y(\mathbf{x}')}_{u_x(\mathbf{x}')} \right). \tag{5.16}$$

Every other entry of the block is this same object with a different pair of velocity components, so the steps below are the recipe for the whole matrix.

By the bilinearity property of the covariance and the independency of A_x, A_y and A_z, Equation 5.16 becomes:

$$K_{xx} = \partial_y \partial_{y'} \left[\operatorname{cov}\left(A_z(\mathbf{x}), A_z(\mathbf{x}')\right)\right] + \partial_z \partial_{z'} \left[\operatorname{cov}\left(A_y(\mathbf{x}), A_y(\mathbf{x}')\right)\right]. \tag{5.17}$$

Substituting the shared kernel,

$$K_{xx} = \partial_y \partial_{y'} k(\mathbf{x}, \mathbf{x}') + \partial_z \partial_{z'} k(\mathbf{x}, \mathbf{x}'). \tag{5.18}$$

The kernel is stationary, depending on the separation $\mathbf{r} = \mathbf{x} - \mathbf{x}'$ with components such as $r_y = y - y'$. Using the chain rule:

$$\begin{aligned} \partial_y k(\mathbf{x}, \mathbf{x}') &= \partial_{r_y} k(\mathbf{x}, \mathbf{x}') \cdot \partial_y r_y = \partial_{r_y} k(\mathbf{x}, \mathbf{x}') \\ \partial_{y'} k(\mathbf{x}, \mathbf{x}') &= \partial_{r_y} k(\mathbf{x}, \mathbf{x}') \cdot \partial_{y'} r_y = -\partial_{r_y} k(\mathbf{x}, \mathbf{x}'), \end{aligned}$$

so a derivative with respect to a primed coordinate has the opposite sign of the corresponding unprimed one. Applying this to the mixed derivatives gives:

$$K_{xx} = -\partial_y \partial_y k(\mathbf{x}, \mathbf{x}') - \partial_z \partial_z k(\mathbf{x}, \mathbf{x}'). \tag{5.19}$$

Every one of the nine entries is built in the same way: pick the two velocity components, expand by bilinearity, drop the cross terms that pair independent potential components, and apply the stationary sign rule. The diagonal entries inherit the negative, summed, second-derivative pattern of K_{xx}. The off-diagonal entries keep a single surviving covariance and reduce to one mixed second derivative. Collecting K_{xx} and its eight sibling into the velocity covariance block for the point pair $(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})$ gives

$$K_{ij} = \begin{bmatrix} -H_{yy} - H_{zz} & H_{xy} & H_{xz} \\ H_{yx} & -H_{zz} - H_{xx} & H_{yz} \\ H_{zx} & H_{zy} & -H_{xx} - H_{yy} \end{bmatrix}. \tag{5.20}$$

The resulting block is exactly the curl-of-curl operator $\partial_a \partial_b\, k - \delta_{ab} \nabla^2 k$ applied to the scalar kernel, and is divergence-free by construction. The off-diagonal entries couple the velocity components, which is what enforces $\nabla \cdot \mathbf{u} = 0$. The construction requires the kernel to be at least twice differentiable for the second derivatives to exist, motivating the use of a sufficiently smooth Matérn kernel ($\nu \geq 5/2$).

Figure **??** shows the effect of using the divergence free kernel GPR on the divergence of the flow. This 3D democase uses a prior mean of $\mathbf{0}$ and only one training point in the origin for simplicity.

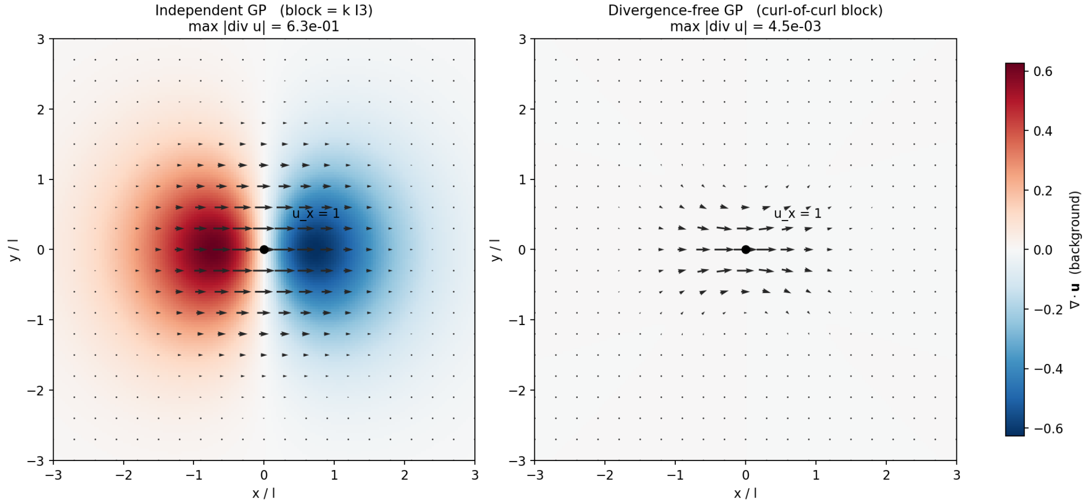

Figure 5.1: xy slices of a 3D demo with a prior mean of **0**, with one training point at the origin that measures a velocity in the x-direction of $1m/s$. The velocity field is expressed with the vectors, the divergence is shown with colour. On the left the velocity components are modelled independently, showing a lot of divergence. On the right the Helmholtz decomposition is used, showing little divergence because of finite-difference errors.

On the left in Figure **??**, the velocity field was modelled independently, resulting in a $K_{ij} = kI_3$. On the right, the Helmholtz decomposition was used, using the covariance matrix as expressed in Equation 5.20. The plot shows that for the independent GPR, the field becomes divergent, as artificial sources and sinks are being replicated. On the right, there is only little divergence due to finite difference errors.

Velocity GP implementation

For velocity prediction, the divergence-free Helmholtz kernel constructed from an anisotropic Matérn 5/2 scalar kernel is used. The Gaussian process is trained on the time-averaged velocity measurements at the training points, and the posterior mean is evaluated at the test points. The kernel hyperparameters were optimized as described in Section 5.4. In contrast to the pressure model, the prior mean is supplied by the incompressible potential flow panel solver, $\mathbf{m}(\mathbf{x}) = \nabla\phi(\mathbf{x})$, as derived in Subsection 5.2.2. This places the irrotational content in the deterministic prior and restricts the learned correction to the rotational wake, while the curl of curl kernel structure guarantees that every posterior sample is divergence free by construction.

5.3. Base Kernel Selection

The aim of this section is to outline the choice of the base kernel for the pressure and velocity GPs, explain the rationale behind the choice, and provide recommendations for future development.

The base kernel chosen for both the pressure and velocity GP is the Matérn 5/2 kernel, obtained by evaluating the general formulation of Matérn kernels for the smoothness parameter $\nu = 5/2$. The Matérn family of kernels was chosen over RBF or fourth order polynomial based on a proven track record of effectiveness on various sparse datasets [**aero**] and limited testing during the 2D feasibility study. It was hypothesized that the slight advantage is because it does not over-smooth, but this claim remains to be verified.

Half-integer smoothness was chosen because Matérn kernels simplify to a much less computationally expensive form for half-integer smoothness values. $\nu = 5/2$ was chosen because, as outlined in **??**, the divergence-free GP formulation requires the base kernel to be at least twice continuously differentiable, which Matérn 5/2 is but Matérn 3/2 is not [38].

The kernel is in the form shown in Equation 5.21. The anisotropy of the kernel is implemented by introducing three separate length scales (l_x, l_y, l_z), with the covariance function defined as:

$$k(x, x') = \sigma^2 \left(1 + \sqrt{5}\, r + \frac{5r^2}{3}\right) \exp\left(-\sqrt{5}\, r\right) \tag{5.21}$$

where

$$r = \sqrt{\frac{(x - x')^2}{l_x^2} + \frac{(y - y')^2}{l_y^2} + \frac{(z - z')^2}{l_z^2}}. \tag{5.22}$$

The length scales determine how far the effect of training points spreads over the domain. Small length scales indicate that a function rapidly changes in the given direction while large values impose high smoothness and slow change. The selection mechanism for the hyperparameters is described in Section 5.4.

5.4. Hyperparameter tuning

As outlined in Section 5.3, the base Matérn 5/2 scalar kernel $k(x, x')$ models the covariance of the underlying vector potential components (A_x, A_y, A_z), with the assumption that covariances between different components are zero as A_x, A_y, A_z are independent. As such, in order to formulate the Gaussian Process Regressor, the base scalar kernel needs to be defined by tuning its hyperparameters.

The covariance matrix based on anisotropic Matérn 5/2 kernel is based on the following hyperparameters:

$$\theta = [l_x, ly_, l_z, \sigma^2, \sigma_n^2].$$

l_x, l_y, l_z represent the physical length scales in the three global dimensions, σ^2 is the signal variance (controlling the amplitude of the kernel), and σ_n^2 is the noise variance (outlined in **??**). The hyperparameters were optimized by minimizing the negative marginal log likelihood:

$$-log(P(\mathbf{y}_{\mathbf{train}}|\mathbf{X}_{\mathbf{train}})) = \frac{1}{2}\, \mathbf{y}_{\mathbf{train}}{}^{\top} K^{-1} \mathbf{y}_{\mathbf{train}} + \frac{1}{2} \log\det K + \frac{3n}{2} \log(2\pi) \tag{5.23}$$

as was done before in this exact context by Berlinghieri et al. [2] and , also proposed by Rasmussen et al. [37, p. 19]. In Equation 5.23, the covariance matrix K is dependent on θ and n is the number of samples. Besides having precedent, this approach was chosen for several reasons: firstly, minimizing negative likelihood is equivalent to maximizing positive likelihood. Secondly, the likelihood was optimized in logarithmic space because it ensures that for all real outputs, hyperparameters must remain positive. Moreover, minimizing in log space is equivalent to minimizing in linear space because the logarithm is a monotonically increasing function, meaning the location of the minima is the same.

The hyperparameters minimizing the negative marginal log likelihood (Equation 5.23) were found using the L-BFGS-B optimizer (as implemented in scipy.optimize [42]) with 6 restarts from starts chosen with Latin Hypercube Sampling within empirically determined hyperparameter bounds (to prevent degenerate solutions).

5.5. Uncertainty Calculation

One of the major advantages of a GP is that it is able to return a measure of confidence in its predictions. At each test point, a posterior variance $\sigma^2(\mathbf{x}_*)$ can be provided that acts as an error bar: it is small where the prediction is heavily supported by measurements.

This uncertainty measure, the posterior variance at any testpoint $\mathbf{x}^*$

$$\sigma^2(\mathbf{x}^*) = \underbrace{K(\mathbf{x}^*, \mathbf{x}^*)}_{\text{prior variance}} \underbrace{- K(\mathbf{x}^*, \mathbf{X}) \left[K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 I\right]^{-1} K(\mathbf{X}, \mathbf{x}^*)}_{\text{reduction due to measurements}}, \tag{5.24}$$

The first term, $K(\mathbf{x}^*, \mathbf{x}^*)$, is the prior variance: the uncertainty in the field at $\mathbf{x}^*$ before any data is taken into account. The second term quantifies how much that uncertainty is reduced once the measurements are included. The second term is always positive, so observing data can only increase certainty.

The magnitude of the reduction is governed by the cross-variance $K(\mathbf{x}^*, \mathbf{X})$, which indicates how strongly a test point correlates with the training points through the kernel. When $\mathbf{x}^*$ is close to the measurements the correlation is high, the reduction term is large, and the posterior variance is small. When $\mathbf{x}^*$ is far from every measurement the cross-variance approaches zero, the reduction disappears and the posterior variance returns its prior value. Thus, the uncertainty grows with distance from the date, which is the expected behaviour. The inverse term $[K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 I]^{-1}$ accounts for correlations among the measurements themselves, preventing closely spaced and largely redundant observations from being counted as independent reductions in uncertainty.

Measurement noise enters through the term $\sigma_n^2 I$ added to the training covariance. For LAWSS, the velocity and pressure sensors carry some inherent uncertainty, shrinking the reduction term as it is propagated to every test point.

5.6. Force Estimation

5.6.1. Momentum Integral and Origin of Force

Estimating the aerodynamic force on a structure is a general objective of LAWSS. This section will discuss how to obtain that force in all directions, but for the purposes of this project the force in the skywards direction (Z) is disregarded. This is done because for the structures of interest, such as building, flettner rotors and wind turbines, the in plane forces are far more important. Since the force cannot be measured directly, it needs to be computed from the momentum equation in a control volume that contains the body, but large enough to lie outside the viscous boundary layer. The general integral formulation of the momentum equation in 3D for a control volume V with surface area A is:

$$\frac{\partial}{\partial t} \iiint_V \rho \mathbf{u}\, dV + \iint_A \rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n})\, dA = -\iint_A p\mathbf{n}\, dA + \iint_A \tau \cdot \mathbf{n}\, dA + \iiint_V \rho \mathbf{g}\, dV. \tag{5.25}$$

The following simplifying assumptions are made:

1. **Viscous stress neglect:** the full surface includes pressure, viscous and velocity contributions, however assuming that the measurements are taken far enough from the building, the viscous term can be neglected as pressure forces are far larger. This cannot be applied to the ground viscous layer which will be discussed further in Subsection 5.6.4
2. **Incompressibility** the flow is assumed to be incompressible which for the scale of 100m and velocities under 20 m/s is well within the accurate range of incompressibility and temperature homogeneity. Important note is that the density on different days and different atmospheric(temperature) conditions can be significantly different.
3. **Excluded gravitational forces applied to fluid** To make the CFD results simpler, the gravitational body force acting on the fluid was omitted. This drops the term $\iiint_V \rho \mathbf{g}\, dV$, but could be reinstated by modifying the pressure term to include a gravity term $p^* = p + \rho g z$. However, this is determined to be outside the scope of the system as the ground simplification, which will be discussed Subsection 5.6.4 makes the vertical force estimation completely incomplete, and this pressure term will only contribute to the vertical force as it varies only with z.

With these assumptions in mind, Equation 5.25 reduces to a description of the relation between the unsteady flow, the momentum flux and the pressure force,

$$\rho \frac{\partial}{\partial t} \iiint_V \mathbf{u}\, dV + \rho \oiint_A \mathbf{u}(\mathbf{u} \cdot \mathbf{n})\, dA = -\oiint_A p\mathbf{n}\, dA. \tag{5.26}$$

Following a similar derivation as in the midterm report, the aerodynamic force on the building can be expressed as [43]

$$\mathbf{R} = -\oiint_{A_{\text{out}}} p\,\mathbf{n}\,dA - \rho \oiint_{A_{\text{out}}} \mathbf{u}(\mathbf{u}\cdot\mathbf{n})\,dA - \rho\frac{\partial}{\partial t}\iiint_V \mathbf{u}\,dV. \tag{5.27}$$

5.6.2. Time Averaging of the Force

Section 6.1 elaborates on data handling required for the functioning of LAWSS, and it's explained that the drones send averages to the computational subsystem, which are the same quantities produced by the RANS simulations. Every flow quantity is decomposed into a time mean (represented by the overline) and a zero mean fluctuation (represented by the apostrophe):

$$\mathbf{u} = \overline{\mathbf{u}} + \mathbf{u}', \qquad p = \overline{p} + p', \qquad \overline{\mathbf{u}'} = \mathbf{0}, \qquad \overline{p'} = 0, \tag{5.28}$$

and the average is applied to the whole Equation 5.27. There are three properties of time averages used in the following derivations:

1. Time average is linear, commuting with the volume/surface integrals, with $\partial/\partial t$ and the constants ρ, $\mathbf{n}$, dA and dV pass through it.
2. A mean is constant through time, acting as a constant, e.g. $\overline{\overline{f}} = \overline{f}$.
3. As described in Equation 5.28, the mean of a lone fluctuation is $\overline{f'} = 0$.

The unsteady term is linear in $\mathbf{u}$, averaging leaves only the mean,

$$\overline{\rho\frac{\partial}{\partial t}\iiint_V \mathbf{u}\,dV} = \rho\frac{\partial}{\partial t}\iiint_V \overline{(\overline{\mathbf{u}} + \mathbf{u}')}\,dV = \rho\frac{\partial}{\partial t}\iiint_V \overline{\mathbf{u}}\,dV, \tag{5.29}$$

the fluctuation contributing $\overline{\mathbf{u}'} = \mathbf{0}$.

Reynolds Stresses

The momentum flux, $\mathbf{u}(\mathbf{u}\cdot\mathbf{n})$, is the only nonlinear term. Implementing the Einstein notation, let u_i represent the velocity vector and n_i the outward unit normal vector and $i, j \in \{1, 2, 3\}$ are the three dimensions. The dot product $\mathbf{u}\cdot\mathbf{n}$ is expressed as the scalar notation $u_j n_j$. Hence, the ith component of the momentum flux vector is written as:

$$[\mathbf{u}(\mathbf{u}\cdot\mathbf{n})]_i = u_i u_j n_j \tag{5.30}$$

Factoring the time independent ρ and n_j out of the average leaves $\overline{u_i u_j}$. Substituting the decomposition and expanding the product into all four pairings gives

$$\overline{u_i u_j} = \overline{(\overline{u_i} + u_i')(\overline{u_j} + u_j')} = \underbrace{\overline{u_i}\,\overline{u_j}}_{\text{means}} + \underbrace{\overline{u_i}\,\overline{u_j'}}_{=0} + \underbrace{\overline{u_j}\,\overline{u_i'}}_{=0} + \underbrace{\overline{u_i' u_j'}}_{\text{Reynolds stress}}. \tag{5.31}$$

The first term is a product of means and survives unchanged. The two cross terms both have a factor of a mean of a deviation, essentially multiplying a mean with zero. The fourth term is a product of two fluctuations with nothing to factor out, so it can not be reduced. This fourth term is called the Reynolds stress. In vector form the averaged flux is

$$\overline{\rho\,\mathbf{u}(\mathbf{u}\cdot\mathbf{n})} = \underbrace{\rho\,\overline{\mathbf{u}}(\overline{\mathbf{u}}\cdot\mathbf{n})}_{\text{mean flux}} + \underbrace{\rho\,\overline{\mathbf{u}'(\mathbf{u}'\cdot\mathbf{n})}}_{\text{Reynolds stress flux}}. \tag{5.32}$$

The pressure term is linear again in p, and since the mean of the fluctuation is zero: $\overline{p'} = 0$, it commutes to the mean of the pressure.

Hence, from the three terms, only the momentum flux produces a new object: the Reynolds stress. This is the equation where turbulence enters as a term in its own rather than just shaping the mean.

Simplifying Assumptions
With the averaged equation in hand, the following assummptipns are made.

1. **Statistical stationarity:** Even though the flow is unsteady, under fixed inlet conditions its statistics are independent of time, making the averaged unsteady vanish:

$$\frac{\partial}{\partial t}\iiint_V \rho\overline{\mathbf{u}}\,dV = 0. \tag{5.33}$$

   Important to note is that this is not an approximation of an unsteady field by a steady one but a consequence of working with stationary statistics.

2. **Reynolds stress neglected:** The reconstruction only supplies the mean of its measurements. The CFD simulation is not time resolved, but time averaged. For the proof of concept, to obtain this Reynolds stress, the turbulent viscous term and the k terms would have to be decomposed into the Reynolds stresses. Since this would be a very synthetic operation and not accurate to real life, these stresses would be calculated through in flight time data as discussed in Section 6.1. The RANS simulation ground truth bypasses the Reynolds stress by integrating at the object's walls. Since with LAWSS this is not feasible as the drones can't be directly next to the building, nor can GPR or any method interpolate accurately enough that far away from the sampling points, a bias is likely introduced.

5.6.3. Resulting Equation

Applying the specified assumptions,the unsteady, viscous and gravitational terms drop and the pressure and momentum contributions reduce to their means. Retaining the Reynolds stress symbolically, the aerodynamic force on the enclosed body is the inverse of the net mean momentum and pressure leaving the outer boundary,

$$\overline{\mathbf{R}} = -\oiint_{S_{\text{out}}} \overline{p}\,\mathbf{n}\,ds - \rho\oiint_{S_{\text{out}}} \overline{\mathbf{u}}(\overline{\mathbf{u}}\cdot\mathbf{n})\,ds - \underbrace{\rho\oiint_{S_{\text{out}}} \overline{\mathbf{u}'(\mathbf{u}'\cdot\mathbf{n})}\,ds}_{\text{not evaluated}}. \tag{5.34}$$

Because the reconstruction provides only the mean field, the final Reynolds stress flux is dropped, leaving the expression actually evaluated by the integration routine,

$$\overline{\mathbf{R}} \approx -\oiint_{S_{\text{out}}} \overline{p}\,\mathbf{n}\,ds - \rho\oiint_{S_{\text{out}}} \overline{\mathbf{u}}(\overline{\mathbf{u}}\cdot\mathbf{n})\,ds. \tag{5.35}$$

5.6.4. Ground Interference Simplifications

Large errors in the 3D momentum force estimations come from the effects of the ground. These were not present in the feasibility study since the building region itself could be fully enclosed without any difficulties. When extended to 3D, for the non-idealistic case of non-floating buildings this becomes impossible due to the connection of the building to the ground.

To address this problem, the ground is excluded from the region as this also solves the problem with having to collect samples on the ground itself, where data is highly inaccurate.

If the ground is perfectly flat as it is in Section 5.9, several simplifications are valid. The no-penetration condition of velocity never having a component in the direction of the ground, alongside the pressure component always acting on the area vector which points along the z axis means that from the equationEquation 5.35, the terms are acting on the R_z component of the force, which we are choosing to disregard as unimportant. However, the problem is not fully resolved as the viscous term that is on the ground but not included in the equation does contribute to the forces in x and y. Thus, some error is expected along it.

For a non-flat ground this limitation has to be considered as a genuine limitation of the system - the only mitigation possible is to disregard the lowest few meters that are too close to the ground and accept that the force estimation cannot be done for the entirety of the building.

Further study will have to be done on what happens when the wind profile is physically accurate and logarithmic, since the viscous boundary layer extends to over 100 meters in the real world, a CFD with this profile will be needed. Matching the ground roughness to the inlet profile is going to take considerable time, so this is considered outside the scope of the project, the expected effect on the force should still be small, considering the layer has fully developed before the building.

5.6.5. Mesh Construction

In order to accommodate multiple meshes, if a more complex geometry is determined necessary for more accurate flow reconstruction, which is not known pre-sampling, a method of ball-pivoting was implemented, which splits the geometry defined by the points into triangles, based on some maximal distance allowed between the points (upon which the radii of the ball pivoting method depend), so that surfaces where holes are meant to be, such as the bottom, where the ground is do not get filled up. For the purposes of this project a pre-determined cylindrical shape was used for each test-case, greatly simplifying and making the determination of the surface far more robust, since the ball pivoting method needs precise definitions of radii. Both with the ball pivoting method and the direct surface construction, to make sure normals are in the correct direction, a reference point in the middle of building is taken, and a constraint that the dot product between the normal and the vector to the face should be positive: $\mathbf{n_i} \cdot \mathbf{v_i}$, where $\mathbf{v_i}$ is the vector from the centre of the building to the middle of the face. The resulting mesh is demonstrated in Figure 5.2, where a highly exaggerated low resolution surface is shown to better illustrate the geometry.

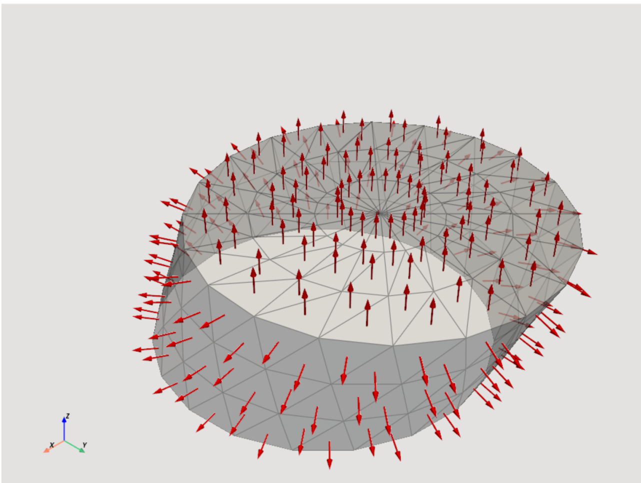

Figure 5.2: Exaggerated low-resolution mesh with normal vectors used by the momentum equation

5.6.6. Numerical Integration

Since the GPR can be sampled anywhere on the region, an extremely fine mesh can be used to converge the force, so a simplistic method was taken in discretizing the integral. The method consists of averaging the velocities and the pressures over each one of the three points in every triangle, and then applying that with respect to the area and normal of the triangle they belong to. The result is the following discretized force integral:

$$\overline{\mathbf{R}} \approx -\sum_{k=1}^{N_t} \left(\bar{p}_k \, \mathbf{n}_k + \rho \, \bar{\mathbf{u}}_k \left(\bar{\mathbf{u}}_k \cdot \mathbf{n}_k\right)\right) \Delta A_k, \tag{5.36}$$

where N_t is the total number of triangles in the mesh, $\mathbf{n}_k$ is the outward unit normal of triangle k, and ΔA_k is its area. The averaged pressure $\bar{p}_k$ and velocity $\bar{\mathbf{u}}_k$ at each triangle are computed from its three vertices,

$$\bar{p}_k = \frac{1}{3}\sum_{i=1}^{3} p\left(\mathbf{x}_k^{(i)}\right), \qquad \bar{\mathbf{u}}_k = \frac{1}{3}\sum_{i=1}^{3} \mathbf{u}\left(\mathbf{x}_k^{(i)}\right), \tag{5.37}$$

with $\mathbf{x}_k^{(i)}$ denoting the i-th vertex of triangle k. The first term inside the sum is the pressure force contribution of the triangle and the second is its momentum flux contribution.

A convergence study on the actual data from the CFD Section 5.9 was done, which showed that the force in the Y direction of the domain (in the windward direction) converges at around 100 000 points, to a force of 214.6 kN (shown in Verification in Figure 10.1). When comparing this to the value from the Subsection 5.9.1, which is F_y =208,647 kN. The differences in this force are considered to be due to the ground interference and the Reynold stresses as per Subsection 5.6.2, Equation 5.6.2. The full convergence study is outlined in Verification and Validation Chapter 10.

5.7. Initial Sampling Strategy

The momentum-integral force evaluation needs the reconstructed velocity on a surface enclosing the building, and the GP must be trained from drone measurements before any field information exists. The initial pass therefore places the first batch on a cylinder enclosing the obstacle, instead of a simple box, for its edge-free side surface on which the normal is everywhere continuous. The cylinder is additionally tilted downstream, so each horizontal ring is shifted streamwise with height, separating drones at different heights horizontally to prevent vertical stacking and avoid mutual prop-wash interference on the measurements.

The cylinder is built in a horizontal frame aligned with the freestream. The inflow $\mathbf{v}_\infty$ defines a downstream unit vector $\hat{\mathbf{s}}$ and a cross-wind unit vector $\hat{\mathbf{c}} = (-\hat{s}_y, \hat{s}_x, 0)$. Its radius and height scale to the building characteristic size L_c, the larger of the building's cross-wind width or its vertical height, as $R = f_r L_c$ and $H = f_h L_c$. The tilt is applied by shifting the ring centres. The rings remain horizontal and of equal radius, but each is translated streamwise in proportion to its height,

$$\mathbf{P}(z) = \mathbf{P}_0 + \tan(\theta)\,(z - z_{\mathrm{mid}})\,\hat{\mathbf{s}}, \tag{5.38}$$

where the tilt angle θ sets the streamwise shift per unit height and $\mathbf{P}_0$ is the horizontal center of the building. A surface point at height z and azimuth ϕ is then

$$\mathbf{x}(\phi, z) = \mathbf{P}(z) + R\big(\cos\phi\,\hat{\mathbf{s}} + \sin\phi\,\hat{\mathbf{c}}\big), \tag{5.39}$$

with $\phi = 0$ pointing into the wake. Besides the prop-wash separation it provides by construction, the shear keeps the sampling surface following the downstream-deflected wake at every height.

Points are placed by Latin Hypercube Sampling in (ϕ, z) for low-discrepancy coverage, with a fraction f_{front} drawn in a wedge $|\phi| \le \phi_{\mathrm{half}}$ about the wake to concentrate measurements where gradients are steepest. An optional top cap places measurements on the disk at the cylinder's top, drawing the radial coordinate as $\rho = R\sqrt{u}$ with $u \sim \mathrm{LHS}[0,1]$ for area-uniform density. The bottom of the cylinder is left open, since the building meets the ground there and the surface cannot be closed; the consequences of this for the force integral are discussed in Subsection 5.6.4.

5.8. Adaptive Sampling

5.8.1. Principle

The initial cylinder distributes drones with no knowledge of the flow, spending as many measurements on the inflow as on the wake (phase 0). However, the reconstruction error is far from uniform. It concentrates in the wake behind the building, where the shear layers and recirculation zone are hardest to recover, while the surrounding field is captured well. With only a limited number of drones, concentrating them where the reconstruction is harder, rather than spreading them uniformly, uses the budget more effectively. Adaptive sampling does this by placing only an initial batch blindly and then using the model trained on the data so far to decide where the next measurements are most needed, repeating this after every phase [26]. A closely related gradient- and uncertainty-driven strategy was applied to Gaussian-process reconstruction of underwater-vehicle wake fields from sparse data [45], from which the present approach is adapted.

Adaptive sampling criteria balance two competing aims [26]. *Exploration* places points where the model is most uncertain, resolving regions not yet captured. *Exploitation* places points where the response varies most strongly, concentrating measurements on the structures hardest to reconstruct. Using only one is deficient. Pure exploration chases posterior variance, which is largest far from existing samples, so it drifts toward the domain edges rather than the physically interesting flow. Pure exploitation never ventures beyond the regions already known to be difficult. The present strategy combines both, using the GP posterior variance for exploration and flow-feature indicators for exploitation, and adds points over successive phases that each re-train the GP.

5.8.2. Difficulty score

After each pass a difficulty score is evaluated, combining one exploration term and two exploitation terms. The exploration term is the GP posterior standard deviation $\hat{\sigma}(\mathbf{x})$, which is large where measurements are sparse. The exploitation terms are the magnitude of the speed gradient $\|\nabla\|\mathbf{u}\|\|$, large across shear layers and separation, and the vorticity magnitude $\|\nabla\times\mathbf{u}\|$, large in the rotational wake. Using these two rotation-invariant feature magnitudes, rather than a single lumped sum of all velocity derivatives, lets the score distinguish shear from rotation and target each separately. The vorticity term in particular captures the cross-stream shear that dominates a bluff-body wake without committing to a hand-chosen derivative component. Both are quantities long used as refinement indicators in feature-based mesh adaptation.

The three terms differ greatly in magnitude, so each is normalised to $[0, 1]$ over the domain before they are combined into the score

$$D(\mathbf{x}) = w_{\text{var}}\,\hat{\sigma}(\mathbf{x}) + w_{\text{grad}}\,\overline{\|\nabla\|\|\mathbf{u}\|}(\mathbf{x}) + w_{\text{vort}}\,\overline{\|\nabla\times\mathbf{u}\|}(\mathbf{x}), \tag{5.40}$$

where the overbar denotes the normalised field and the weights $w_{\text{var}}, w_{\text{grad}}, w_{\text{vort}}$ set the exploration and exploitation balance. The gradient and vorticity are evaluated by finite differences directly on the GP posterior. Rather than building a grid, the posterior is evaluated at each proposed point and at its six neighbours one step $\pm\Delta x, \pm\Delta y, \pm\Delta z$ away, so the derivatives are formed exactly where the score is needed, without interpolating from a precomputed grid.

The variance term alone is not sufficient. As noted above, maximising posterior variance pushes points toward the edges of the sampled region, where the GP is least constrained, rather than toward the physically interesting flow. Adding the gradient and vorticity terms anchors the score to the wake structures that actually drive the reconstruction error.

5.8.3. Candidate generation and selection

A batch is proposed in three steps: a large candidate pool is generated, scored, and reduced to a well-spread set that respects the inter-drone spacing.

Candidate pool

Candidates are drawn over the same sheared cylinder as the initial pass, but through a thick-walled shell spanning $[(1-t_{\text{in}})R,\ (1+t_{\text{out}})R]$ rather than the single surface, so points probe inward and outward of the nominal radius. The pool is generated by Latin Hypercube Sampling in the shell coordinates of azimuth, height, and radius for low-discrepancy coverage, with the same downstream-wedge bias as the initial cylinder so candidates concentrate on the wake side. The pool is oversampled well beyond the phase budget to give the selection room to choose. Candidates lying within the building are removed before scoring.

Scoring and importance resampling
Each candidate is scored by Equation 5.40, and the pool is reduced to a working set by drawing candidates with probability proportional to D^β. The exponent β sharpens the bias toward high-score regions. Drawing by probability rather than simply keeping the highest-scoring candidates is deliberate. Retaining the top-scoring points would cluster the entire batch in the single most difficult region and leave the rest of the wake unmeasured, whereas probabilistic selection still favours difficult regions while preserving coverage elsewhere. Setting $\beta = 0$ removes the bias entirely and recovers uniform sampling over the shell, which serves as a control for whether the scoring helps at all.

Spacing and exclusion
From the working set the final batch is chosen by greedy farthest-point selection under an anisotropic distance metric, following the maximin space-filling principle [21]. Starting from the highest-scoring candidate, each subsequent point is the one maximising the minimum distance to all points already chosen. The metric is an ellipsoid wider vertically than horizontally, which serves two purposes at once. It spreads the batch so the points are not redundant, and it enforces a hard minimum separation, since drones flown together must stay clear of one another's prop-wash. Any candidate falling within the exclusion ellipsoid of an existing or already-accepted point is rejected, so the spacing constraint is satisfied by construction. The budget is split across three radial bands, on the nominal surface, just inside, and just outside, so coverage is not drawn entirely onto the surface.

5.8.4. Phased refinement
The proposed batch is added to the accumulated training set and the GP is re-trained on the full set, yielding a sharper posterior and an updated difficulty score for the next phase. Repeating this produces a sequence of phases that progressively concentrate measurements where the reconstruction remains weakest, while every phase respects the same inter-drone spacing. The strategy exposes several free parameters: the score weights, the sharpening exponent β, the shell thickness, the radial budget split, and the number of phases against the points placed per phase. The optimal and the study that motivated it, is presented in Section 7.5.

5.9. CFD Case Studies
Publicly available experimental measurements around full-scale large structures are limited, especially when detailed wake fields and aerodynamic force data are required. Therefore, two CFD case studies were produced to provide controlled reference flow fields for evaluating the GPR reconstruction method and the momentum-deficit-based force estimation. The selected cases represent the two main application areas considered in this project: flow around a large bluff body, represented by the Aerospace High-rise building at TU Delft, and flow through an energy-extracting rotor, represented by the NREL 5 MW offshore reference wind turbine. The following subsections describe the setup, assumptions, and limitations of each simulation.

These CFD results should not be interpreted as exact representations of the real flow physics. Both simulations were performed as steady-state cases, meaning that unsteady wake behaviour, vortex shedding, and transient turbulent structures are not resolved directly. In addition, turbulence is modelled rather than fully resolved, so the accuracy of the results depends on the chosen turbulence model, mesh resolution, and near-wall treatment. A further limitation is the finite size of the computational domain, which means that the boundary conditions may influence the flow around the object and its wake. Ideally, a domain-independence study would be performed by comparing simulations with different domain sizes and confirming that the main results remain unchanged. However, due to the available time and computational resources, this verification step was not included. As a result, the CFD cases are mainly used as practical test cases for the reconstruction methodology, rather than as fully validated simulations of the real structures. In addition, besides the fact that the GPR kernel enforces a divergence free field, the GPR method is not physics informed, so it is never going to produce inaccurate results because of the limitations of the simulation.

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5.9.1. First Proof-of-Concept (Aerospace High-rise)

The first CFD case study considers the flow around the Aerospace High-rise building at TU Delft. For the purpose of this proof-of-concept, the building geometry was simplified to a rectangular bluff body with dimensions of approximately $49.1 \times 13.28 \times 52.6$,m. The computational domain was defined with a length of 500,m, a width of 200,m, and a height of 250,m, as seen in Figure 5.4. The incoming flow was prescribed with a velocity magnitude of 13.6,m/s and was directed at an angle of 30° relative to the building, allowing the wake reconstruction method to be tested under non-axis-aligned flow conditions.

The simulation was performed as a steady-state RANS case using the k-ω SST turbulence model. The mesh contained approximately 7 million cells, with a near-wall resolution corresponding to a representative y^+ value of approximately 3. A velocity inlet boundary condition was applied at the upstream boundary, while a pressure outlet was used at the downstream boundary. Symmetry boundary conditions were imposed on the two lateral boundaries and the top boundary to reduce artificial wall effects. The building surfaces and the ground were modelled as no-slip walls. This setup provides a controlled CFD reference field for testing the GPR-based wake reconstruction and the subsequent momentum deficit integration around a large bluff body. The calculated forces are $F = [155,433; 208,647; 72,586]$ N which is expressed in the axes system from Figure 5.4.

Figure 5.3: TU Delft Aerospace building (source: TU Delft) [15].

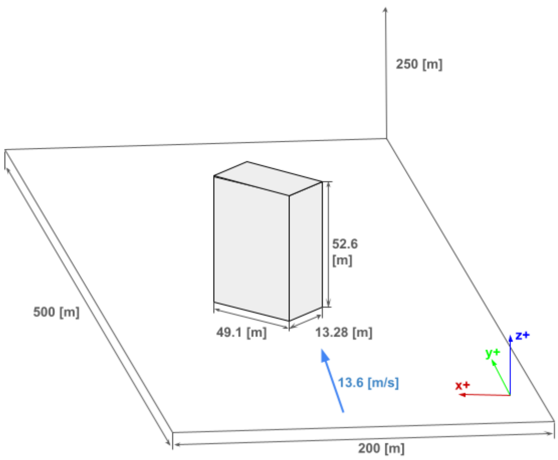

Figure 5.4: Domain definition for the ground truth of the TU Delft Aerospace building proof-of-concept. The building is rotated +30° with respect to the free stream.

5.9.2. Second Proof-of-Concept (NREL 5MW Offshore Wind Turbine)

The wind turbine was represented using a volume actuator-disk formulation rather than a zero-thickness fan boundary condition. This was done so that the rotor forcing could be distributed over a finite fluid region and so that additional source terms for swirl and turbulent kinetic energy could be included. The reference turbine used for the model was the NREL 5 MW offshore baseline wind turbine [23]. The actuator disk was defined as a cylindrical fluid zone of thickness $\delta = 5$ m, outer radius $R = 63$ m, and excluded hub radius $R_h = 3$ m. In the present coordinate system the incoming flow is aligned with the positive y direction, and the actuator disk is therefore normal to the y axis.

Table 5.1: Parameters used in the actuator-disk UDF derivation.

Parameter	Symbol	Value	Source
Rated electrical power	P_e	5.0×10^6 W	Jonkman et al. [23]
Generator efficiency	η_g	0.944	Jonkman et al. [23]
Rated mechanical rotor power	P_{mech}	5.2966×10^6 W	P_e/η_g
Rotor diameter	D	126 m	Jonkman et al. [23]
Rotor radius	R	63 m	$D/2$
Hub diameter	D_h	3 m	Jonkman et al. [23]
Excluded hub radius	R_h	3 m	Model geometry
Rated wind speed	U_∞	11.4 m s^{-1}	Jonkman et al. [23]
Rated rotor speed	n	12.1 rpm	Jonkman et al. [23]
Angular rotor speed	Ω	1.267 rad s^{-1}	$2\pi n/60$
Air density	ρ	1.225 kg m^{-3}	CFD material setting
Rated thrust coefficient	C_T	0.77	Actuator-disk model input
Actuator-disk thickness	δ	5 m	Model geometry
Disk centre	(x_0, y_0, z_0)	$(0, -6, 93)$ m	Model geometry
Base TKE fraction	β	0.02	Model assumption
Sensitivity TKE fraction	β_{sens}	0.10	Sensitivity case

The axial momentum source was obtained from the rated thrust coefficient. At the rated operating point, the mean pressure drop associated with the actuator disk is

$$\Delta p = \frac{1}{2}\rho C_T U_\infty^2 \tag{5.41}$$

Substituting the values from Table 5.1 gives

$$\Delta p = \frac{1}{2}(1.225)(0.77)(11.4)^2 = 61.29 \text{ Pa.} \tag{5.42}$$

Since the actuator disk is represented as a finite volume of thickness δ, the corresponding streamwise momentum source is obtained by distributing this pressure drop over the disk thickness,

$$S_y = -\frac{\Delta p}{\delta} \tag{5.43}$$

The negative sign is used because the turbine extracts momentum from the incoming positive-y flow. The central hub region was excluded from the forcing region, so a small correction was applied to preserve approximately the same total thrust over the reduced annular area,

$$S_y = -\frac{\Delta p}{\delta}\frac{R^2}{R^2 - R_h^2}. \tag{5.44}$$

With $R = 63$ m, $R_h = 3$ m, and $\delta = 5$ m, this gives

$$S_y = -\frac{61.29}{5}\frac{63^2}{63^2 - 3^2} = -12.29\ \mathrm{N\,m^{-3}} \tag{5.45}$$

This value was implemented in the UDF as the constant axial body-force coefficient,

$$\texttt{FY\_CONST} = -12.29 \tag{5.46}$$

The swirl source was derived by matching the rated mechanical torque of the NREL turbine. The rated mechanical power is obtained from the rated electrical power and generator efficiency,

$$P_{\text{mech}} = \frac{P_e}{\eta_g} = \frac{5.0 \times 10^6}{0.944} = 5.2966 \times 10^6\ \mathrm{W} \tag{5.47}$$

The rated angular velocity is

$$\Omega = \frac{2\pi n}{60} = \frac{2\pi(12.1)}{60} = 1.267\ \mathrm{rad\,s^{-1}} \tag{5.48}$$

The corresponding rated low-speed-shaft torque is therefore

$$Q = \frac{P_{\text{mech}}}{\Omega} = \frac{5.2966 \times 10^6}{1.267} = 4.18 \times 10^6\ \mathrm{N\,m} \tag{5.49}$$

A smooth axisymmetric tangential body-force distribution was then prescribed inside the actuator disk,

$$F_\theta(r) = K_{\text{swirl}}\, r \left(1 - \frac{r^2}{R^2}\right) \tag{5.50}$$

where r is the radial distance from the rotor axis in the x-z plane. This form was chosen because it is zero at the centre, increases through the blade region, and returns to zero at the blade tip. The torque generated by this tangential force distribution is

$$Q = \int_V r F_\theta(r)\, dV \tag{5.51}$$

For an actuator disk of thickness δ, this becomes

$$Q = 2\pi\delta K_{\text{swirl}} \int_{R_h}^{R} \left(r^3 - \frac{r^5}{R^2}\right) dr \tag{5.52}$$

Evaluating the integral gives

$$Q = 2\pi\delta K_{\text{swirl}} \left[\frac{R^4}{12} - \frac{R_h^4}{4} + \frac{R_h^6}{6R^2}\right] \tag{5.53}$$

The torque-matched swirl coefficient is therefore

$$K_{\text{swirl}} = \frac{Q}{2\pi\delta \left[\frac{R^4}{12} - \frac{R_h^4}{4} + \frac{R_h^6}{6R^2}\right]}. \tag{5.54}$$

Using $Q = 4.18 \times 10^6$ N m, $R = 63$ m, $R_h = 3$ m, and $\delta = 5$ m gives

$$K_{\text{swirl}} = 0.10136\ \text{N}\,\text{m}^{-4} \tag{5.55}$$

This value defines the base, torque-matched swirl case. A sensitivity case was also considered in which the swirl coefficient was multiplied by a factor of five.

$$K_{\text{swirl,sens}} = 5K_{\text{swirl}} = 0.50680\ \text{N}\,\text{m}^{-4} \tag{5.56}$$

This amplified case was not intended to represent the rated NREL turbine directly. It was used only to assess the sensitivity of the reconstructed velocity and force fields to stronger wake rotation. The tangential body force was decomposed into Cartesian source terms for Fluent. For a cell centroid at (x, y, z), the radial distance from the rotor axis is

$$r = \sqrt{(x - x_0)^2 + (z - z_0)^2} \tag{5.57}$$

For rotation about the positive y axis, the tangential unit vector is

$$\mathbf{e}_\theta = \left(\frac{z - z_0}{r}, 0, -\frac{x - x_0}{r}\right). \tag{5.58}$$

The swirl source terms implemented in the UDF are therefore

$$S_x = F_\theta(r)\frac{z - z_0}{r} \tag{5.59}$$

$$S_z = -F_\theta(r)\frac{x - x_0}{r}. \tag{5.60}$$

The sign of the imposed rotation can be reversed by multiplying both components by -1. In the UDF this was controlled by the parameter `SWIRL_SIGN`.

A turbulent kinetic energy source was also included to represent the unresolved turbulence generated by the rotor. This term was not taken directly from the NREL data, since the actuator-disk model does not resolve the blade boundary layers, blade-tip vortices, or blade-passing structures. Instead, the source was scaled as a fraction of the local axial power extraction per unit volume,

$$S_k = \beta \left|S_y u_y\right|, \tag{5.61}$$

where u_y is the local streamwise velocity and β is a modelling parameter. For the base case, $\beta = 0.02$ was used. At the rated wind speed, this gives the approximate scale

$$S_k = 0.02\,|(-12.29)(11.4)| = 2.80\ \text{W}\,\text{m}^{-3}. \tag{5.62}$$

For the amplified sensitivity case, the TKE fraction was increased to $\beta = 0.10$, giving

$$S_{k,\text{sens}} = 0.10\,|(-12.29)(11.4)| = 14.01\ \text{W}\,\text{m}^{-3}. \tag{5.63}$$

The final source-term coefficients used in the base actuator-disk model were therefore

$$S_y = -12.29\ \text{N}\,\text{m}^{-3}, \qquad K_{\text{swirl}} = 0.10136\ \text{N}\,\text{m}^{-4}, \qquad \beta = 0.02. \tag{5.64}$$

For the sensitivity case, both the swirl coefficient and the turbulent kinetic energy source fraction were increased by a factor of five. The axial thrust source was left unchanged so that the mean thrust loading remained consistent with the rated actuator disk estimate. The sensitivity coefficients were therefore

$$\texttt{FY\_CONST} = -12.29, \qquad \texttt{K\_SWIRL} = 5(0.10136) = 0.50680, \qquad \texttt{TKE\_FRACTION} = 5(0.02) = 0.10. \tag{5.65}$$

This sensitivity case is not torque-matched to the NREL rated rotor. It was used only to evaluate how stronger wake rotation and turbulence production influence the reconstructed velocity field and the resulting force estimate. The physically motivated base case remains the one with `K_SWIRL` = 0.10136 and `TKE_FRACTION` = 0.02.

6 Data Processing, Control, and Communication

6.1. Data Processing

The LAWSS system will be measuring flow velocity and pressure within Earth's atmospheric boundary layer (ABL) which includes many perturbations (of different kinds) that will add noise to the velocity and pressure measurements. This section will focus on the meteorology theory and sampling theory used to process the raw incoming data from the sensors onboard the drones. It will take the reader through the derivation for the running mean variance which will dictate one of the four conditions for convergence used by the drones to determine when the running mean has converged to an acceptable degree of accuracy. It will detail the remaining convergence conditions which triggers each drone to be dispatched to the next sampling point using a Hungarian algorithm which will be elaborated in Section 6.3.

6.1.1. Atmospheric Boundary Layer (ABL) Theory

The atmospheric boundary layer is similar in concept to a boundary layer found upon the surface of an airfoil. The velocity across the ABL varies greatly with respect to altitude but unlike the boundary layer found on the surface of an airfoil, the velocity does not decrease gradually in a quadratic manner with respect to distance from the ground. The velocity has fluctuations and varies unpredictably. There are two fundamental fluctuations that add to the uncertainty of the measured velocities.

Firstly, there is the random fluctuation variance (σ_u) which is high frequency fluctuations on top of the average velocity at a measured point. These fluctuations are essentially the signal noise that the sensors will read when sampling. The random fluctuation is independent of the sampling duration because there will always be random noise in the collected velocity measurements irrespective of sampling duration. This is inherent to unstable wind conditions that are always present. This random fluctuation will be necessary in the calculation of the running mean variance.

The other type of fluctuation is the low frequency atmospheric variation itself where the mean wind speed and mean wind direction change over long periods of time. These can be compared to the tidal variations of oceans over the course of long time intervals. Both of these fluctuations need to be taken into account when processing the raw velocity when determining whether the running mean has converged to a satisfactory accuracy level in order to trigger the drone to be dispatched to the next sampling point. However, the convergence criteria must be selected carefully since a sampling duration that is too long (> 20 min) will result in the atmospheric variations being absorbed into the running mean calculation. A sampling duration that is too short will not give the running mean enough time to converge, making this a balance.

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we need to justify our choic of 24 min utes mea surement time here

Taylor's Frozen Turbulence Hypothesis

When truly random noisy data is sampled, each additional data point is often assumed to be independent of one another. However, due to wind gusts and lulls, this assumption is invalid and the correlation between sampling points belonging to the same wind gust must be taken into account. The integral time scale $\mathcal{T}_{int}$ is a quantity describing the average length of time across which measurements are correlated. The exact mathematical derivation of this quantity will be detailed in Subsection 6.1.3 and the numerical approach to calculating it onboard the drones using incoming data will also be explained. To estimate the integral time scale, sampled data and the lag between them must be known which takes some time to stabilize when the drone starts sampling. However, the integral time scale is necessary to estimate the running mean variance and the effective independent intervals making it necessary to have a prior estimate of the integral time scale until sufficient data has been sampled for the lag-1 correlation estimation of the integral time scale to kick in and provide more accurate and continuous estimation of the integral time scale. This prior estimation will be estimated using Taylor's frozen turbulence hypothesis.

Taylor's frozen turbulence hypothesis is a theory often applied to atmospheric turbulence where eddies of large scales are found. The hypothesis states that if a turbulent eddy advects past a fixed point in space faster than it has time to change shape, then the passage of this eddy can be approximated as a fixed eddy of unchanging spatial pattern moving at constant speed across the point [29]. In other words, if the turbulence intensity is small enough with respect to the mean wind velocity, then the turbulent eddies can be approximated as fixed patterns across time frames. Since a wind gust is essentially the passage of a strong eddy, the time it takes for a large turbulent eddy to pass across a fixed point can be approximated as the integral time scale since measurements within the same eddy will be correlated [29]. In this context, the fixed point is the sensor onboard the drone swarm measuring the wind velocities and the time it takes for the gust to pass over the sensor can be approximated using Taylor's hypothesis as follows:

$$\mathcal{T}_{int} \approx \frac{L_u}{\bar{U}} \tag{6.1}$$

where L_u is the size of the turbulent eddy passing the drone and $\bar{U}$ is the mean velocity measured at the location of the sensor.

- Important to note the distinction between theoretically derived $\mathcal{T}_{int}$ and numerically derived $\hat{\mathcal{T}}_{int}$.
  - For theoretically derived $\mathcal{T}_{int}$, Taylor's Frozen Turbulence Hypothesis holds true and is applicable.
- Explain Taylor's Hypothesis: $\frac{L_u}{\bar{U}} \approx \mathcal{T}_{int}$ throughout.
- Explain that this can't be used as a priori since this might not hold true at the specific location where the drone is sampled due to high turbulence intensity, and cannot estimate autocorrelation accurately.
  - So, Taylor's Hypothesis is used as a prior **only** for burn in time to decide when the drone has been at the location long enough (enough independent intervals) for the sampling to be considered reliable and prevent sampling from stopping prematurely.
- Explain conditions for Taylor's hypothesis and why it is appropriate in this scenario, for determining "burn time" a priori.

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6.1.2. Velocity Measurement: Running Mean vs True Mean

With the above definitions of random fluctuation, the raw velocity data sampled by the sonic anemometers can be decomposed into a true mean velocity ($\bar{U}$) superimposed with random noise error ($u'(t)$). Together, the raw measured velocity can be written as shown in Equation 6.2

$$u(t) = \bar{U} + u'(t) \tag{6.2}$$

It is important to note that this is a theoretical decomposition since the true mean of the measured velocity can never be accurately known. Therefore, the mean is instead estimated using a running mean (U_T) which stabilises as more raw sampling points are added. From classic signal analysis, the time averaged value of a signal is found using Equation 6.3.

$$U_T = \frac{1}{T}\int_0^T \bar{U} + u'(t)dt \tag{6.3}$$

Equation 6.3 can be simplified to find the difference between the running mean and the true mean which would effectively be quantifying the error from the noisy fluctuations in the sample raw velocity which is shown below:

$$
\begin{aligned}
U_T &= \frac{1}{T}\int_0^T \bar{U} + u'(t)dt \\
&= \frac{1}{T}\int_0^T \bar{U}dt + \frac{1}{T}\int_0^T u'(t)dt \\
&= \bar{U} + \frac{1}{T}\int_0^T u'(t)dt. \\
\text{error} = U_T - \bar{U} &= \frac{1}{T}\int_0^T u'(t)dt
\end{aligned} \tag{6.4}
$$

Since the true mean is unknown in reality, the running mean is taken as the best current estimate of the "true mean". This assumption becomes more justified as the sampling time T increases because the longer the sampling time is, the closer this running mean converges to the true mean. This convergence will be later imposed in the form of a condition to ensure that the running mean does not vary by more than $x\%$ over N independent samples.

6.1.3. Running Mean Variance Derivation

Calculating the running mean of the measured velocities is irrelevant unless it can be determined when the running mean has converged to a sufficiently accurate degree to allow the drone to transit to the next sampling point. The first condition that will be imposed on the running mean convergence is that the variance in the running mean must fall within a certain percentage (i.e. 5%) of the running mean, called the acceptable error Epsilon ϵ, and once this is satisfied (along with other conditions), the running mean is considered to have converged. To this extent, the variance of the running mean must be calculated which will be derived in this subsection. Using the definition of variance as the expected value of the error of a function, and how the sampled error is defined in Equation 6.4, the variance of the running mean can be defined as shown in Equation 6.5.

$$
\sigma_{U_T}^2 = E[\text{error}^2] = E\left[\left(\frac{1}{T}\int_0^T u'(t)dt\right)^2\right] \tag{6.5}
$$

Then:

$$
\left(\frac{1}{T}\int_0^T u'(t)\,dt\right)^2 = \frac{1}{T^2}\left(\int_0^T u'(t)\,dt\right)\left(\int_0^T u'(t)\,dt\right),
$$

The definite integral evaluates to the same value irrespective of the variable of integration. Hence, the variable t can be replaced by two different variables t_1 and t_2 which denote the random noise error at two different time instances:

$$
\begin{aligned}
&\frac{1}{T^2}\left(\int_0^T u'(t)\,dt\right)\left(\int_0^T u'(t)\,dt\right) \\
&= \frac{1}{T^2}\left(\int_0^T u'(t_1)\,dt_1\right)\left(\int_0^T u'(t_2)\,dt_2\right) \\
&= \frac{1}{T^2}\int_0^T u'(t_1)\left(\int_0^T u'(t_2)\,dt_2\right)dt_1, \\
&= \frac{1}{T^2}\int_0^T \int_0^T u'(t_1)u'(t_2)\,dt_1 dt_2
\end{aligned}
$$

Therefore,

$$\sigma^2_{u_T} = \left(\frac{1}{T}\int_0^T u'(t)\,dt\right)^2 = \frac{1}{T^2}\int_0^T\int_0^T u'(t_1)u'(t_2)\,dt_1dt_2$$

$$\sigma^2_{U_T} = \frac{1}{T^2}\int_0^T\int_0^T E[u'(t_1)u'(t_2)]dt_1dt_2 \tag{6.6}$$

Now, to evaluate the integrand, the expected value of the random noise fluctuations at the two time instances of t_1 and t_2 must be evaluated. When $t_1 = t_2$, the operand of the expected value operator simply becomes $E[u'(t) \cdot u'(t)]$ which is the square of the random error at a single time instance. This was defined earlier in Equation 6.4 to be equal to the variance. Therefore, the variance of the random fluctuation (σ_u) is defined in Equation 6.7.

$$\sigma^2_u = E[u'(t) \cdot u'(t)] \tag{6.7}$$

To evaluate the integrand now where $t_1 \neq t_2$, the influence that the random fluctuation at one time instant has on the fluctuation at another time instant must be derived. This is where the autocorrelation function is used to relate the fluctuations at two different time instances. In introductory signal analysis courses such as AE2235-II, raw sample data is considered independent of the data at every other instant making the data truly random. However, within the ABL, there are wind gusts/lulls and sample points within a single gust are not independent. This is because if one were to measure the wind speed at $t = t_1$ and $t = t_2$ where both t_1 and t_2 are time instants within the same gust, then if the wind speed is high at $t = t_1$, it is highly likely that it will also be high at $t = t_2$. Therefore, measurements within the same gust or lull will be correlated and this is accounted for by the autocorrelation function. Since only the separation in time between the two time instances matters and not the actual time instances themselves, the time lag τ is the core input for the autocorrelation function which defines how much lag there is between t_1 and t_2, as shown below:

$$\begin{aligned} t_1 &= t \\ t_2 &= t + \tau \\ \tau &= t_2 - t_1 \end{aligned}$$

The autocorrelation function represents how related the wind velocity and turbulence measurements at a specific time instant are correlated with each other across different time instances [17]. An autocorrelation function uses the distance in either time or space between two inputs to output the similarity or correlation between those two inputs [17]. Since the wind speed fluctuations are being measured at the same location but at different time instances, the autocorrelation will be using the 'distance' across time as the input to determine how correlated the measurements are. Dosio et al. investigated the use of different autocorrelation functions including using an exponential autocorrelation to model the relation between velocity measurements within the turbulent atmospheric boundary layer [6]. It was concluded that the autocorrelation of the horizontal velocity measurements within the ABL was related using an exponential function which is characteristic of stochastic turbulent motion [6]. However, the autocorrelation for the vertical velocity component was more complex and in their simulations, it exhibited "wave-like" motion due to the ABL boundaries, and the integral time scale calculated was zero [6]. This is where the simulation used to calculate convergence times in the LAWSS system deviates from the research study. In the wake survey system simulation of the running mean calculation, the three dimensional components of velocity at the inlet are assumed to exhibit the simplified exponential autocorrelation behaviour for the scope of this analysis. Therefore, for the inlet simulation, the exponential autocorrelation function is used exclusively for all velocity components and is shown in Equation 6.8.

$$\rho(\tau) = e^{\frac{-\tau}{T_{int}}} \tag{6.8}$$

Equation 6.8 indicates how the correlation in velocity between measurements from two time instances decreases exponentially to zero as the phase lag increases. This is intuitive since the longer the phase lag between two measurements, the greater the likelihood that they belong to different wind gusts and motions making them uncorrelated. The autocorrelation function also indicates a perfect autocorrelation of 1 when the two measurements are from the same time instant ($\tau = 0$), which is also intuitive since any measurement will be perfectly correlated with itself.

Recommencing from Equation 6.6, Stephen B. Pope in his textbook, 'Turbulent Flows', defines the auto-covariance function $R(\tau)$ when measuring the velocity of a turbulent flow as the expected value or the covariance of the fluctuations at two time instances [35]. The autocorrelation function $\rho(\tau)$ is then defined as the normalisation of this with the variance of the random fluctuations σ_u^2 [35].

$$R(\tau) \equiv E[u'(t) \cdot u'(t+\tau)]$$

$$\rho(\tau) \equiv \frac{R(\tau)}{\sigma_u^2}$$

This gives the following:

$$E[u'(t_1)u'(t_2)] = E[u'(t_1)u'(t_1+\tau)]$$

$$\begin{aligned}
\sigma_{U_T}^2 &= \frac{1}{T^2}\int_0^T \int_0^T E[u'(t) \cdot u'(t+\tau)]dt_1 dt_2 \\
&= \frac{1}{T^2}\int_0^T \int_0^T R(\tau) dt_1 dt_2 \\
&= \frac{1}{T^2}\int_0^T \int_0^T \sigma_u^2 \rho(\tau) dt_1 dt_2 \\
&= \frac{\sigma_u^2}{T^2}\int_0^T \int_0^T \rho(\tau) dt_1 dt_2.
\end{aligned}$$

The time instances t_1 and t_2 themselves are irrelevant because only the time lag between them τ matters meaning the autocorrelation function applies in the same manner to any time instant t and another time instant $t+\tau$ with a time lag of τ. Therefore, the differential terms dt_1 and dt_2 can be replaced by change of variables to dt and $d\tau$. This results in a transformation of the domain of integration due to change of variable which is shown in Figure 6.1.

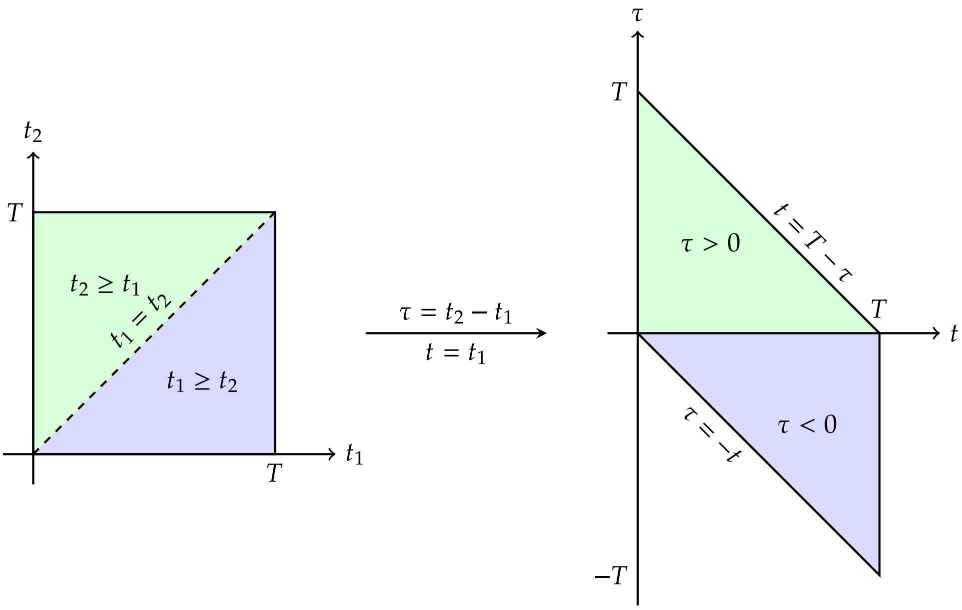

Figure 6.1: Integration Domain Transformation due to Change of Variables

Please note that integrating over the entire domain with respect to the new variables (i.e. both green and purple triangles) is equivalent to evaluating the double integral across the domain of a single triangle (i.e. green) and doubling this result since the autocorrelation function is an even function and the areas of the two triangles are equivalent [5]. Intuitively, the autocorrelation function is an even function, so $\rho(\tau) = \rho(-\tau)$. This is why the integral with respect to the variable t has domain from $0 \to T - \tau$ and is multiplied by factor 2 which is brought outside the double integral as the coefficient of σ_u^2. When replacing the differential terms with the new variables, the Jacobian must be taken into account for scaling the area of integration between the two variable coordinates:

$$J = \begin{bmatrix} \frac{\partial t_1}{\partial t} & \frac{\partial t_1}{\partial \tau} \\ \frac{\partial t_2}{\partial t} & \frac{\partial t_2}{\partial \tau} \end{bmatrix} \quad \text{where} \quad \begin{matrix} t_1 = t \\ t_2 = t_1 + \tau \\ t_2 = t + \tau \end{matrix}$$

$$J = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \to \text{Det}(J) = (1 \cdot 1) - (0 \cdot 1) = 1 \to dt_1 dt_2 = dt d\tau$$

So, applying the change of variables gives the following:

$$\sigma_{U_T}^2 = \frac{2\sigma_u^2}{T^2} \int_0^T \int_0^{T-\tau} \rho(\tau) dt d\tau$$

Since the autocorrelation is independent of the exact time instant t and only depends on time lag τ, it can be brought outside the integral with respect to the variable t as a constant. Evaluating the inner integral then yields the following:

$$\sigma_{U_T}^2 = \frac{2\sigma_u^2}{T^2} \int_0^T \rho(\tau) \cdot \left(\int_0^{T-\tau} dt \right) d\tau$$

$$\sigma_{U_T}^2 = \frac{2\sigma_u^2}{T^2} \int_0^T \rho(\tau) \cdot (T - \tau) d\tau$$

Factoring out the sampling duration T results in

$$\sigma_{U_T}^2 = \frac{2\sigma_u^2}{T}\int_0^T \left(1-\frac{\tau}{T}\right)\rho(\tau)d\tau$$

In the simulation, it is assumed that gust lengths will on average be much shorter than sampling lengths ($\tau \ll T$). This is also reflective of the real world case, where the drones will be measuring for durations of minutes whereas the gusts are estimated to be seconds to maximum tens of seconds:

$$\sigma_{U_T}^2 = \frac{2\sigma_u^2}{T}\int_0^T \rho(\tau)d\tau$$

The integral time scale is defined as $\mathcal{T}_{int} \equiv \int_0^\infty \rho(\tau)d\tau$ which captures the approximate average length of gusts. It can also be thought of as the average length of time in which wind velocities are correlated with one another. This simplifies the integral to the following:

$$\sigma_{U_T}^2 = \frac{2\sigma_u^2}{T}\cdot\mathcal{T}_{int}.$$

This ultimately gives the final form (Equation 6.9) of the running mean variance $\sigma_{U_T}^2$ as a function of the inherent velocity fluctuation variance σ_u^2, the integral time scale $\mathcal{T}_{int}$, and the sampling duration T. This equation is derived formally in the textbook, 'The Structure of Atmospheric Turbulence' by Lumley and Panofsky and is often referred to as the Lumley-Panofsky relation of the error variance of the time averaged mean of velocity within the atmospheric boundary layer [27]. There, σ_u^2 is referred to as ensemble variance which reflects sensitivity to small fluctuations in measured velocity with respect to the true mean velocity which aligns with the previous definition earlier in this section [27].

$$\boxed{\sigma_{U_T}^2 = \frac{2\sigma_u^2\mathcal{T}_{int}}{T}.} \tag{6.9}$$

Equation 6.9 is very important for determining the convergence criteria for the drones to decide when to move to the next sampling time. However, instead of using the variance as a convergence criteria, the standard deviation of the running mean estimate will be used instead as shown in Equation 6.10.

$$\sigma_{U_T} = \sqrt{\frac{2\sigma_u^2\mathcal{T}_{int}}{T}} \tag{6.10}$$

Notes on the Running Mean Variance Relation

Before using the Lumley-Panofsky relation to determine the convergence time of the sampled raw velocity data, it is important to take note of the relation between the variables in the relation as well as how it reflects the real world wind speeds within the ABL. The following trends are reflected in the Equation 6.9 noted below.

Firstly, variance of the time averaged mean is clearly dependent on the random fluctuations captured by σ_u^2. It makes intuitive sense that if the random fluctuations of the wind velocity is high, meaning the sampled raw velocities are very noisy, the variance of the running mean itself will also be high. This is accentuated by the squared relation between σ_u and σ_{U_T}. Furthermore, the fact that random fluctuations do not diminish with increased sampling duration T also reflects how the running mean will have a large variance σ_{U_T} even with longer sampling intervals if the random fluctuations are large in magnitude.

Secondly, the running mean variance is influenced not only by the random noise and velocity fluctuations but also by the low frequency 'tidal' motions of velocity which is characteristic of the ABL.

These low frequency tidal changes are caused by gusts and large scale eddies in the ABL and they are captured by the $\mathcal{T}_{int}$ term in Equation 6.10.

Additionally, although the random fluctuations do no subside with increased sampling duration since this is categorised as aleatory uncertainty, sampling for a longer time period T, will result in a decrease in the running mean variance. This is due to the law of large numbers where the running mean will converge to a constant value with some variance as T increases. The tidal variations eventually cancel each other out resulting in a convergence of the running mean.

One of the more important trends in Equation 6.9 is between $\mathcal{T}_{int}$ and T. For longer gusts (larger $\mathcal{T}_{int}$), the velocities are correlated with one another over longer time periods meaning these sinusoidal trends in the velocity have longer periods. This would require longer sampling time for the low frequency variations to cancel each other out and for the running mean to converge, hence it requires a longer sampling time T. For long integral time scales, the sampling algorithm might assume that the running mean has converged whereas in reality, it might simple be within a long gust or lull and this results in an incorrect premature termination of sampling.

To account for the above phenomena, one of the convergence criteria discussed in Subsection 6.2.1 will take into account the number of independent or effective sampling intervals N_{eff} which is equal to the sampling duration T divided by the integral time scale. This is because measurements with time lag exceeding the integral time scale are considered independent while measurements with a time lag less than the integral time scale are said to be correlated. As a result, the most important parameters that will be swept in a Monte Carlo Simulation of the sampling algorithm in different atmospheric conditions will be the integral time scale $\mathcal{T}_{int}$, turbulence intensity I_u, and the effective independent sampling intervals N_{eff}. This will be detailed in Subsection 6.2.2.

6.1.4. Numerical Approach to Integral Time Scale and Running Mean

The theory introduced in Subsection 6.1.3 is an analytical approach to understanding the behaviour of wind speeds and impact of atmospheric parameters on the statistics of the measured wind speeds. However, this needs to be converted to a numerical approach for the computers onboard the drone swarm to calculate the running mean and its variance completely online as raw velocity data is incoming in order to determine when the running mean has converged to a satisfactory level. To this extent, this subsection will detail how the running mean, its variance, and the integral time scale is calculated and improved on as new data is sampled by the sensors. Please note that the python code implementation of Welford's algorithm explained below and the lag-1 estimation of the autocorrelation function to estimate the integral time scale was done with the help of Anthropic Claude Sonnet Model. It took as input the theory derived by hand in the previous section regarding the Lumley-Panofsky relation, exponential autocorrelation functions, and running mean calculation. The authors however take responsibility for the assumptions, inaccuracies and the system model as a result of this AI use.

CITE AP- PROPRI- ATELY THE USE OF AI IN THE CRE ATION OF THE DATA SAM- PLING MODEL

Welford's Algorithm and Running Mean Calculation

Calculating the mean and standard deviation of a complete dataset is straightforward. However, in the case of the wake survey system, the sensors onboard the drone will be collecting velocity measurements in three dimensions and feeding it to the onboard computer as raw data. This constant in flow of raw data will need to be converted into a running mean and standard deviation but this is difficult if the computer's onboard memory gets full rapidly with large amounts of incoming data. Therefore, Welford's algorithm will be used to calculate the running mean and standard deviation of the incoming velocity measurements. Welford's algorithm is highly suited for this application since it only requires the previous mean $\bar{u}_{n-1}$ and current data point u_n and current total number of data points n to update the running mean. Therefore, this algorithm is not memory intensive and does not require large memory or computational power onboard the drone. The algorithm is shown below in Equation 6.11 as to how the running mean is updated with each new sampled raw data point.

$$\begin{aligned}\bar{u}_n &= \frac{\bar{u}_{n-1}(n-1)+u_n}{n}\\ &= \frac{\bar{u}_{n-1}n-\bar{u}_{n-1}+u_n}{n}\end{aligned}$$

$$\bar{u}_n = \bar{u}_{n-1} + \frac{u_n - \bar{u}_{n-1}}{n} \tag{6.11}$$

The variance of the random fluctuations σ_u^2 is also updated using a similar Welford's algorithm where the sum-of-squared-deviations of each additional sampling point with respect to the updated mean from Equation 6.11 is updated. This update is divided by the current number of sampling points $n-1$ to get the variance using the definition of variance in Equation 6.12 where the numerator is the updated sum-of-squared-deviations.

$$\sigma_u^2 = \frac{\sum_{i=1}^{n}(u_i - \bar{u}_n)}{n-1} \tag{6.12}$$

This is all updated with each additional sampled velocity measurement in the code in a function called `update_welford` as follows:

```
s.wf_n    += 1               # update number of sample points 'n'
d          = u - s.wf_mean    # deviation of current sample point from old mean
s.wf_mean += d / s.wf_n      # update mean using welford algorithm
s.wf_M2   += d * (u - s.wf_mean)            # update sum of squared deviations
var_u      = s.wf_M2 / max(s.wf_n - 1, 1)   # calculate variance of fluctuation
```

Thus the Welford algorithm allows for calculation of the running mean as new sample points are added, as well as calculating σ_u^2 which is needed to calculate the variance of the running mean $\sigma_{U_T}^2$ as shown in Equation 6.9.

Autocorrelation Function and Integral Time Scale Estimation

Along with the random noise variance σ_u^2, the integral time scale is also necessary to estimate the running mean variance $\sigma_{U_T}^2$. It is a very crude approach to estimate the integral time scale to be constant through the sampling process for two fundamental reasons. Firstly, gusts and lulls will not be identical across the entire 100x100m sampling area and secondly, the gusts will not be of the same length throughout the sampling duration. Hence, it is more appropriate to estimate and update the integral time scale continuously throughout the sampling period similar to the running mean.

The integral time scale is estimated and updated continuously in the simulation using a similar Welford algorithm. The autocorrelation is estimated using the current measured velocity and the measured velocity from the previous time. It essentially updates a list of the product of the current velocity and the velocity from the previous time instant (hence the name 'lag-1') as well as the variance which is just the difference between the mean of the sampled velocities squared and the square of the mean. Once the autocorrelation ρ_1 is found, it uses the fact that the auto-correlation function is estimated to be an exponential function as described in Subsection 6.1.3, to estimate the integral time scale and update this estimation continuously. This is done as follows:

$$\rho(\tau) = e^{\frac{-\tau}{\mathcal{T}_{int}}}$$

The numerical version is as follows:

$$\rho(\tau) = e^{\frac{-\Delta t}{\mathcal{T}_{int}}}$$

$$\mathcal{T}_{int} = \frac{-\Delta t}{\ln(\rho)}$$

Since the autocorrelation is calculated at every time instant, it can be treated as an unknown variable and $\ln(\rho)$ can be approximated to its first order form using the Taylor's series for the function $\ln(x)$. Moreover, since the autocorrelation is being estimated with a time lag of one time step, which is really small, it can be approximated to 1 since consecutive measurements are highly likely to be correlated. This results in the following:

$$\mathcal{T}_{int} = \frac{-\Delta t}{\frac{\rho-1}{\rho}}$$
$$\mathcal{T}_{int} = \frac{\Delta t \cdot \rho}{1-\rho} \tag{6.13}$$

The above equation is used to update the integral time scale with each additional sampled velocity measurement and together with the running mean and fluctuation variance explained earlier, they are used to calculate the variance of the running mean $\sigma^2_{U_T}$ completely online.

6.2. Measurement Convergence Analysis for Drone Measurements

The theory behind velocity fluctuations within the ABL, the running mean calculation and the associated variance detailed in Subsection 6.1.3 is important, but how is it actually used to determine whether the running mean calculated onboard the drone has converged? This section will explore the four convergence criteria in detail that determine whether the running mean that is constantly updated with incoming raw data, has indeed converged, which will trigger the drone to hover to a new sampling location. This section will also explain the simulation procedure used to determine confidence interval thresholds for different turbulence intensities and integral time scale lengths used for convergence which was estimated using mean convergence time.

6.2.1. Determining Convergence Time within ABL

Overall, there are three convergence criteria used in the inlet sampling algorithm to determine when the running mean from the measured wind velocities has stabilised. There is an additional condition added to the three for sampling in the wake which will also be discussed in this subsection.

Condition 1: Running Mean Standard Deviation is Acceptable

The running mean standard deviation is defined as previously in Equation 6.10. Then, the first condition imposed to ensure convergence is that the standard deviation is within a certain percentage of the running mean. This is defined as the Epsilon 'ϵ' parameter in the evaluation, defined as

$$\epsilon = \frac{\sigma_{U_T}}{U_T}.$$

By default, the epsilon is set to:

$$\epsilon = 0.08,$$

However, as described further, this parameter will become a variable in order to trade-off the convergence time and measurement accuracy, since the system is the most sensitive to this parameter.

Condition 2: Running Mean Stability

The variance in the running mean is influenced by the random fluctuations but also by the general low frequency 'tidal' motions and variations of the velocity measured.

This is defined as the 'delta δ' parameter in the evaluation. By default, the delta is set to:

$$\delta = 0.05.$$

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Condition 3: Running Mean Variance Confidence Interval

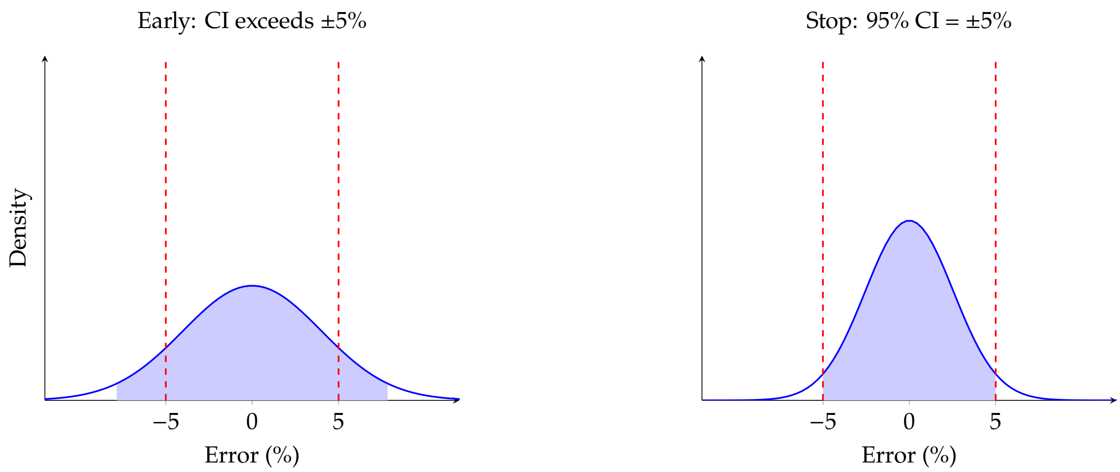


Figure 6.2: The running mean's 95% confidence interval narrows as samples are collected. Sampling stops when the confidence interval reaches the required ±5% error tolerance.

During sampling of the running mean and the computation of its variance, a normal distribution may be assumed. unfinishe

This is defined as the 'z-score' parameter in the evaluation. By default, the z-score is set to:

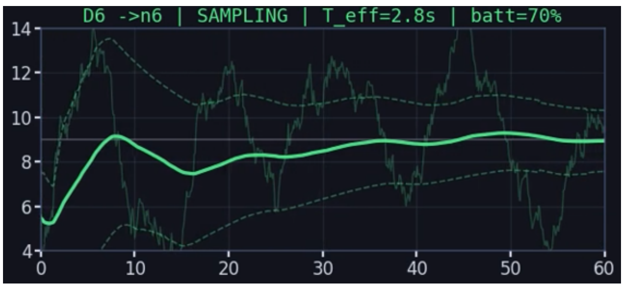
$$\text{z-score} = 1.645 \text{ (90\% confidence interval).}$$

Condition 4: Minimum Effective Independent Sample Intervals

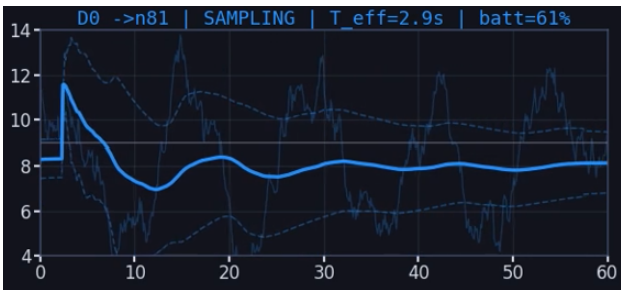
In order to not introduce bias during the measurement interval, several independent samples should converge within the same time-series to declare convergence. This is defined as the 'N_{eff}' parameter in the evaluation. By default, this parameter is set to: guessing here lol

$$N_{eff} = 6.$$

Condition 5: Minimum Shedding Cycles (Wake Only)



(a) Inlet Sampling Dashboard View



(b) Second view

Figure 6.3: Wake Sampling Dashboard View

This is defined as '$N_{shed_{min}}$' parameter in the evaluation. By default, this parameter is set to:

$$N_{shed_{min}} = 5$$

6.2.2. Mean Convergence Time Simulation

To fulfil the measurement node requirement of 100x4 set by GPR flow reconstruction in Section 7.5, convergence times for velocity samples at different atmospheric mean convergence variables had to be studied. This would provide an understanding of whether the sampling strategy is feasible in terms of number of drones required so that this quantity can be frozen. Sets of recommended atmospheric parameters for various wind conditions are provided as a result of the parameter sweep. On top of this, this parameter sweep will offer insight into the total measurement time required to fulfil the 4x100 configuration for point measurements, which will inform hardware design by confirming the required hover time for the drone is sufficient to result in sufficient measurement accuracy.

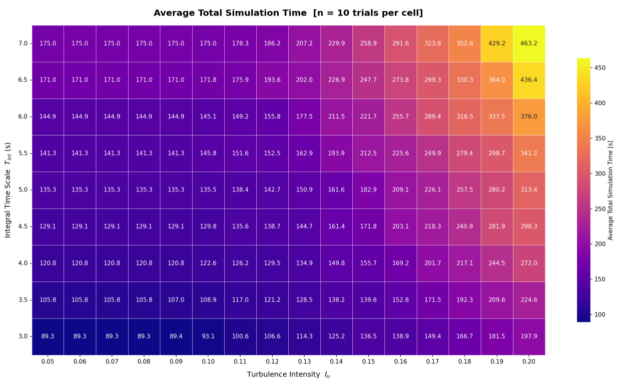
A python routine was developed to simulate random fluctuations in the atmosphere, as well as the drones that sample them. The measurement nodes' positions are generated randomly, and the drones move independently of each other to complete all the required points that must be sampled. At each node, every drone collects running means of velocity measurements and evaluates conditions for convergence as described in Section 6.2. Note that velocity measurements are generated synthetically, using a ground constant with white noise applied on top . Kinematic equations for the movement of drones are considered to estimate the travel time of drones between measurement points as well. Figure 6.4 shows the available dashboard for the drone measurement sampling simulation.

parameter for noise?

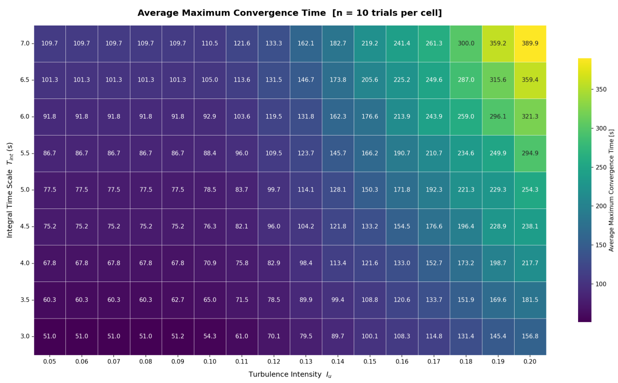
list equations?

Figure 6.4: Drone Velocity Measurement Sampling Simulation Dashboard

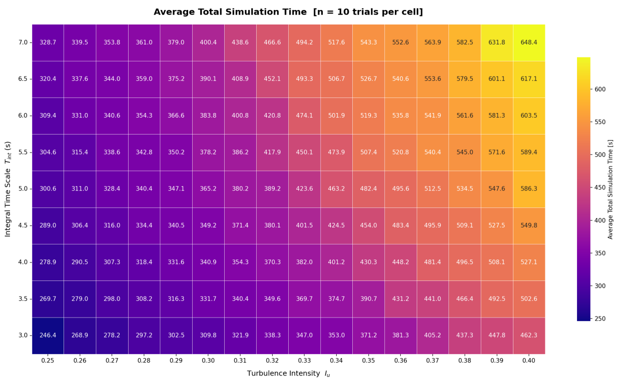
Then, for each combination of turbulence intensity and integral time scale with constant epsilon, delta, z-score, $Neff$ and $N_{shed_{min}}$, 10 trials of the simulation are run and averaged. The average total simulation time and maximum convergence time for both the inlet and wake conditions are then tabulated, resulting in heat maps in Figure 6.5.



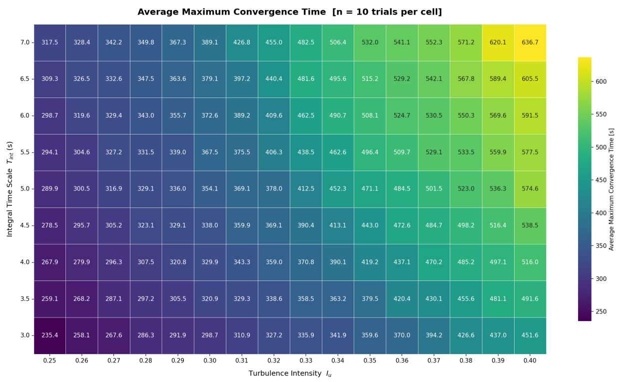
(a) Inlet Total Simulation Time [s]



(b) Inlet Mean Convergence Time [s]



(c) Wake Total Simulation Time [s]

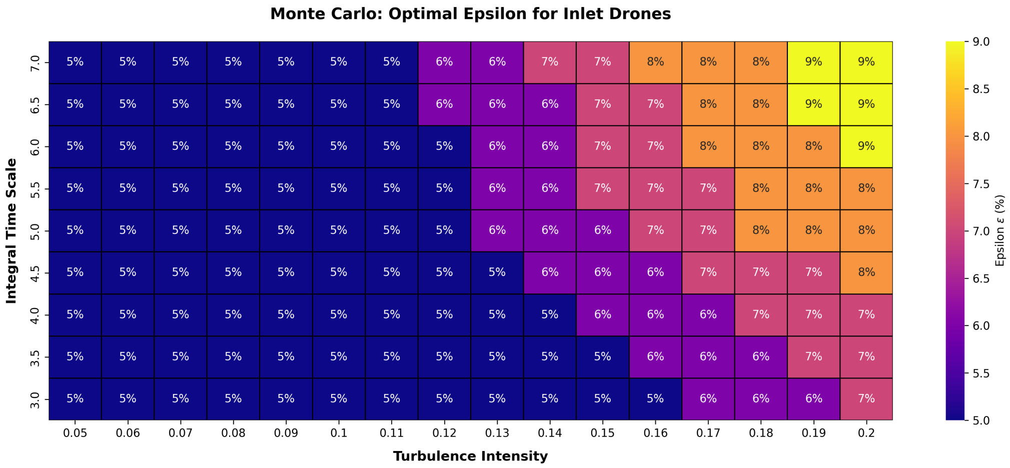


(d) Wake Mean Convergence Time [s]

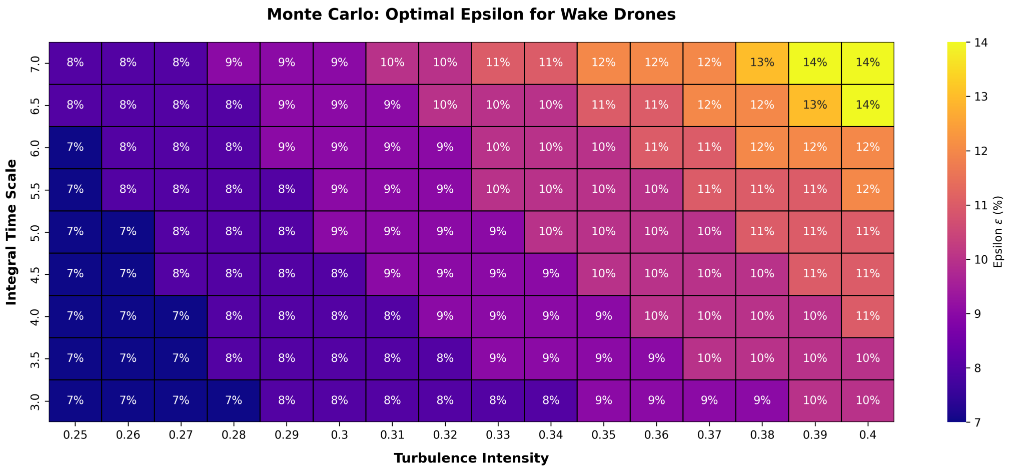
Figure 6.5: Total Simulation Time and Mean Convergence Time for drones at the inlet and wake for a grid of combinations of integral time scale T_{int} and turbulence intensity I_U. 10 trials were taken at each combination and averaged. (epsilon ϵ = 8%, delta δ = 5%, z-score = 1.645 (90% confidence), N_{eff} = 6, number of drones = 15 in inlet and 70 in wake, number of targets = 30 in inlet 70 in wake, phase hard-cap = 360/600 [s])

Taking into account the required maximum hover time of 24 minutes (split into 6 minute phases), the simulation informs whether it is possible to measure the target points in time. Since the phase time for measurements is fixed, the operators need to understand with what measurement error they can get away at given atmospheric conditions.

To produce a meaningful reference table for operators to understand this, a Monte Carlo simulation was run where the maximum acceptable error (Epsilon ϵ) was varied. In the inlet, the parameter was varied between 5% and 10% and 7% to 14% in the wake, both in increments of 1%. For each combination of Integral Time Scale and Turbulence Intensity, 10 trials were run and the average was taken. This totals 1440 trials per selected Epsilon ϵ, which sums to 8640 trials for the inlet and 11520 trials for the wake. With a given requirement for Total Simulation Time set to 6 minutes, the the highest possible turbulence intensity and integral time scale that still fulfils this requirement are recorded. This process is repeated for every epsilon ϵ, which results in the heatmaps shown in Figure 6.6.



(a) Inlet Confidence Interval Error Percentage Overview



(b) Wake Confidence Interval Error Percentage Overview

Figure 6.6: Comparison of Minimum Confidence Interval for Turbulence Intensity and Integral Time Scale Lengths

Figure 6.6 is extremely valuable because it informs the operators' decision for the trade-off between measurement accuracy and measurement convergence time. Depending on estimated weather on the day of measurements, the operators can select an appropriate epsilon to determine their relative error a priori. Separate decisions on the inlet and wake conditions must be made since the downstream is naturally more turbulent than the inlet.

In the worst case, the integral time scale T_{int} and turbulence intensity I_U were set to 7 and 0.2 in the inlet and 7 and 0.4 in the wake respectively, which can be attributed to be the poorest atmospheric conditions due to long-lasting gusts and extreme fluctuations usually observed in slower flows in the context of urban environments, where airflow is moving around chaotic building geometries. Alternatively, in wind farms or otherwise, such conditions can be an inherent characteristic of airflow in perturbed conditions, potentially found in the wake of the building. In the best case, the integral time scale T_{int} was set to 3, which depicts faster developed gusts, and turbulence intensity I_U was set to 0.05, which depicts short gust durations potentially seen at higher wind speeds. Generally speaking, the turbulence intensity in undisturbed wind conditions offshore (at a height of 90 meters) is found to reduce with wind speed non-linearly, in the range of 0.08 at 3 [m/s] to 0.05 at 13 [m/s], which justified the range [9, p.807]. However, in disturbed flow, turbulence intensity will spike significantly and therefore will be the limiting condition. ref???

Figure 6.6 also informs the sensitivity of the system to measurement conditions. Higher measurement epsilon ϵ seen at poorer measurement conditions in the wake will be assimilated globally, since σ_n is a variable inherent to GPR, the method takes into account how much to trust the incoming training points in the reconstruction. In the showcase, this is deemed acceptable due to constraints on drone sizing, particularly hover time being 24 minutes, such that the drones may operate at the maximum operating wind speeds of Beaufort 6. It is important to note that the system's standard operating conditions will likely be significantly below Beaufort 6 for most drones most of the time. According to KNMI, in the Netherlands, wintertime wind speeds average around 6 meters per second (90% confidence interval), equivalent to conditions of Beaufort 4 [24, p.29]. In the summertime, wind speeds have a mean of 4.5 meters per second (90% confidence interval), equivalent to Beaufort 3 conditions [24, p.29]. Therefore, the provided hover time of 24 minutes is an absolute minimum hover time required at the worst conditions, and epsilon values of 5% in the inlet and 8-9% in the wake are far more realistic during nominal operations.

6.2.3. Finalized number of drones and sampling configuration

Following the analysis of the mean convergence time to deem measurements of sufficient accuracy for flow reconstruction, the optimal sampling strategy for the system can now be described. In order to fulfil the worst possible measurement conditions, the system is elected to have 85 drones in total. To cover 100 measurements every 6 minutes, the inlet and wake drones follow slightly different strategies:

- The 15 inlet drones will sample a total of 30 targets over 2 measurement rounds. This way, the inlet drones must have converged measurements of sufficient accuracy in less than 3 minutes.
- The 70 wake drones will sample a total of 70 targets over a single measurement round. This way, the wake drones must have converged measurements of sufficient accuracy in less than 6 minutes.

The measurement error epsilon ϵ must be set according to Figure 6.6, with other parameters set in accordance with Table 6.1.

Table 6.1: Inlet and Wake Measurement Variables

Variable	**Value**	Unit
delta δ	5	[%]
z-score (confidence)	1.645 (90%)	[-]
N_{eff}	6	[-]
$N_{shed_{min}}$ (wake only)	5	[-]
phase hard-cap (inlet)	360	[s]
phase hard-cap (wake)	600	[s]

Note that this sampling strategy assumes that flow conditions do not change significantly between rounds of measurements. More globally, it is assumed that atmospheric and flow conditions do not change significantly within the 24 minute measurement time. These results remain valid if the proportion of measured nodes per drone does not change in the inlet and wake. Using the total measurement time of 24 minutes, the drone sizing may now be converged.

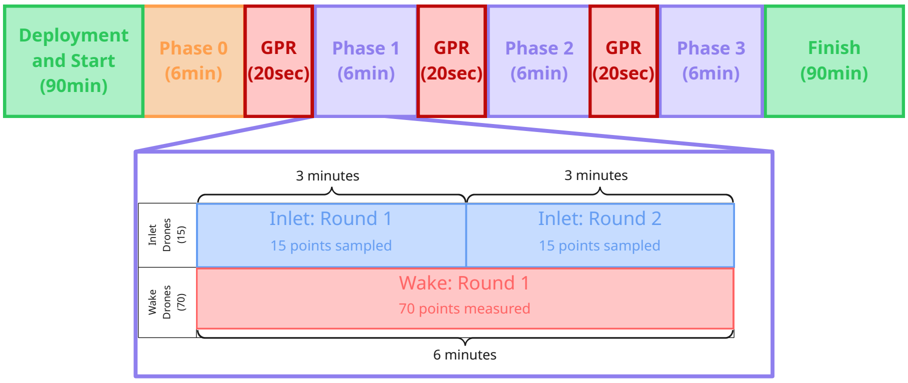


Figure 6.7: Optimal drone measurement configuration required to reach 100x4 measurement configuration specified by GPR software. Orange block indicates that the sampling points for this phase must be preset. Purple blocks indicate that the sampling points are adaptive, based on the results of the previous phase.

6.3. Control

Picture and look at ORCA

6.3.1. Control Hierarchy

The control architecture is organised in two connected loops that run on different timescales (Section 6.8). The outer loop runs on the ground station and decides where the drones should be. The inner loop lives on each drone and decides how that drone flies to and holds the position it has been given. The two loops only meet each other at the target position goal. The coordinate position travels down the link and the status and measurement data travels back up. No part of the ground station commands a drone's motor or attitude directly.

The outer loop is mainly driven by the force and flow reconstruction. GPR runs on the ground station whilst measurements arrive, it updates its estimate of the flow reconstruction and indicates which regions are least certain. These regions make up a new list of requested measurement points. The Drone Location Planner takes this list, together with the live status of every drone, and assigns each drone to a new point to measure such that the total straight-line travel distance is as small as possible. Each drone is then sent only its own target position. Once the new measurements have been gathered and returned, the reconstruction updates again and the cycle repeats. This framework is handled via an architecture developed using ROS2 [32].

The inner loop runs continuously on each drone at a much higher rate. Given a target position, the drone estimates its own state, plans its own path to that target, avoids its neighbours and tracks the path with its flight controller, all without further instruction from the ground. The position and attitude control loops run at 400 to 1000 Hz which is several orders of magnitude faster that the outer loop. This is the reason the drone will only communicate averaged data and its estimated position to avoid high bandwidth requirements. The inner loop is handled using PX4 autopilot software [28].

6.3.2. Ground Station control

The ground station is the supervisory part of the architecture. It runs the GPR, the planner and gives an interface to the crew. It does not control any how any drone flies but it decides where the drones should be and maintains an overview of their position.

The planning cycle starts during the reconstruction, as he GPR updates its estimate of the wake. To eliminate large uncertainties the adaptive sampling step picks out the regions where the estimate is least certain and turns them into a list of requested measurement points. The planner receives the list and passes it to the drone position manager, this component assigns drones to points. It does this with the Hungarian algorithm, which finds the assignment that minimises the total straight-line distance to reach the new points. Solving for the minimum total distance keeps the repositioning time and the energy spent low. Furthermore it tends to avoid drones having to cross each other on the way which eases the load on the collision avoidance. Once the assignment is completed each drone is sent only its own target position down the link.

The drone position manager also takes the live status of every drone, which is continuously reported up the link and shown in Section 6.8 as the feedback into the manager. This status is used to tell the manager where each drone currently is and which drones are actually available such that it can solve the assignment algorithm. It also tells the telemetry of the drones such that a drone with a low battery status returns on time to the ground station.

The crew works through the operator console to start running the campaign, hold or interrupt it. They never command an individual drone and will always have a full overview of the swarm status. The remaining two ground-station blocks in Section 6.8 sit alongside the planner rather than inside it. The RTK base broadcasts the positioning corrections that each drone's estimator uses and the data collection block receives the measurement packets coming up from the swarm which is then fed into the reconstruction.

6.3.3. Onboard flight control

Each drone runs its own flight control stack and this is the inner loop, which is previously presented. The drone is given a target position from the ground station and nothing more. From that single point it estimates its own state, decides how to get there and holds the position once it arrives. This is all done without further instructions and the inner control loop.

State estimation comes first because everything else depends on knowing where the drone is. The sensors feed an extended Kalman filter which is indicated as the estimator in the figure. This estimator outputs the position, velocity and attitude estimate based of the inertial measurement unit, barometer and RTK-GNSS receiver. The inertial measurement unit gives the fast motion and rotation that the controller needs. The RTK-GNSS receiver corrects the drift induced by the inertial measurement unit and anchors the estimate to a fixed frame. This is done using a correction stream broadcast by the ground station's RTK base. When this correction stream is used, the position estimate is accurate to centimetre level. This accuracy matters because the measurement points the planner assigns are only meaningful if each drone can be placed and held at its point precisely. Finally, the position estimate is also required for the collision avoidance, which needs a reliable relative positions to keep the drones apart.

The flight control itself runs on the PX4 autopilot, shown as the PX4 controller driving the airframe and rotors. It runs the position and attitude control loops at 400 to 1000 Hz which is significantly faster than the planning cycle on the ground. The onboard software generates a smooth path to the target, given the target position and the current state estimate. It also passes the corresponding setpoints to the autopilot which tracks them while keeping the drone stable. The loop closes through the airframe, whose motion is sensed again and fed back into the estimator. This allows the drone to continuously correct itself toward its target and holds it stable once the target is reached. Because all of this runs onboard, the link only has to carry the slow target updates rather than high-rate commands, which removes any potential communication constraints.

While the inner loop runs, each drone reports its state and health back up the link as the drone status shown leaving the block. This telemetry is what the ground station uses to keep its live picture of the swarm, feeding into the drone position manager and providing the central tracking required by the system requirements.

6.3.4. Onboard collision avoidance

The path a drone flies to its target is not a straight line, since they have to remain at a safe distance from each other. This is handled onboard by each drone running its own collision avoidance, rather than running it on the ground station planning. Planning conflict-free routes would put the high-rate and safety-critical work back on the ground station which is preferably avoided. Instead each drone runs a local software of Optimal Reciprocal Collision Avoidance (ORCA), shown as the onboard ORCA block between the estimator and the controller in Section 6.8. It takes the velocity the drone wants in order to head towards its target and adjusts it just enough to stay clear of its neighbours. This adjusted velocity is then passed on to the flight controller.

The adjustment works on velocities rather than fixed paths. Each drone takes the positions and velocities of its nearby neighbours and extrapolates their motion forward over a short time window. Any neighbour whose extrapolated path would bring it within the set safety distances becomes a constraint. For each such neighbour ORCA builds the set of velocities that would keep the two drones apart. Every drone in a pair takes half of the correction needed to clear the other so the two of them resolve the conflict between them. The drone then takes the intersection of all these allowed sets, one per neighbour, and picks the velocity inside it that is closest to the direction it actually wanted to travel. The result is a velocity that still drives the drone towards its target while guaranteeing it stays clear of every neighbour it can see.

ORCA needs current neighbour states to do this and those come from the other drones over the link rather than from the ground station. Each drone broadcasts its own position and velocity and listens for the same from its neighbours, shown as the neighbour to the block in the figure. Each drone keeps only the neighbours within a short radius on the order of ten to twenty meters and ignores the rest. This filtering is needed so that the per-drone load stays flat whether the amount of drones in the swarm increases or decreases. The avoidance runs every control time step on the latest neighbour states so it reacts continuously as the drones move rather than committing to a route up front.

Because the avoidance is local and reciprocal, no central component has to coordinate it and no global path plan has to be computed. This keeps the ground station on its slow supervisory loop and leaves the fast, safety-critical reaction onboard. The safety distance and the length of the extrapolation window are the two parameters that set how cautious the avoidance is.

6.3.5. Control verification in simulation

The control architecture was tested in a software simulation to check that the control design behaves as intended and to expose integration problems. The point of the simulation is to exercise the control logic rather than testing the link budget or sensor performance.

The simulation is built around the same drone structure as the deployed system. Each drone runs as a separate PX4 instance in a Gazebo Harmonic world. Each one also has its own autopilot and its own ROS 2 namespace, bridged to ROS 2 through a Micro XRCE-DDS agent. A ground station ROS2 node stands in for the central planner (Ground Station in the actual system) and assigns each drone a target position. A second node stands in for the estimation and data path. Because each drone is a separate vehicle running the real PX4 flight flight software, the inner loop is exercised on the actual autopilot and not on a simplified motion model. Each drone is commanded only a target position in its own frame and then holds stationary. The run described here used nine drones to show that the multi-vehicle structure works on a control level. The simulation confirms that a target position is a sufficient command for the architecture to work. The drones arm, climb and hold their assigned positions on their own flight control with the ground control only sending the targets. This work would also serve as the basis for final software integration with other subsystems, as shown in Section 6.8.

The collision avoidance runs in the same framework, with each drone resolving conflicts from the neighbour states it receives rather than from the central planner. The behaviour to verify is that the swarm keeps clear of itself while repositioning measured against the 3 to 4 m inter-drone separation enforced as hard constraint and avoidance reacts inside the 50 ms target set by the risk analysis.

The simulation runs the real flight and avoidance software against simulated dynamics which makes it a strong check of the control behaviour. However, it is a synthetic result and not flight-ready. Additionally, the radio link is carried over local DDS rather than the HaLow WiFi mesh. In the current

Gazebo simulation world, the wind is described as scene context and does not physically force the airframe, so the stationkeeping error under wind gusts was not exercised.

add recommendations at the end about this

are we adding any cool figures of the control animation???

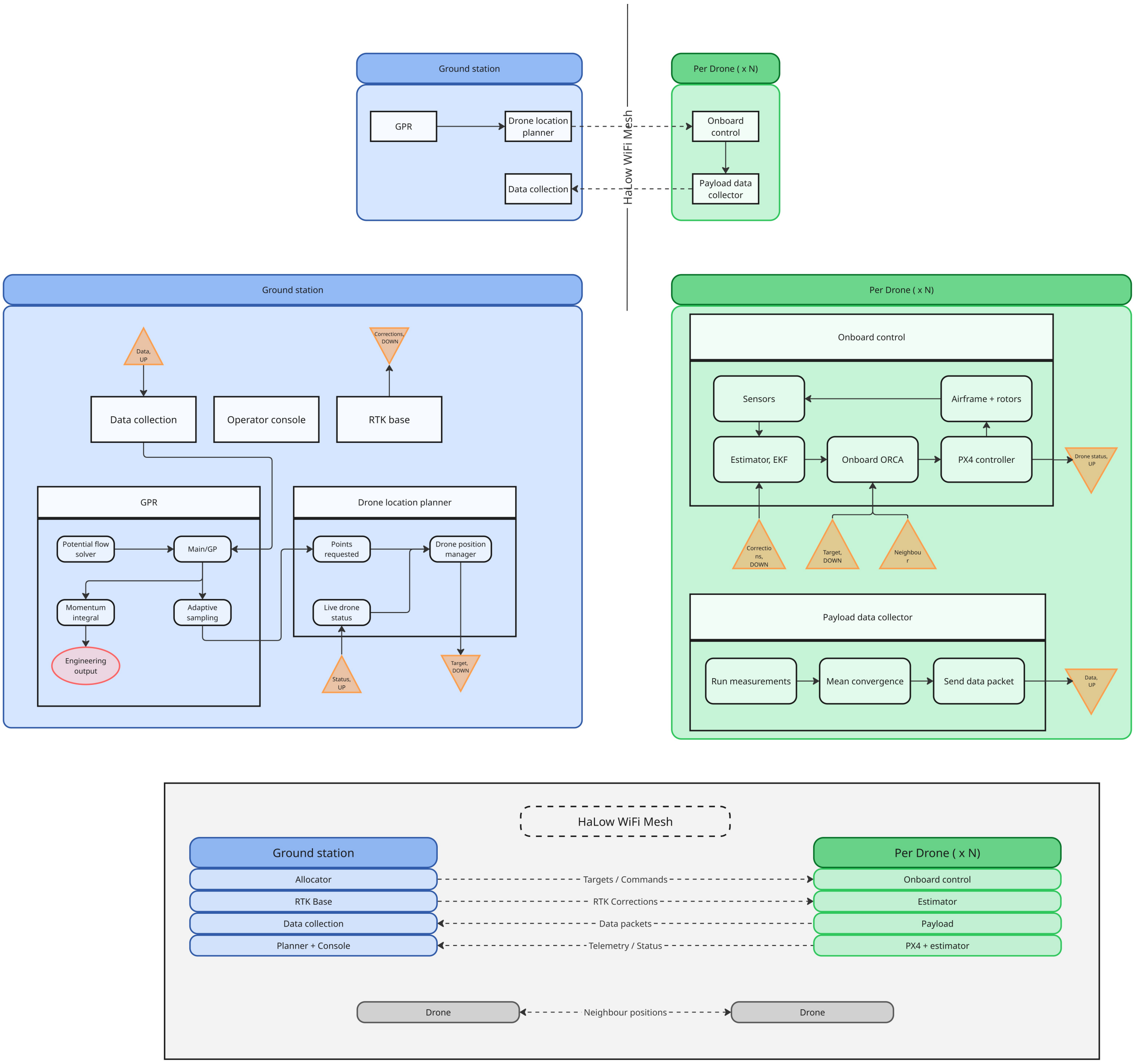


Figure 6.8: Control Architecture

6.4. Communications

7 Results and Discussion

In this chapter, the samples-to-force pipeline of Chapter 5 is evaluated on two CFD case studies and establishes a quantative result needed in Chapter 6: the number of test points required to estimate the aerodynamic force within 5% target. Like the previous chapter, each test point is treated as returning the true instantaneous mean velocity at its location. The results here therefore isolate the characteristics of the reconstruction method. This is done deliberately, since the test points count established here is the quantity that the following chapter converts into a hover time and a required number of drones.

The discussion follows the same pipeline as the one from the samples-to-forces. Firstly, Section 7.1 first assesses the reconstructed velocity field against the CFD simulation. This is done by comparing the recovered wake structure, field error and the divergence of the reconstruction. Additionaly, the pressure field is examined in the same section

Throughout this chapter the force is benchmarked against the momentum integral evaluated on the CFD truth field, rather than against the ANSYS force directly. The reconstruction feeds the same momentum integral, so the closest it can come is the value that integral gives on the true field. That value differs from the ANSYS force by the neglected Reynolds-stress flux (Equation 5.34), a known offset of the mean-field method that could be recovered in deployment as described in Section 15.2. Benchmarking against the momentum integral therefore isolates the reconstruction error from that offset.

7.1. Field reconstruction with 200 test nodes

In this section, a first reconstruction of the building wake is discussed to establish a baseline for any optimisation. The aim of this section is to show that the GPR reconstruction improves on the bare potential-flow prior, recovering the wake structure that the prior cannot represent. It will use a configuration of 200 sample nodes, placed on a domain as explained in Section 5.7. The count of 200 was an initial estimate rather than a derived figure.

The velocity field is reconstructed by evaluating the GPR posterior on a denser regular grid of testing points and is shown in Figure 7.1, on a horizontal slice at 25 m height through the building. The dashed circle marks where the cylinder integration surface cuts this slice, and since the force integral is evaluated on this surface, the reconstruction error along the ring is the most relevant indicator of force-estimate quality. With just 200 points, only the freestream and inlet regions are reconstructed with low error. The GPR already improves on the potential-flow prior, which has no wake at all, by reconstructing the low-velocity region behind the building. However, the error is large at the wake, and largest at the shear layer and close to the building, where the posterior is too smooth to capture the sharp velocity gradient of the CFD truth. This is because the widely spaced samples force large fitted length scales, so the kernel is smoother than the true field at the shear layer. On the integration surface the error is already low around most of the ring, except where the ring crosses the wake.

HaLow WiFi network is technically not mesh, in our case it will be a star net work wit 1 GS aces point scheduling everything. While mesh capabilities exist they are used when range/pe natration is a limiting factor and compromise data rate at high number c nodes...

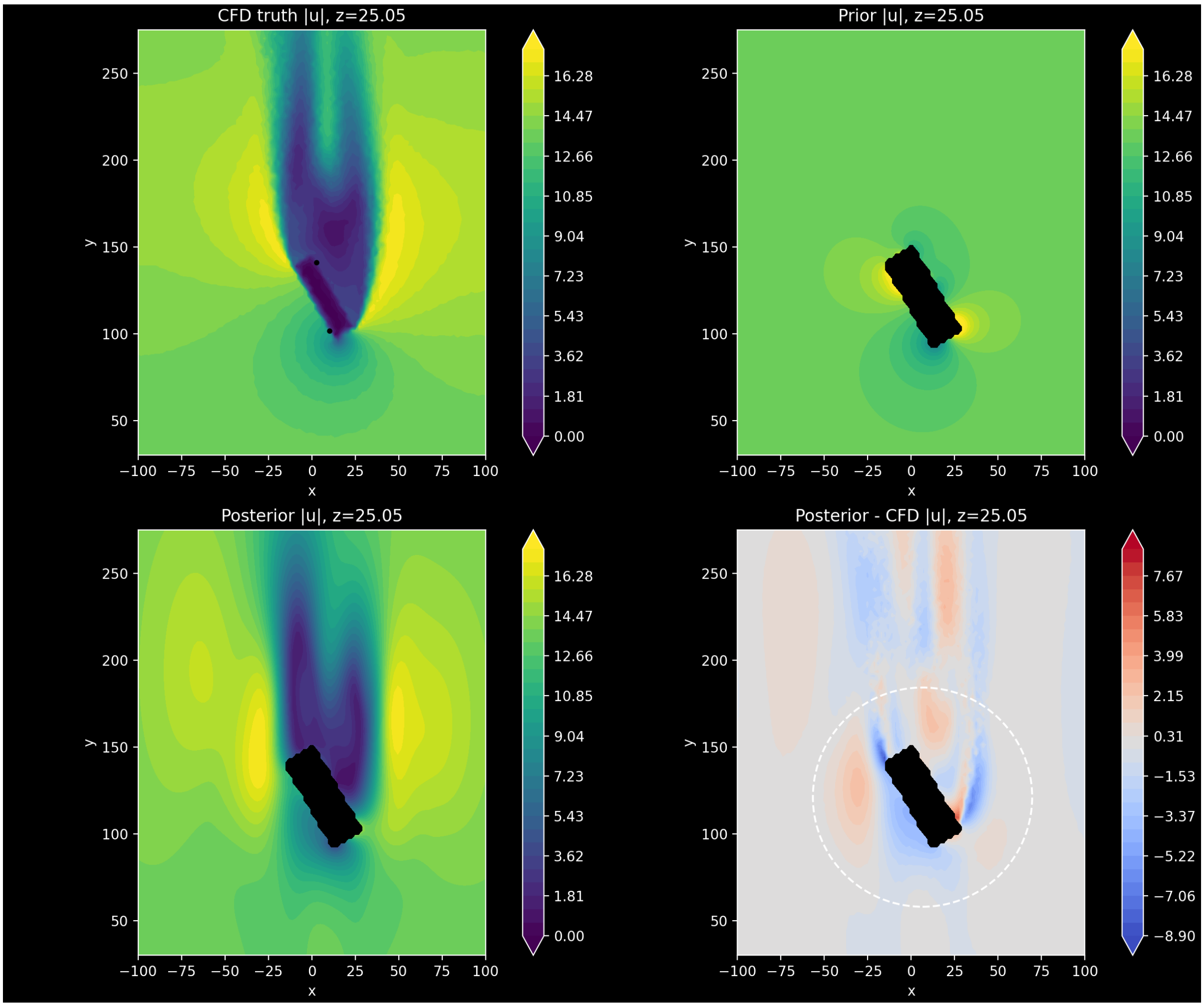


Figure 7.1: Velocity field posterior after the initial sampling step. Top left to right: CFD truth, potential prior. Bottom left to right: GPR posterior, GPR posterior difference with CFD truth.
Points: 200 **Domain RMSE:** 1.04 m/s F_x: 151.8 kN F_y: 270.5 kN

Unlike the divergence-free, potential-flow-primed velocity GP, the pressure field uses a plain scalar Matérn-5/2 with no physical prior, so it reconstructs pressure purely by regressing between samples. The pressure reconstruction after the initial phase is shown in Figure 7.2. Its general shape is less accurate than the velocity reconstruction, with large errors at the building and in the far wake. The error is largest at the building, where the CFD pressure transitions sharply from high pressure on the inflow face to strong suction on the wake face, a jump the smooth GP reconstructs as a gradual transition. However, along the integration surface, the pressure error remains low, even in the region where it crosses the wake.

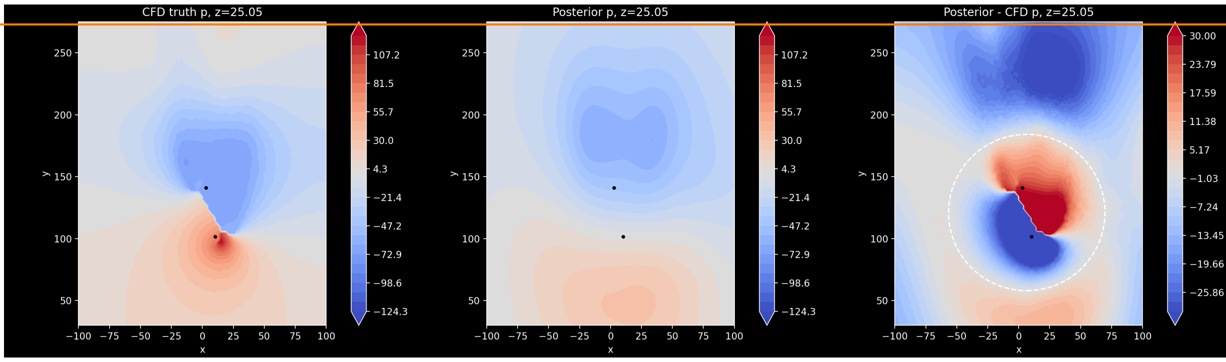


Figure 7.2: Pressure field posterior after the initial sampling step. Left to right: CFD truth, GPR posterior, and their difference.
Points: 200 **Domain RMSE:** 15.0 Pa

In Section 5.9, the ground-truth force vector on the building was given to be $\mathbf{F_{truth}} = [155.433, 208.647, 72.586]$ kN. The y follows the flow, z points upward, and x completes a right handed system. As described in Section 5.6, the vertical component is not of interest, restricting the comparison to $\mathbf{F_{truth}} = [155, 433; 208, 647]$ kN.

The fixed 200 nodes configuration estimates this force as $\mathbf{F_{fix,200}} = [151.800, 270.500]$ kN. This gives relative errors of: $\epsilon = [-2.34, 29.64]\%$. The side force F_x is recovered within the 5%. This holds for every configuration examined later in this chapter,so F_x is not a driving factor and is omitted from the remaining discussion. The axial force F_y is the driving quantity: at nearly 30% error, the 200 node fixed configuration is well outside the target. This error is what motivates the investigation of the node count in the following section.

7.2. Node count for single-phase sampling

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This section determines how many nodes a single-phase, fixed distribution requires to meet the 5% force target. The central result is a sweep of the force error against the number of nodes. This identifies the point at which the estimate first satisfies the target. With that count found, the section then considers whether placing all nodes in a single pass is the configuration the system should adopt.

This section determines how many nodes a single-phase, fixed distribution requires to meet the 5% force target. To do this, a single sampling step is run with the nodes distributed over the same thick cylinder shell that an adaptive step would fill, but without any scoring, since with only one step there is no prior reconstruction to score against. The number of nodes is increased run by run to find where the force estimate first meets the target. With that count found, the section then considers whether placing all nodes in a single pass is the configuration the system should adopt.

The nodes fill the shell by the spacing and exclusion rules alone, with no feedback from the reconstruction. The node count is increased from 80 to 560 over 24 seeds. Figure 7.3 shows the resulting drag force F_y against the momentum integral on the CFD truth.

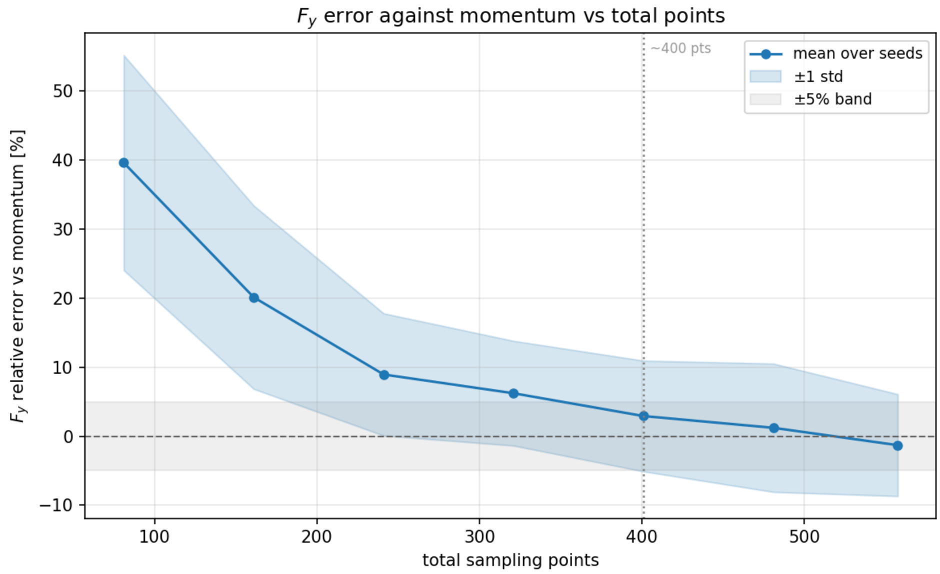


Figure 7.3: Drag force F_y error against the momentum integral on the CFD truth for a single non-adaptive batch, against the number of nodes. The line is the mean over 24 seeds, the band ±1 standard deviation, and the grey strip the ±5% target.

The mean error falls steadily as nodes are added, from +39.6% at 80 nodes to +8.9% at 240 and +6.2% at 320, first meeting the 5% target at 400 nodes (+2.9%). Beyond 400 the mean only drifts within the seed spread, so the estimate has effectively converged and more nodes add noise rather than accuracy. The standard deviation stays around 8% across the whole range and does not shrink with node count, since without adaptive feedback the placement cannot be steered toward the wake. A single pass brings the mean within the 5% target at 400 nodes but not the spread, so individual runs can still fall outside it. 400 nodes is therefore the smallest count at which the mean reliably meets the target.

This node count is also consistent with the operational budget. With a measurement window of about 24 minutes, an average measurement time of 6 minutes in the wake and 3 minutes at the inlet, a 70:30 wake-to-inlet split, and a fleet of 80 to 90 drones, around 400 point measurements can be collected (see Section 6.1, Subsection 6.2.2). The accuracy requirement and the operational limit point to the same figure, so 400 nodes is adopted as the budget for the configuration study that follows, where adaptive phasing is used to reduce the spread that a single pass cannot.

7.3. Number of phases and nodes per phase

The previous section fixed the budget at 400 nodes and showed that a single pass cannot bring the spread within target. This section divides that budget across adaptive phases, to see how the error depends on the number of phases and to select a split. The total is held at 400 so every split is compared at equal cost, with the number of nodes per phase not limited by the fleet size, since depending on how the time budget is divided a single drone can sample more than one point.

Table 7.1 and Figure 7.4 compare splits from a single phase of 400 up to six phases of 67, judged on the final-phase drag force F_y against the momentum integral on the CFD truth, since F_x is accurate for every split and F_y is the component of interest.

I really don't like how we sometime name the measure-ment points "nodes"...

why is the 8.37% different from the

Table 7.1: Final-phase drag force F_y error against the momentum integral on the CFD truth, for each budget split, over 18 seeds.

Split	mean F_y error [%]	σ [%]
1×400	+8.37	10.94
2×200	−4.51	6.87
3×133	−0.91	4.96
4×100	+0.66	3.37
5×80	+1.28	2.81
6×67	+2.39	4.36

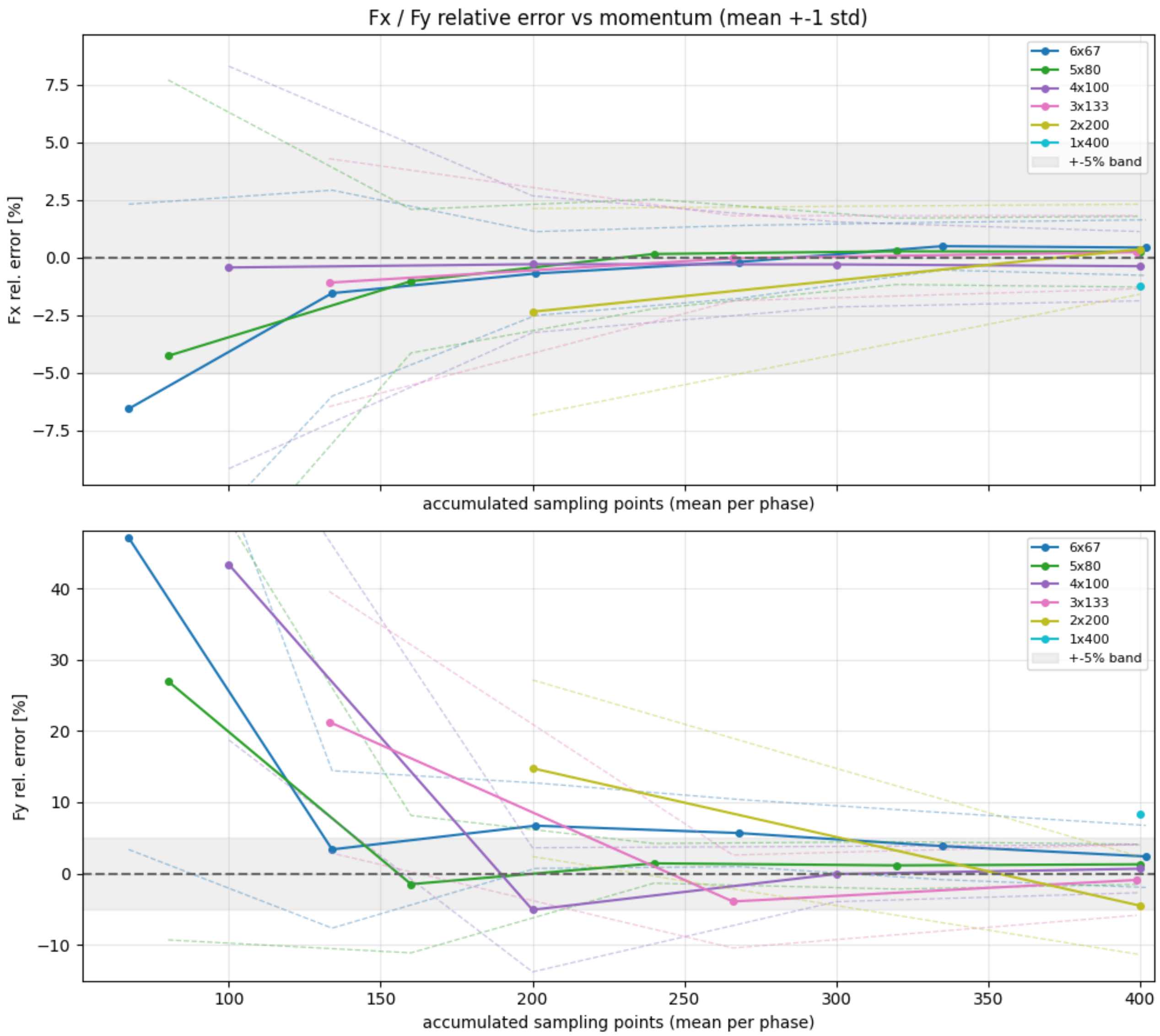


Figure 7.4: Side force F_x and drag force F_y error against the momentum integral on the CFD truth, as points accumulate, for each budget split of 400 points. Lines are the mean over 18 seeds, dashed lines ±1 standard deviation, grey the ±5% band.

The two extremes perform worst. A single phase of 400 points places every drone on the initial cylinder with no adaptive feedback, reaching only +8.4% ± 10.9%, and two phases of 200 is the next worst. The intermediate splits all converge to within a few percent of the momentum integral on the CFD truth, with 4×100 giving the lowest mean error (+0.66%) and 5×80 the lowest spread (±2.8%).

A balance between phases and points per phase is expected, rather than a preference for either extreme. The difficulty score that guides each phase is computed from the current reconstruction, so it is only informative once enough points have been placed to reconstruct the field reasonably. Too few points per phase, as in the 6×67 configuration, means the early phases adapt on a poor reconstruction and spend refinement steps on unreliable scores. Too few phases, as in the 2×200, results in little adaptive feedback, so not all difficult areas are covered by measurement points. The optimum places enough points per phase for the score to be trustworthy while keeping enough phases for the refinement to iterate.

The 4×100 split was therefore selected as optimal. It gives the lowest mean error and a spread close to the best, and 100 measurements per phase can be covered by 85 drones, 70 in the wake sampling one point each and 15 in the inlet sampling two points each per phase, within the available hover time (see Subsection 6.2.2).

7.4. Scoring Sharpness (Adaptive vs. non-Adaptive phases)

This section checks whether the adaptive scoring improves the estimate at all, and if so finds a good value for its sharpness β.

The difficulty score D from Equation 5.40 does not select points directly. Each candidate from the Latin hypercube pool is drawn with probability $p \propto D^{\beta}$, where the exponent β sets how strongly the selection is biased towards high-score regions. Increasing β sharpens the bias, while $\beta = 0$ makes all points equally probable to be selected.

At $\beta = 0$ the selection is purely geometric. The candidates are still drawn over the wake-biased shell and placed outside each other's exclusion zones, but the exploitation terms (velocity gradient and vorticity) and exploration term (posterior variance) have no influence on where the points go. The case therefore isolates the contribution of the physics-based scoring: any improvement over $\beta = 0$ is due to the score, while $\beta = 0$ itself retains only the adaptive geometry and the accumulation of points across phases.

Table 7.2 and Figure 7.5 compare β from 0 to 8, on the final-phase F_y against the momentum integral on the CFD truth. The scoring clearly helps: $\beta = 0$ is the worst on both mean error and spread, and increasing β improves both up to $\beta = 4$, which gives the lowest mean error (+0.54%) and a spread close to the tightest. Beyond this, $\beta = 8$ is slightly worse, as over-sharpening concentrates the batch too narrowly. Even $\beta = 0$ converges to within a few percent, so the adaptive geometry and phasing contribute on their own, but the physics-based scoring roughly halves the spread relative to the unscored case. A value of $\beta = 4$ was selected as optimal.

Table 7.2: Final-phase drag force F_y error against the momentum integral on the CFD truth, for each scoring sharpness β, over 24 seeds.

β	mean F_y error [%]	σ [%]
0	+4.78	7.54
1	+3.08	5.88
2	+1.69	3.46
4	+0.54	3.57
8	+0.87	4.16

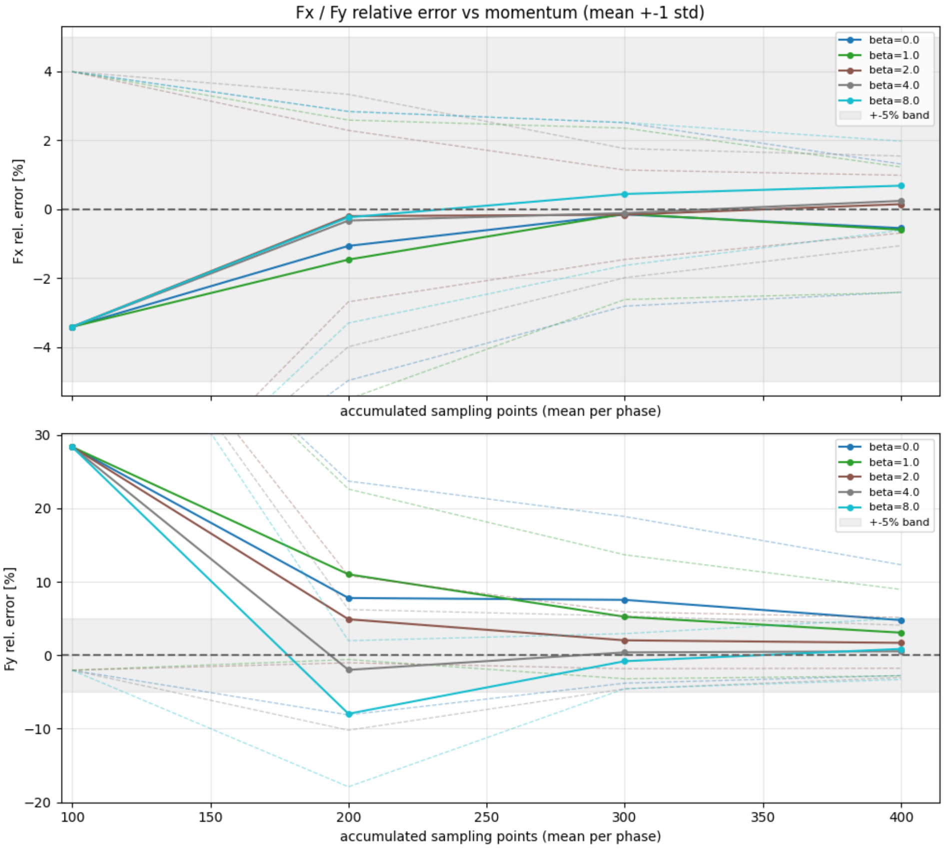


Figure 7.5: Side force F_x and drag force F_y relative error against the momentum integral on the CFD truth as points accumulate, for scoring sharpness β from 0 to 8. Lines are the mean over 24 seeds, dashed lines ±1 standard deviation, grey the ±5% band.

7.5. Sensitivity study

The configuration of Section 7.6 was selected as a baseline, but it should not depend on a precise tuning to work. This section checks how robust the force estimate is to changes in that configuration, varying one parameter at a time around the adopted values and comparing the final-phase drag force F_y against the momentum integral on the CFD truth. The variations cover the integration radius (r), the cylinder height (h), the bias towards the wake of the adaptive shell (`front_frac`), the shell thickness, and the radial budget split across inner and outer regions and cylinder surface (`regions even`), each over 16 seeds.

Table 7.3: Final-phase drag force F_y error against the momentum integral on the CFD truth, for variations around the adopted configuration, over 16 seeds.

Configuration	mean F_y error [%]	σ [%]
baseline	+1.50	2.87
$r = 1.0$	−5.79	4.12
$r = 1.4$	+0.32	4.50
$h = 1.2$	+1.68	4.88
`front_frac` = 0.3	+2.38	4.01
`front_frac` = 0.7	+0.13	3.53
thick shell	−3.18	3.61
regions even	−0.89	3.97

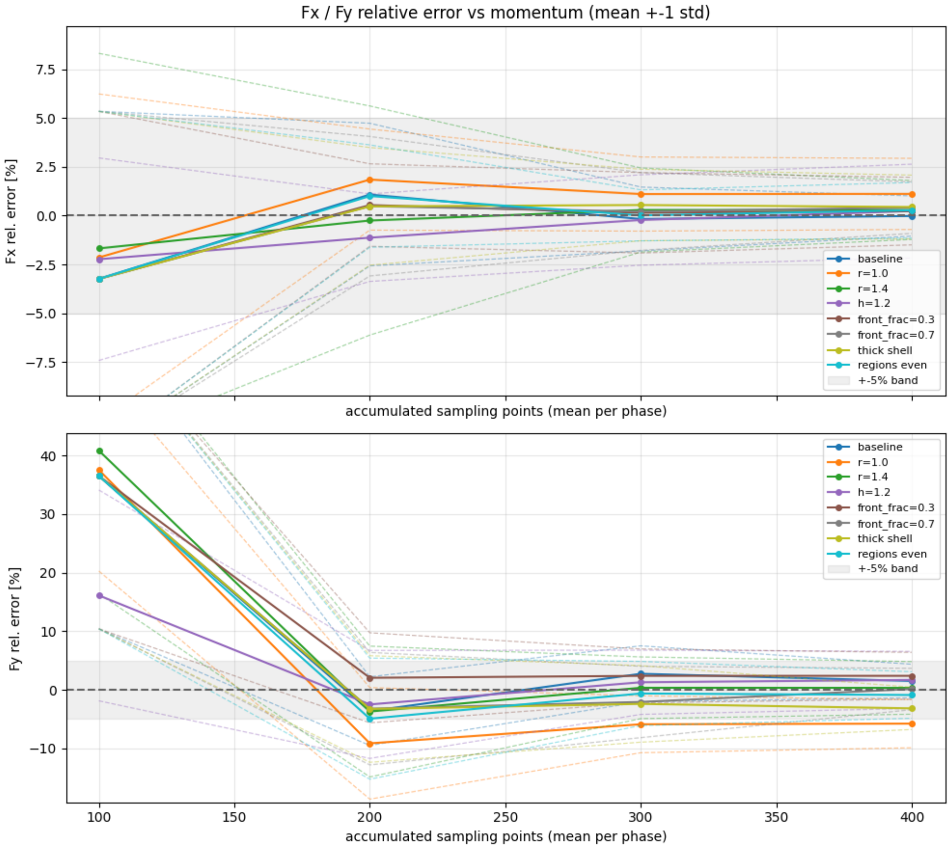


Figure 7.6: Side force F_x and drag force F_y error against the momentum integral on the CFD truth, as points accumulate, for variations around the adopted configuration. Lines are the mean over 16 seeds, dashed lines ±1 standard deviation, grey the ±5% band.

The estimate is robust to every change tested. All variations converge to within a few percent of the momentum integral, with final mean errors between −5.8% and +2.4% and spreads of 3 to 5%, so no single parameter has to be set precisely for the method to work. The geometric parameters move the result most, while the adaptive parameters have a smaller effect. The largest shift comes from reducing the integration radius ($r = 1.0$), the only variation whose mean reaches the edge of the target, since a smaller radius places the integration surface in the steep near-body gradients where the reconstruction is least accurate. The larger radius ($r = 1.4$) stays well within target, consistent with the radius being the main case-specific parameter identified in Section 7.6.

Other parameters were varied during development, including the scoring weights ($w_{\text{var}}, w_{\text{grad}}, w_{\text{vort}}$), but are not reported here. The weights changed the result far less than the sharpness β of Section 7.4, which was the dominant scoring parameter, so the weight variations add little beyond what the geometric and β studies already show.

7.6. Optimal Configuration

Based on the sensitivity studies above and on broader experimentation with the parameters, the configuration in Table 7.4 was adopted for the reconstruction and force results. The initial cylinder is sampled with 100 points, followed by three adaptive phases of 100 points each, for 400 points in total, with the adaptive scoring using a sharpness of $\beta = 4$ and a 1.0 m finite-difference step.

Table 7.4: Adopted sampling configuration.

Stage	Parameter	Value
Initial cylinder	radius factor r	$1.2\,L_c = 63.1\,\mathrm{m}$
	height factor h	$1.5\,L_c = 78.9\,\mathrm{m}$
	tilt	$10°$
	wake fraction	0.25
	top cap points	20
	total points	100
Adaptive	phases	3 (×100 points)
	score weights ($w_{\mathrm{var}}, w_{\mathrm{grad}}, w_{\mathrm{vort}}$)	0.2, 0.4, 0.4
	sharpness β	4
	shell thickness fraction (in, out)	0.20, 0.20
	region split	0.70, 0.15, 0.15
	finite-difference step	$1.0\,\mathrm{m}$

Within the ranges tested the force estimate is robust to these parameter choices, with similar configurations all converging to within a few percent of the momentum integral on the CFD truth.

This configuration is tuned for the current proof-of-concept case (Aerospace High-rise building at TU Delft, see Subsection 5.9.1). For a different structure, the sampling geometry could be set from a preliminary CFD study rather than carried over directly. The main parameter to determine is the cylinder radius, which balances two competing effects. Too small a radius places the surface in the steep near-body gradients, where the reconstruction is hardest and the measurements will be noisiest, while too large a radius lowers the measurement density on the surface and requires a thicker sampling shell to achieve good accuracy. A smaller radius is therefore preferable where the reconstruction can support it. This is discussed further in Chapter 15.

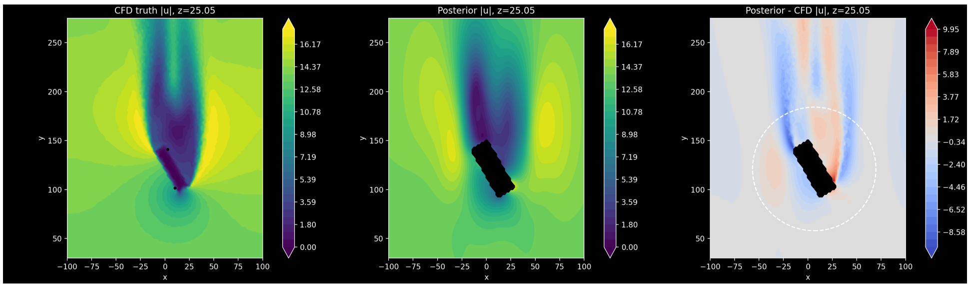
7.7. Results for Optimal Configuration

This section reports the reconstruction and force for the adopted 4×100 configuration of Section 7.6. The velocity and pressure fields are first examined as they evolve across the adaptive phases, then the force convergence is assessed against both the ANSYS force and the momentum integral on the CFD truth, with the latter used as benchmark for the reasons given in the chapter introduction.

7.7.1. Reconstruction Across Adaptive Phases

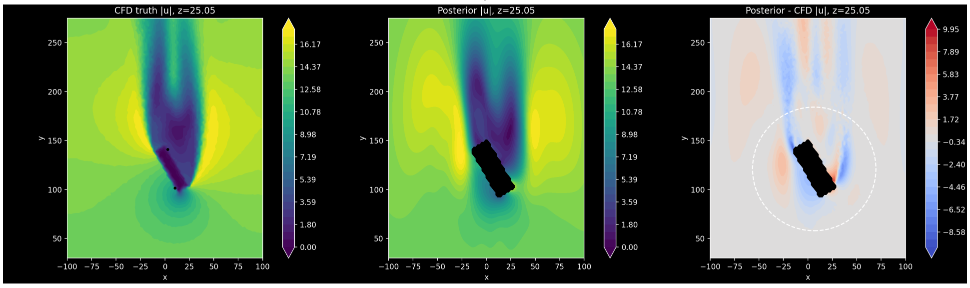
After each phase, the reconstructed field is used to evaluate the difficulty score (posterior variance for exploration, velocity gradient and vorticity for exploitation) as described in Section 5.8. The score selects the points for the next phase and the GP is retrained on the accumulated set.

Figure 7.7 shows the reconstruction across adaptive phases, with the error reducing as points are added. The whole-domain RMSE reduces most on the first adaptive phase, from 1.37 to 1.02 m/s, then more slowly to 0.94 and 0.91 m/s. The error along the integration surface improves most between phases 1 and 2, in particular where the surface crosses the wake and shear layer, with almost no change in phase 3. As more closely spaced points are added the fitted length scales shorten, so the kernel reproduces the sharp shear layer transition more faithfully. The largest errors remain close to the building and in the far wake, but the error along the cylinder ring is small, and the low-error region around it thickens as the adaptive phases place samples inward and outward of the cylinder surface, rather than only on it. These results are for a single seed and configuration. The phase at which the integration surface reaches low error varies between runs, as the thin grey per-run lines in Figure 7.11 show, with some reaching the target by the first adaptive phase and others only by the third.



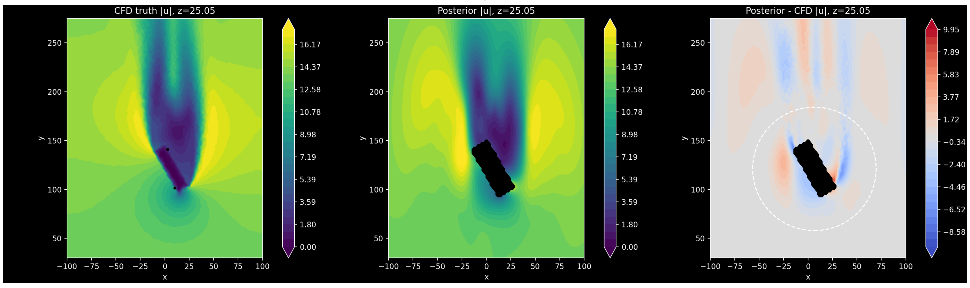
(a) Phase 0 **Points:** 100 **Domain RMSE:** 1.37 m/s **Length scales** (ℓ_x, ℓ_y, ℓ_z): 30.8, 81.8, 44.0 m

↓



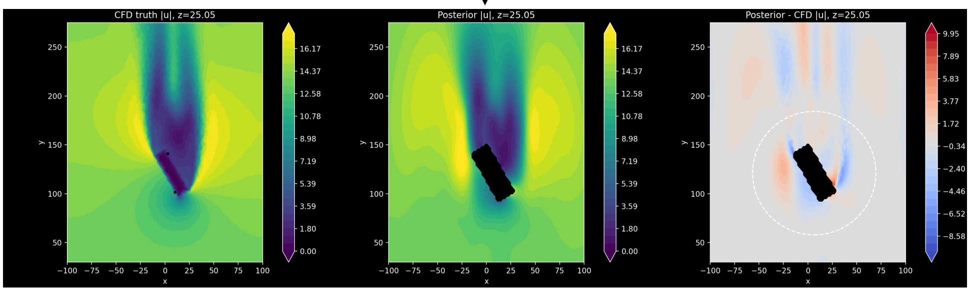
(b) Phase 1 **Points:** 200 **Domain RMSE:** 1.02 m/s **Length scales** (ℓ_x, ℓ_y, ℓ_z): 20.9, 64.7, 40.9 m

↓



(c) Phase 2 **Points:** 300 **Domain RMSE:** 0.94 m/s **Length scales** (ℓ_x, ℓ_y, ℓ_z): 19.7, 59.7, 35.3 m

↓



(d) Phase 3 **Points:** 400 **Domain RMSE:** 0.91 m/s **Length scales** (ℓ_x, ℓ_y, ℓ_z): 20.6, 61.3, 35.5 m

Figure 7.7: Evolution of the velocity reconstruction across phases on a horizontal slice at z = 25 m. Each row shows, left to right, the CFD truth, the GPR posterior, and their difference, with the dashed circle marking the cylinder integration surface.

The posterior variance across phases is shown in Figure 7.8, where a low-variance region forms and tightens around the sampled cylinder as points accumulate. The absolute level is not directly comparable across phases, since the kernel is refitted each phase. The shorter length scales fitted in the later phases reduce the correlation length, which raises the posterior variance away from the measurements. The maps are therefore read as a within-phase indicator of where the reconstruction is data-constrained, rather than as a measure of convergence across phases.

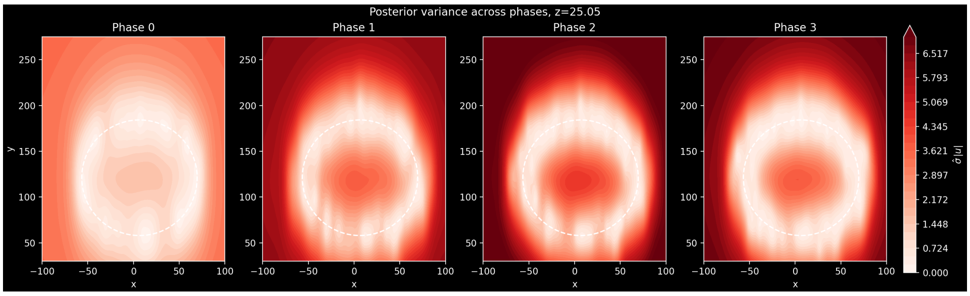
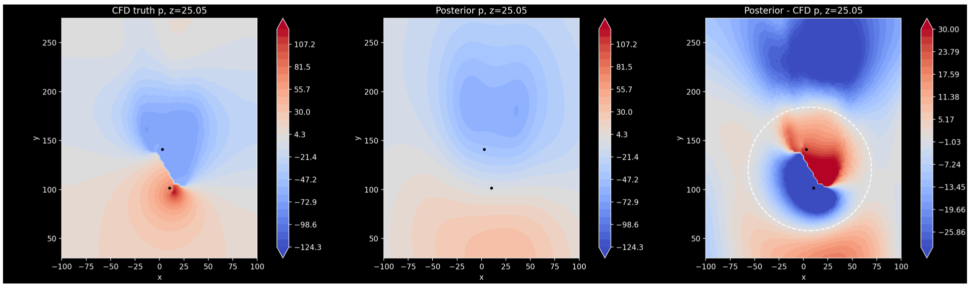
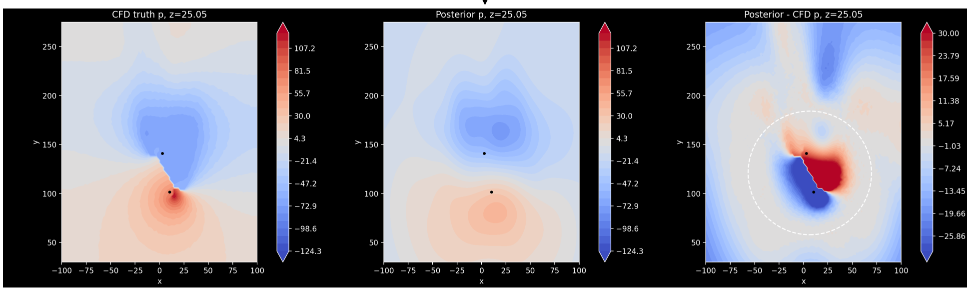


Figure 7.8: Variance across phases.

The pressure reconstruction also improves with the adaptive phases. Figure 7.9 compares the pressure GP in phase 0 and phase 3. Away from the building the error is reduced and the reconstructed pattern is more faithful to the ground truth in phase 3. Close to the building the error remains high: even with many training points the GP cannot reproduce the steep pressure jump between the inflow and wake faces.



(a) Phase 0 **Points:** 100 **Domain RMSE:** 16.3 Pa



(b) Phase 3 **Points:** 400 **Domain RMSE:** 11.37 Pa

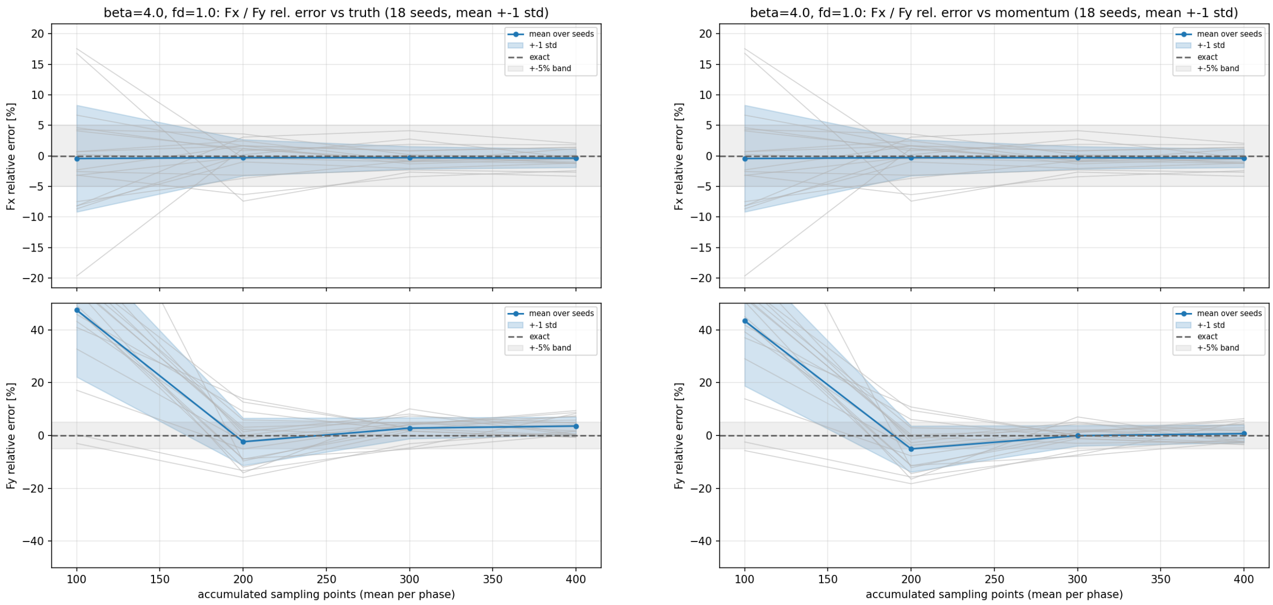
Figure 7.9: Comparison of the pressure reconstruction between the first and last phases on a horizontal slice at z = 25 m. Each row shows, left to right, the CFD truth, the GPR posterior, and their difference, with the dashed circle marking the cylinder integration surface.

7.7.2. Force Estimation

The force is calculated as described in Section 5.6, with ≈32000 points on the integration surface, where the GP posterior and prior mean are evaluated. As discussed earlier, the result varies measurably with the random seed used in the Latin hypercube sampling, both for the initial and the adaptive phases, so the result presented is the mean over 18 seeds. Figure 7.11 shows the convergence against both the ANSYS force and the momentum integral on the CFD truth.

The side force F_x has almost no mean error (<1%) in every phase, with the adaptive phases improving only its standard deviation, falling from ±8.7% in phase 0 to ±1.5% in phase 3. The mean error measures the accuracy of the seed-averaged estimate, while the standard deviation measures the run-to-run spread, so a single early-phase flight can be several percent off even where the mean is exact.

The drag force F_y is much harder to converge. After the initial phase both its mean error (+47%) and its spread (25%) are large. The first adaptive phase removes most of the mean error, dropping it to -2.4%, after which it stays within a few percent. The spread improves more gradually, falling steadily over the remaining phases, so that by the final phase F_y is estimated to $+3.54\% \pm 3.46\%$.



(a) vs. CFD truth force reported in ANSYS
Final phase mean F_y error: +3.54% σ: ±3.46%

(b) vs. CFD truth momentum integral
Final phase mean F_y error: +0.67% σ: ±3.37%

Figure 7.10: Force convergence analysis compared to CFD truth and momentum integral on the CFD truth. Averaged over 18 runs with different seeds for the Latin hypercube sampling, each grey line is the result of a single run.

Evaluating against both references confirms the expectation set out in the chapter introduction. The GPR force converges to the momentum integral on the truth field, with a final F_y error of only +0.67%, rather than to the ANSYS force, against which it sits at +3.54%. Since the GPR feeds the same momentum integral, the momentum value is the target it can actually reach, and the larger gap to ANSYS is the neglected Reynolds-stress flux (Equation 5.34), not reconstruction error. This offset could be recovered in deployment by measuring the fluctuation statistics $\overline{u_i' u_j'}$ in flight (Section 15.2). Table 7.6 gives the per-phase error against both references.

Table 7.5: Drag force F_y mean error and standard deviation over 18 seeds, per phase, against the CFD truth (ANSYS) and the momentum integral on the truth.

	vs. CFD truth (208.6kN)		vs. mom. int. (214.6kN)	
Phase	mean [%]	σ [%]	mean [%]	σ [%]
0	+47.5	25.3	+43.4	24.6
1	−2.4	9.0	−5.1	8.7
2	+2.8	4.0	−0.1	3.9
3	+3.5	3.5	+0.7	3.4

7.8. Future application of proof of concept: wind turbine

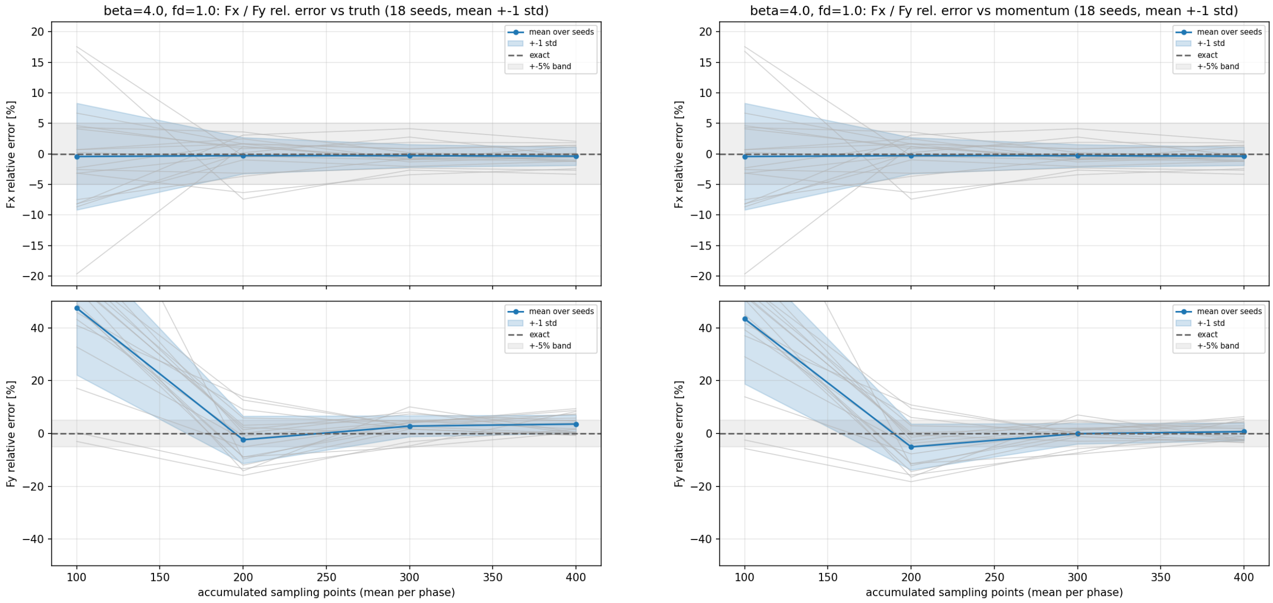
Finally, the pipeline is applied without re-tuning to the wake of the NREL 5MW offshore wind turbine [**jonkman2009definition**], a flow with a different topology from the bluff-body building case.

7.9. Force Estimation

The force is calculated as described in Section 5.6, with ≈32000 points on the integration surface, where the GP posterior and prior mean are evaluated. As discussed earlier, the result varies measurably with the random seed used in the Latin hypercube sampling, both for the initial and the adaptive phases, so the result presented is the mean over 18 seeds. Figure 7.11 compares these results against two references: the CFD ground-truth force reported by ANSYS, and the converged value of the momentum integral evaluated on the CFD truth field with 20 million points (Figure 10.1).

The side force F_x has almost no mean error (<1%) in every phase, with adaptive phases improving only its standard deviation, falling from ±8.7% in phase 0 to ±1.5% in phase 3. The mean error measures the accuracy of the seed-averaged estimate, while the standard deviation measures the run-to-run spread, so a single early-phase flight can be several percent off even where the mean is exact.

The drag force F_y is much harder to converge. After the initial phase both its mean error (+47%) and its spread (25%) are large. The first adaptive phase removes most of the mean error, dropping it to −2.4%, after which it stays within a few percent. The spread improves more gradually, falling steadily over the remaining phases, so that by the final phase F_y is estimated to +3.54% ± 3.46%.



(a) vs. CFD truth force reported in ANSYS
Final phase mean F_y error: +3.54% σ: ±3.46%

(b) vs. CFD truth momentum integral
Final phase mean F_y error: +0.67% σ: ±3.37%

Figure 7.11: Force convergence analysis compared to CFD truth and momentum integral on the CFD truth. Averaged over 18 runs with different seeds for the latin hypercube sampling, each gray line is the result of a single run.

The GPR force converges much closer to the momentum integral on the truth field (F_y error +0.67%) than to the ANSYS force (+3.54%). This is expected, as the GPR force is obtained by evaluating the same momentum integral, only on the reconstructed field rather than the true one, so the best it can reproduce is the value that integral gives on the true field. The remaining gap between the momentum integral and the ANSYS force is the neglected Reynolds-stress flux (Equation 5.34), which the mean-field integral drops but which ANSYS corrects for. The GPR reconstruction should therefore be benchmarked against the momentum integral on the truth, not against the ANSYS force, since it has no access to the Reynolds-stress correction. In a real deployment this correction could be recovered by measuring the fluctuation statistics $\overline{u_i' u_j'}$ in flight, as discussed in Section 15.2. Table 7.6 gives the per-phase F_y error against both references. The final-phase GPR mean sits at +3.5% against the ANSYS force but only +0.7% against the momentum integral, so almost all of the error relative to ANSYS is the Reynolds-stress offset rather than reconstruction error.

Table 7.6: Drag force F_y mean error and standard deviation over 18 seeds, per phase, against the CFD truth (ANSYS) and the momentum integral on the truth.

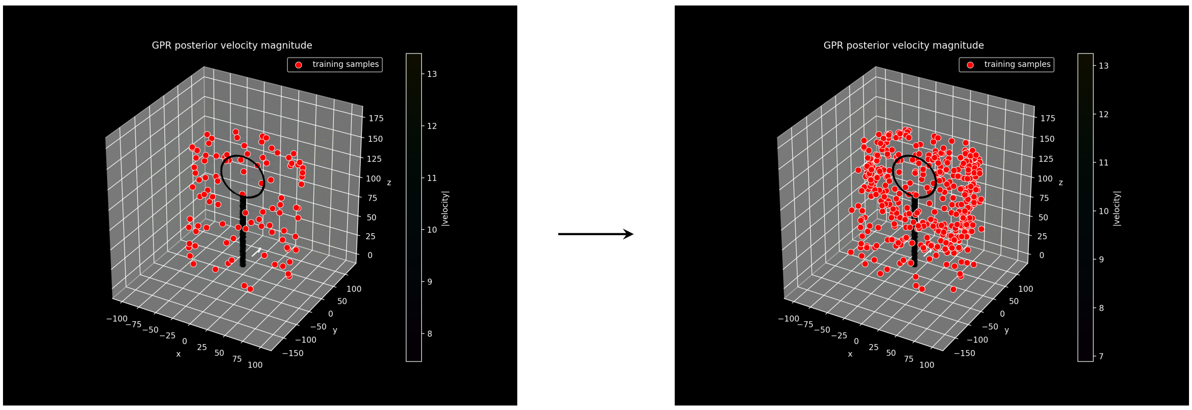
	vs. CFD truth (208.6kN)		vs. mom. int. (214.6kN)	
Phase	mean [%]	σ [%]	mean [%]	σ [%]
0	+47.5	25.3	+43.4	24.6
1	−2.4	9.0	−5.1	8.7
2	+2.8	4.0	−0.1	3.9
3	+3.5	3.5	+0.7	3.4

7.10. Future Application Proof of Concept: Wind Turbine Results

The aim of this section is to outline the results obtained by applying the reconstruction and force estimation methods presented in this design report to the synthetic wind turbine CFD case presented in **??**. Importantly, the case was tried as a benchmark for generalizability of LAWSS to test cases for which is has not been optimized – as such no dedicated sampling strategy was devised for wind turbines. Because of the difficulty of modelling wind turbines, two separate prior means were considered, computed and compared: an inviscid ANSYS Fluent CFD simulation prior mean with the turbine modelled as an actuator disk and a simple prior mean assuming uniform freestream velocity flow everywhere. This comparison shall allow for the determination of the extent of the importance of a good prior mean function for force determination and flow reconstruction. Furthermore, it shall showcase the applicability of LAWSS in the wind energy sector and applicability beyond cases it was optimized for.

7.10.1. Sampling

The sampling for the wind turbine case was taken directly from the aerospace building case (skewed cylinder with adaptive sampling) for lack of available development time. The adaptive sampling criteria were also exactly analogous to the previous implementation. An example final distribution of three dimensional distribution of samples is shown in Figure 7.12.



(a) Phase 0 sampling point distribution

(b) Phase 3 sampling point distribution

Figure 7.12: Evolution in sampling points (from Phase 0 to Phase 3).

7.10.2. Velocity Field Reconstruction

This section compares the velocity field reconstruction results of the two models: inviscid CFD prior (Figure 7.14) and freestream (Figure 7.13). As can be seen in the figures, the peak error in the reconstruction without the prior is much higher than with prior (6.21 > 1.84). This can mainly be attributed to the left-shifted slow turbulent wake which was correctly captured by the CFD prior but diffused and unresolved by the model with freestream prior. Other flow features were captured to a similar extent, though it must be noted that the freestream prior resulted in a more diffused, less realistic wake.

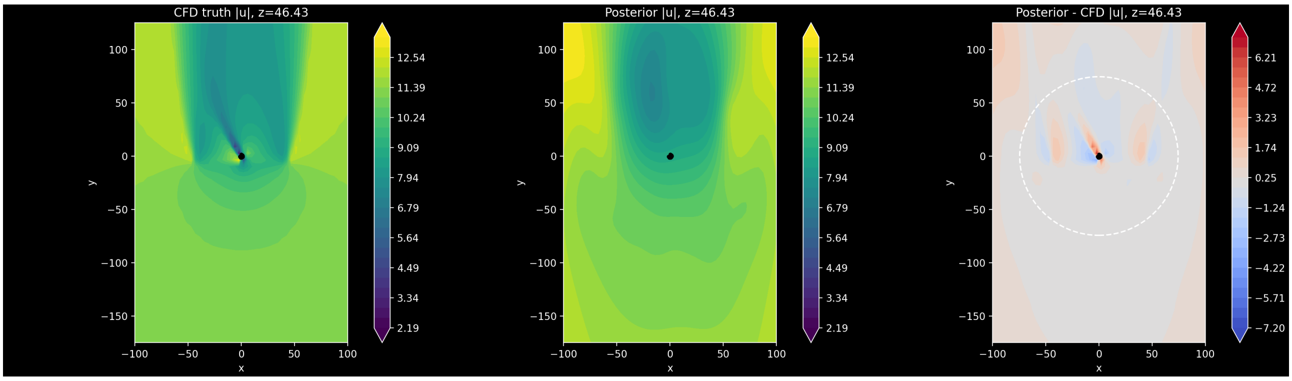


Figure 7.13: Contour plot of velocity magnitude around a wind turbine obtained from divergence-free kernel GPR (N=400, 3 adaptive steps) with uniform freestream velocity prior mean, sliced at z = 46.43 [m]. The three subfigures, left to right, are the CFD truth, the GPR posterior, and their difference, with the dashed circle marking the cylinder integration surface.

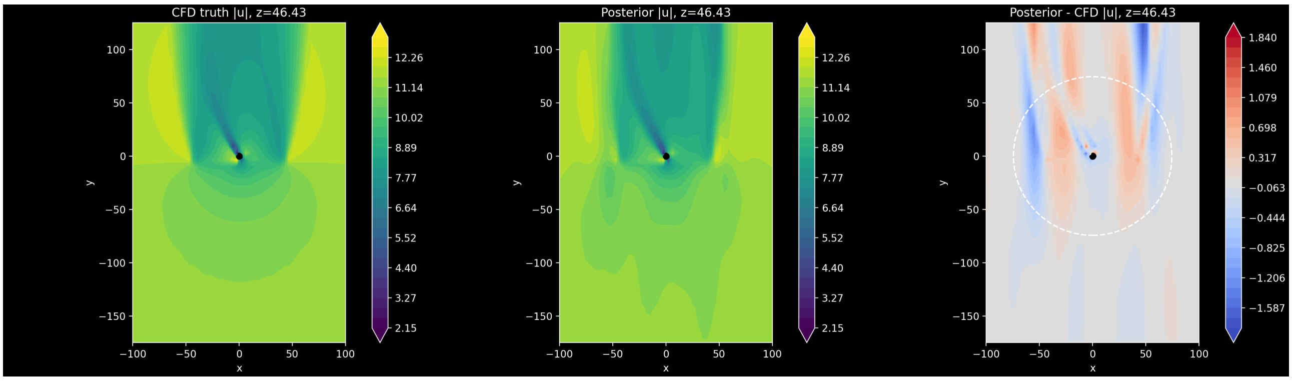


Figure 7.14: Contour plot of velocity magnitude around a wind turbine obtained from divergence-free kernel GPR(N=400, 3 adaptive steps) with inviscid CFD prior mean, sliced at z = 46.43 [m]. The three subfigures, left to right, are the CFD truth, the GPR posterior, and their difference, with the dashed circle marking the cylinder integration surface.

The difference between the two reconstructions can also be seen in the scaling of the RMSE. The change of RMSE in the entire cfd domain as the number of sampling points increases with each adaptive stage is shown in Table 7.7. As the table shows, at every sampling density, the CFD prior outperforms the freestream prior GPR in RMSE. This correlates with the visual similarity in Figure 7.13 and Figure 7.14. Additionally, both RMSEs decrease monotonically.

Table 7.7: Velocity RMSE comparison between two reconstruction methods for differing number of sample points (added through adaptive sampling)

Samples	CFD prior RMSE	Freestream prior RMSE
100	0.5107	0.7634
200	0.4903	0.6723
300	0.4736	0.6522
400	0.4354	0.6466

7.10.3. Pressure Field Reconstruction

The result of the pressure field reconstruction is shown in Figure 7.15. There is only one reconstruction as the pressure prior is set to zero (gauge) on the entire domain for both cases. As can be seen in Figure 7.15, the model effectively determines the existence of the high pressure zone in front of the turbine and low pressure zone behind it, resulting in near-zero pressure error on the surface of the control volume. The greatest errors are exhibited near the surface of the actuator disk where the GPR over-smooths the field where the CFD truth shows a sharp transition between the low and high pressure side. This is expected as the pressure GP is not physics informed and thus struggles away from sampling points. The same is true for far upstream and far downstream of sampling points.

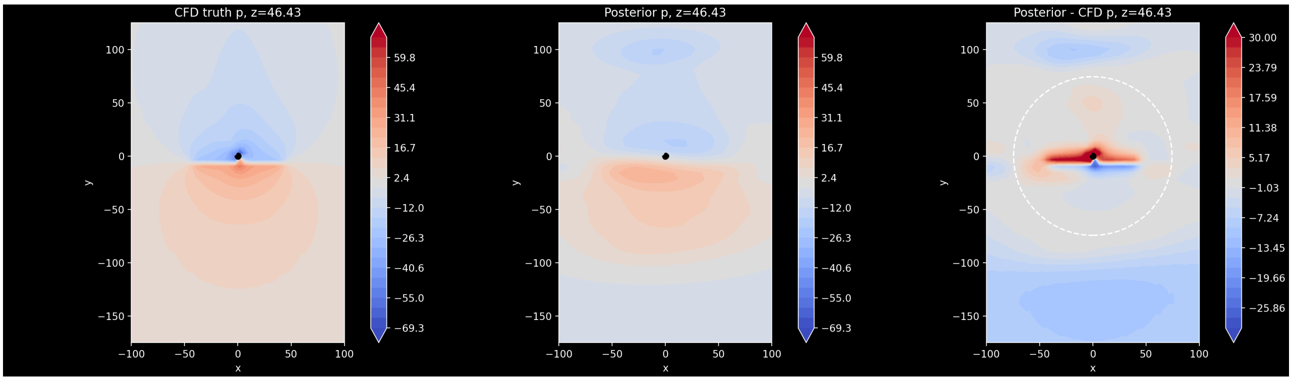


Figure 7.15: Contour plot of GPR static pressure reconstruction (N=400, 3 adaptive steps) around a wind turbine, centred about the ambient freestream static pressure at z = 46.43 [m]. The three subfigures, left to right, are the CFD truth, the GPR posterior, and their difference, with the dashed circle marking the cylinder integration surface.

7.10.4. Force Estimation

In this section, the force estimation performance of the GPR with the two priors is discussed and compared to the ground truth. The force was calculated from the momentum integral applied over the surface of the control volume. Like in **??**, the simplification of neglecting the Reynolds stress contribution was applied, and thus some deviation between the real force and the converged force of the momentum integral is expected. Table 7.8 presents the convergence of the streamwise and lateral forces and compares them to the force obtained from CFD. These forces are the sum of the force generated by the turbine and the drag force of the base of the turbine.

Table 7.8: Comparison of CFD prior and freestream prior methods

Samples	F_x [N]		F_y [N]	
	CFD prior	Freestream prior	CFD prior	Freestream prior
100	29905.4	23019.2	719676	738877
200	29984.2	29316.8	706613	731272
300	30977.1	28073.2	708744	718323
400	29957.8	22468.5	712614	716395

As can be seen in Table 7.8, the streamwise force (F_y) remained consistent between the two GPs across runs, varying by <3.5% at all sample counts. At 400 samples, the difference in F_y between the two GPs was just 0.53%, showing that the choice of prior did not significantly affect this force component. On the other hand, the freestream GP, predicted only 75.0% of the CFD prior's lateral force (F_x). On top of this, unlike the CFD prior GP, the freestream GP's lateral force value significantly oscillates (sometimes being very close to the CFD prior GP, sometimes deviating) which suggests instability and non-convergence (further discussed in Subsection 7.10.5). TALK ABOUT REAL VALUE ONCE WE GET IT.

7.10.5. Discussion and recommendations

This section discusses the findings of the wind turbine proof of concept study and synthesizes them into recommendations for further project development. As shown above, despite the lack of adaptation of the method for the specific challenges of reconstructing the flow around wind turbines, the flow reconstruction and force determination method originally developed for and verified on the aerospace building geometry produce consistent and accurate predictions of the 3D flowfield and aerodynamic forces.

Comparing it to the baseline test case of the aerospace building, the velocity field reconstruction of the wind turbine shows a lower RMSE error in velocity magnitude, both for the inviscid flow prior and for a uniform freestream prior. This is likely due to milder gradients compared to the violent

because thats the mean over man seeds, yo can see the sprea is also ad ditional 11%, also think the 30% erro I quoted in the sin gular run was com-flow separation observed on the sides of the building in the first proof of concept case. When it comes to force determination, the instability of the lateral force component F_x in the freestream prior GP suggests that, as derived in Section 5.2, the $\nabla\phi$ term cannot accurately approximated by the freestream velocity and an accurate prior mean is necessary for accurate and converging force estimation.

The low variation of the F_y component, particularly in the CFD prior GP might be the result of the close proximity of the inviscid prior and CFD ground truth as they both use the same actuator disk model to model the turbulence. Nevertheless, the consistent and accurate performance of the freestream velocity prior GP suggests that this might also be attributed to easy-to-predict smooth gradients across the x direction. Based on this, the study suggests that the drone budget determined for the building is indeed sufficient to for the survey of a wind turbine and the process can likely by optimized further (for example by improving hyperparameter tuning to mitigate the possibility of streaks if initial measurement leads to extreme length scales).

However, it is also clear that further work is required to make the system reliable and deployable in real-world conditions as stability and consistency with reality must be ensured. To accomplish this goal, a list of recommendations is presented and addressed in (Chapter 15).

8 Drone Design

8.1. Study of Drone Wake Effect

The propellers that keep a rotorcraft UAV airborne disturb the surrounding air, potentially corrupting wind measurements. In the present setup, this contamination operates through two mechanisms: the upstream drone perturbs the approach flow before it reaches the building, and the downstream drone operates inside the building wake where rotor wash and wake-induced turbulence interact.

The standard model for a rotor is actuator disk theory, in which the rotor is replaced by a thin disk of area $A = N\pi D^2/4$ (where N is the number of rotors and D the rotor diameter) that imparts a pressure jump on the air passing through it [25]. Applying conservation of mass and momentum to the air column passing through the disk gives the induced velocity at the rotor plane:

$$U_{\text{ind}} = \sqrt{\frac{T}{2\rho A}}, \tag{8.1}$$

where T is the total rotor thrust and ρ is air density. The induced velocity is the primary scaling velocity for all rotor-induced disturbances, and it increases with thrust. Determining the thrust in a crosswind therefore requires an equilibrium analysis of the drone.

Let the coordinate system be defined with x and y horizontal and z vertical (positive upward). The ambient wind vector is

$$\mathbf{U} = (U_x, U_y, U_z), \tag{8.2}$$

where U_x and U_y are horizontal components and U_z is the vertical component of the incoming flow.

A drone in a steady position must generate a thrust vector

$$\mathbf{T} = T\hat{n}, \tag{8.3}$$

where $\hat{n}$ is the unit vector along the rotor axis (normal to the rotor disk). In equilibrium, the thrust balances gravity and the aerodynamic force acting on the vehicle:

$$T\hat{n} = mg\hat{z} + \mathbf{F}_{\text{aero}}. \tag{8.4}$$

Here $mg\hat{z}$ acts downward through the weight term (with $g > 0$ and $\hat{z}$ pointing upward), and $\mathbf{F}_{\text{aero}} = (F_x, F_y, F_z)$ represents the aerodynamic force exerted by the flow on the drone body.

Taking the magnitude of Equation (8.4) gives:

$$T = \left\| mg\hat{z} + \mathbf{F}_{\text{aero}} \right\| = \sqrt{F_x^2 + F_y^2 + (mg + F_z)^2}. \tag{8.5}$$

In the urban wind environment at up to 6 Beaufort considered in this project, all three components of $\mathbf{F}_{\text{aero}}$ are non-negligible: strong horizontal winds drive F_x and F_y, while building-induced updrafts, downdrafts, and flow separation at roof edges produce significant F_z. Closing, Equation (8.5) analytically requires a drag model for $\mathbf{F}_{\text{aero}}$, which depends on the drone geometry, attitude, and local wind vector in a coupled manner. Since the drone attitude itself depends on $\mathbf{F}_{\text{aero}}$ through Equation (8.4), the system cannot be resolved without platform-specific aerodynamic data.

changed from 8

Rather than attempting to model this coupled problem, the thrust is bounded using only the physical limits of the propulsion system. Since the drone must remain airborne and in control at all times, the thrust satisfies:

$$mg \leq T \leq T_{\max}, \tag{8.6}$$

where $T_{\max}$ is the maximum thrust the propulsion system can produce, typically provided by the manufacturer. The lower bound corresponds to calm hover with no aerodynamic load; the upper bound is a hard physical limit independent of flow direction, drone geometry, or drag model. Substituting into Equation (8.1) gives the corresponding range for the induced velocity:

$$\sqrt{\frac{mg}{2\rho A}} \leq U_{\text{ind}} \leq \sqrt{\frac{T_{\max}}{2\rho A}}, \tag{8.7}$$

which bounds the disturbance zone without requiring any assumptions about the wind field or drone aerodynamics.

The spatial extent of the disturbance is characterised by the perturbation velocity $\mathbf{u}^* = \mathbf{u} - \hat{\mathbf{U}}$, the difference between the local velocity and the freestream. Two thresholds define whether a location is disturbed:

- **WMO threshold:** $\|\mathbf{u}^*\| = 0.5\ \mathrm{m\,s^{-1}}$, the maximum acceptable wind measurement error per the World Meteorological Organisation [44].
- **Strict threshold:** $\|\mathbf{u}^*\|/U_0 = 1\%$, more appropriate for structural applications [20].

The disturbance zone scales with the UAV radius R_{UAV} (distance from centreline to outermost rotor tip) and thrust. Page et al. [34] showed that this geometry is largely independent of the number of rotors for quadcopter, hexacopter, and octocopter configurations when normalised by R_{UAV}. The dominant parameters are therefore aircraft size and thrust, not rotor count.

Flight mode has a first-order effect on the disturbance extent [34]:

- **Ascent** produces the smallest disturbance zone (approximately 40% smaller than in hover) as the drone rises into undisturbed air.
- **Hover** is the reference condition, with an approximately axisymmetric disturbance field.
- **Descent** is the most severe case: the drone descends into its own rotor wash, producing a recirculation zone that can extend beyond $6D$ laterally in calm conditions.
- **Ambient wind** advects the rotor wash downstream, partially reducing the disturbance on the windward side, but simultaneously increases $\|\mathbf{F}_{\text{aero}}\|$, raising the required thrust via Equation (8.5) and expanding the disturbance zone.

For disc loading $10\ \mathrm{kg\,m^{-2}}$, the lateral disturbance radius is:

$$r_{0.5} \approx 2.5 R_{\text{UAV}} \frac{m/A}{10\ \mathrm{kg\,m^{-2}}} \tag{8.8}$$

Jin et al. [20] validated CFD simulations against three-Doppler-lidar measurements around a hovering drone ($D = 0.71$ m) and report that under the strict 1% threshold, the disturbance zone on a horizontal plane $1D$ below the drone extends:

$$r_{1\%,\,\text{horiz}} \approx 2.8\,D \tag{8.9}$$

$$r_{1\%,\,\text{vert}} \approx 7.0\,D \tag{8.10}$$

The vertical component is disturbed over a much larger region than the horizontal. These results are collected in Table 8.1, with physical values to be computed once the drone parameters are confirmed.

Table 8.1: Disturbance radii for a hovering rotorcraft UAV at nominal thrust ($T = mg$). Under the wind conditions of this project, actual values will exceed these estimates by a factor up to $\sqrt{T_{\text{max}}/mg}$. Physical values require platform-specific parameters D, R_{UAV}, and T_{max}.

Source	Component	Normalised radius	Physical radius
[34]	Lateral (WMO, 0.5 m s^{-1})	2.5 R_{UAV}	[TBD] m
[20]	Horizontal (1% of U_0)	2.8 D	[TBD] m
[20]	Vertical (1% of U_0)	7.0 D	[TBD] m

Implications for the Measurement Setup

Upstream drone. The upstream drone contaminates the building approach flow if placed too close to the windward face. From Equations (8.9) and (8.10), the conservative minimum stand-off distance is:

$$x_u \geq 7.0\,D, \tag{8.11}$$

driven by the vertical velocity component. If only horizontal components are used for load estimation, this reduces to 2.8D. The resulting error in the facade pressure coefficient from a perturbation $u^*_{\parallel}$ along the mean wind direction is:

$$\Delta C_p \approx \frac{2\,u^*_{\parallel}}{U_0}, \tag{8.12}$$

giving $\Delta C_p \approx 0.2$ at the WMO threshold for $U_0 = 5\ \text{m s}^{-1}$, which is non-negligible relative to typical facade pressure coefficients of order unity.

Downstream drone. Inside the building wake, the local mean velocity is reduced relative to the freestream, which weakens the advection that would otherwise carry the rotor wash away from the drone. The disturbance zone is therefore larger than it would be in open air at the same nominal wind speed. Furthermore, vertical velocity components in the recirculation zone contribute to F_z in Equation (8.5): a downdraft ($F_z < 0$) increases the required thrust and therefore U_{ind}, expanding the disturbance zone beyond the values in Table 8.1. In the most severe case, being a drone descending into a downdraft inside the building wake, the vortex ring state and elevated thrust demand act simultaneously. Data collected during descent in the building wake should therefore be excluded from flow reconstruction.

8.2. Trade-off summary

In the Midterm Report [3] trade-offs were done for the subsystems used on the drones. A summary of the results of the Trade-Offs can be seen in Table 8.2.

Table 8.2: Final Concept - Selected Option per Subsystem

Subsystem	Selected Concept
Software (reconstruction)	Gaussian Process Regression
Hardware platform	Dense Swarm (rel. small drones)
Power supply	Li-Ion Battery
Rotor configuration	Hexa-rotor
Communication	S-Band TDMA Mesh
Positioning	RTK-GNSS
Structural frame	Modular Tubular CF Frame
Velocity sensor	Miniaturised 3D Sonic Anemometer
Pressure sensor	Capacitive Pressure Sensor

wifi LoRa

8.3. Preliminary Sizing

For preliminary sizing of the drones, eCalc's xcopterCalc tool was used [7], at the recommendation of an expert in drone design, Brian Puroja from the Power and Propulsion department of TU Delft. The xcopterCalc tool is an online multicopter performance calculator used to estimate the behaviour of a selected drone configuration based on inputs such as drone mass, rotor configuration, motor and propeller sizing, batteries, ESC and environmental conditions. The platform uses sizing parameters such as thrust, power consumption, but also a proprietary drag calculation to estimate range, endurance and payload. In this project, the xcopterCalc tool was used to converge on a preliminary design and selection of an adequate propulsion system.

The design started with a list of assumptions about the drone design, which can be found in Table 8.3:

Table 8.3: Assumptions used for preliminary drone sizing.

ID	Assumption	Justification / Comment
AS-PRE-01	The combined mass of the frame, sensors, companion computer, flight controller, wiring, mounting hardware and structural accessories was assumed to be 1000 g.	This value was used as an initial non-propulsion mass estimate during the first sizing iterations. It includes the X500 X6 frame, payload and supporting hardware, while leaving the propulsion system and battery to be sized separately.
AS-PRE-02	The usable battery capacity was assumed to be 85% of the nominal battery capacity.	The battery was assumed to be discharged from 100% state of charge to 15% state of charge during operation. The remaining 15% was kept as a safety margin to avoid excessive battery degradation and provide operational reserve.
AS-PRE-03	The current consumption of onboard accessories was initially assumed to be 1 A.	This value was used as a preliminary estimate for the power consumed by low-voltage electronics such as the flight controller, companion computer, sensors, RTK GNSS and communication hardware. In the final sizing iteration, this value was increased to 3 A to better represent the selected onboard electronics.

Table 8.3: Assumptions used for preliminary drone sizing.

ID	Assumption	Justification / Comment
AS-PRE-04	Carbon-fibre propellers were assumed for the preliminary sizing.	Carbon-fibre propellers were selected because of their higher stiffness compared to plastic or glass-fibre alternatives. This reduces blade deformation under load and gives more predictable thrust generation, which is relevant for operation in strong wind conditions.
AS-PRE-05	The preliminary sizing was performed using off-the-shelf components available in eCalc's xcopterCalc database where possible.	Using existing components allowed rapid comparison between feasible motor, propeller, ESC and battery combinations. When an exact component was unavailable in the database, the closest available alternative was used as an approximation.
AS-PRE-06	The frame size was kept fixed during the first sizing iterations.	The X500 X6 frame was selected as the initial design baseline. Keeping the frame fixed constrained the maximum propeller diameter and allowed the propulsion system to be sized around a compact, manufacturable hexacopter configuration.
AS-PRE-07	The drone was sized as a hexacopter.	The hexacopter configuration was selected from the configuration trade-off because it provides improved redundancy and survivability in the case of a single motor or propeller failure compared to a quadcopter.
AS-PRE-08	The eCalc drag and endurance estimates were treated as preliminary estimates only.	The xcopterCalc tool uses simplified and partly proprietary models for drag and flight performance. Therefore, the results were used for preliminary convergence and component selection, not as final validated performance values.
AS-PRE-09	The velocity sensor is placed at the end of a sideways-extending boom.	The literature study presented in Section 8.1 revealed that for the sensor to have at most 1% error due to the drone's influence on the flow, it must be placed at least 7D below it or 2.8D away laterally (where D is the propeller's diameter), with the latter option considered more appropriate.
AS-PRE-10	The propulsion system shall provide a thrust-to-weight ratio of at least 2.5. A ratio of 2.0 is considered the minimum acceptable value.	Based on expert consultation with Brian Puroja, a researcher at TU Delft specialising in drone design. This T/W margin is considered enough to counteract gusts and ensure stable hover in Beaufort 6 conditions.

The preliminary sizing process was not performed as a single-pass calculation, since the frame, propeller, motor, ESC and battery selections are strongly coupled. Additionally, finalized drone sizing offered updated performance characteristics for the software component to consider. Instead, an iterative configuration-level sizing approach was used. Candidate configurations were evaluated in eCalc, after which infeasible combinations were rejected based on endurance, thrust margin, current draw, battery limits, motor loading, packaging constraints and component availability. This sizing process was thus used as a design-space exploration tool, rather than an optimization routine. Several of the important iterative steps can be found in Table 8.4.

Table 8.4: Main design iterations during preliminary drone sizing.

Iteration	Design focus	Main change	Reason for change
PRE-IT-01	Initial baseline sizing	The X500 X6 hexacopter frame was selected as the first sizing baseline, with a 600 mm motor-to-motor distance.	This provided an off-the-shelf hexacopter frame compatible with preliminary eCalc sizing and allowed an initial estimate of propulsion and battery requirements. Here, assumption AS-PRE-01 was used, to estimate initial weight of drone without drive and battery
PRE-IT-02	Container packaging constraint	The original frame size was found to be problematic for fitting the required number of drones inside the container. To comply with requirement the arm length was reduced, resulting in a smaller frame size of approximately 430 mm. An added bonus of this change was the reduction in drag from a smaller frame, but also the reduction in sensor boom size, making the 1000g frame and payload weight more realistic.	The drone had to remain compact enough for storage and transport while still allowing sufficiently large propellers for endurance and wind authority.
PRE-IT-03	Propeller sizing	The reduced frame was matched with 8 x 5.5 inch carbon-fibre propellers.	This propeller size provided a compromise between compactness, thrust capability and efficiency. Carbon-fibre propellers were preferred because of their higher stiffness and more predictable performance under load.
PRE-IT-04	Motor matching	Multiple motors were evaluated together with the selected propeller size until suitable motor-propeller combinations were found.	The motor had to provide sufficient thrust margin without excessive current draw, overheating risk or unrealistic operating points.
PRE-IT-05	ESC selection	Lightweight ESC options were researched and matched to the expected current draw of the selected motors.	The ESC had to support the required peak and continuous current while minimising the mass penalty.

Table 8.4: Main design iterations during preliminary drone sizing.

Iteration	Design focus	Main change	Reason for change
PRE-IT-06	Initial battery selection	LiPo batteries were initially considered due to their high discharge capability.	Although LiPo batteries can provide high current, the tested configurations did not provide sufficient endurance for the mission requirement at an acceptable mass.
PRE-IT-07	Li-ion battery investigation	High-capacity Li-ion cells were investigated, including high-C-rate cells from Amprius.	At high capacities, Li-ion batteries could remain within their allowable C-rate while providing improved specific energy compared to LiPo batteries. This made them more suitable for the endurance-driven mission.
PRE-IT-08	Final battery revision	The battery selection was updated again after further component research and updated mass estimates.	The final battery choice was based on the combined requirements of endurance, discharge current, mass, availability and compatibility with the propulsion system.
PRE-IT-09	Final boom and sensor placement	The sensor-supporting boom is now mounted in front of the drone, instead of it being mounted sideways.	The main reasons behind the change include stability concerns (weathervane effect), improving measurement quality (by placing the sensor in front of the drone) and weight reduction (by removing additional boom).
PRE-IT-10	Final frame design	The frame's central plate is now extended further back to allow aft battery mounting.	This change is done to counteract the forward-shift of the drone's center of gravity caused by PRE-IT-09.

8.3.1. Boom sizing and positioning

As described in Section 8.1,

Table 8.5: Disturbance radii for a hovering rotorcraft UAV at nominal thrust ($T = mg$). Under the wind conditions of this project, actual values will exceed these estimates by a factor up to $\sqrt{T_{\max}/mg}$. Physical values require platform-specific parameters D, R_{UAV}, and $T_{\max}$.

Source	Component	Normalised radius	Physical radius
[34]	Lateral (WMO, 0.5 m s^{-1})	2.5 R_{UAV}	0.829 m
[20]	Horizontal (1% of U_0)	2.8 D	0.569 m
[20]	Vertical (1% of U_0)	7.0 D	1.422 m

8.3.2. Sensor selection

As per the results from the case study in Chapter 5, the necessary measurements needed from each of the datapoints are pressure and velocity.

Pressure Sensor

The main restrictions on the pressure sensor, which is needed to measure the static pressure is that it can measure absolute pressure(a differential pressure needs a reference value, which cannot be connected to the same port for all of the drones) and that it needs to fit well within the drone and be lightweight. The clear choice for this sensor is the BMP581, as it is extremely cheap, readily available and has a relative accuracy of 6Pa. The more restricting part of using the sensor is getting only the static pressure component, without interference from air causing dynamic pressure. To achieve that the sensor is mounted close to the velocity sensor, so that corrections on the pressure estimation can be made, algorithmically correcting for the dynamic pressure. A simplistic implementation of this is the formula $p_{static} = p_{total} - \frac{1}{2}\rho v^2 \cos^2\theta$, where θ is the angle between the sensor and the velocity vector, v is the speed of the air and ρ is the air density. After physical implementation this will be corrected with a interpolation of a sample experiment inside a wind tunnel to calibrate the sensors properly.

FIND REQUIREMENT

Velocity Sensor

Converging on a velocity sensor is more sensitive to the particular requirements of the project. The main options for the velocity sensor are sonic anemometers and differential pitot tube sensors. Comparing the price of the two options, the cost of sonic anemometers is always large, over 1000€ per unit, whereas pitot tubes have the potential of being a cost effective method through developments in custom implementations, such as the one developed by EPFL[41]. The estimated cost of this sensor is closer to the order of magnitude of 150€ per unit. However, all pressure differential based sensors suffer from the problem that at low wind speeds, below 5m/s their accuract is greatly decreased or undocumented even for commercial, expensive ones. ~~This is a restriction that makes them unsuitable~~ for this usecase as the wake of the flow, where the momentum deficit lies has many points with low air velocity, meaning that a lot of data will be lost to the inaccuracies of the sensor, especially considering the need to estimate the fluctuations for the Reynolds stresses. Thus, the realistic choice for the velocity sensor is the cost-ineffective but measurement critical sonic anemometers, of which the Transonica Mini, designed for drones is the perfect candidate. It has both very low mass at around 67 grams and is also very accurate. The only drawbacks are the price and the ±15° in the upwards direction restriction, as it can't measure large vertical components. For our project this is considered enough as the main component of the force is in the in plane direction, and the vertical force is not calculated.

Final Sensor Config

8.3.3. RTK positioning components

8.3.4. WiFi communication components

8.3.5. Battery sizing and selection

As mentioned in Table 8.4, LiPo batteries were disregarded during the preliminary sizing process due to their low endurance capabilities. Thus, Li-Ion batteries became the main focus point of the design, however high capacities were required to be capable of supporting the large current draw of the six motors, companion computer, flight controller, sensors, RTK rover and WiFi antenna.

A market analysis was performed on current off-the-shelf Li-Ion batteries, which revealed that there are currently few options for high capacity, high discharge rate, ready-to-fly batteries, mostly designed for agricultural drones. These options were still slightly too heavy for the drone, resulting in suboptimal endurance, which meant an in-house battery design was required.

Research was carried into State of the Art high capacity Li-Ion cells from manufacturers such as Amprius, AtomFair and Herewin. For each cell, multiple configurations were considered and for the weight, a margin of 10% was added to account for battery packaging, connectors, spot welds and solders. In Table 8.6 some of the trialed battery configurations can be found.

Table 8.6: Battery configurations considered during preliminary sizing.

ID	Cell model	Config.	Cell cap.	Pack cap.	Mass + 10%	Cell dimensions
BAT-01	Amprius SA11	8S1P	30.75 Ah	30.75 Ah	2684 g	198.0 × 96.5 × 7.0 mm
BAT-02	Amprius SA129	8S1P	14.8 Ah	14.8 Ah	1487 g	148.0 × 91.5 × 5.6 mm
BAT-03	Amprius SA504	9S1P	11.05 Ah	11.05 Ah	963 g	125.4 × 42.2 × 8.25 mm
BAT-04	Atomfair NM-C/Gr+SiC 30 Ah pouch cell	6S1P	30.0 Ah	30.0 Ah	1848 g	170×70×12.5 mm
BAT-05	Atomfair 8540140-15 15 Ah pouch cell	6S2P	15.0 Ah	30.0 Ah	1584 g	140 × 40 × 8.5 mm

Each of these battery configurations were tested and matched with appropriate components, and drone performance was tested for each configuration using eCalc approximations. While initially, configuration BAT-01 delivered promising results, it was revealed that the small, high RPM motors required for this application cannot handle the voltage of the configuration. Thus, the BAT-04 configuration was selected in the end.

This configuration was thus further developed, and a bill of materials was drafted to better estimate the cost and weight of the battery, which can be seen in Table 8.7:

Table 8.7: Estimated battery pack component mass and cost.

Component	Specification / Description	Mass [g]	Cost [€]
Battery cells	6S1P, 30 Ah Li-Ion pouch cell configuration	1680	825.0
Main power connector	XT90 connector	15	1.5
Balance connector	JST-XH 7-pin balance connector	10	1.5
Cell insulation	0.38 mm fish paper	7	5.0
Insulation tape	25 mm Kapton tape	2	0.3
Compression plates	Carbon-fibre compression plates	105	9.0
Compression tape	Fibre-reinforced packing tape	5	0.2
Power wires	Estimated wiring mass	5	1.0
Heat-shrink tube	Battery pack heat-shrink sleeve	15	0.8
Total		**1844**	**844.3**

8.3.6. Propeller sizing and selection

The most important influence on propeller selection is the frame size, as a bigger frame can accommodate bigger propellers. Propellers are also directly linked to current draw (bigger propellers require higher torque). Propeller pitch affects required torque, but also thrust-to-weight: higher pitch increases air pushed axially, disk loading and leads to less efficient flying, but increases wind authority, which is important in Beaufort 6 conditions. Finally, the material of the propeller affects its characteristics, as plastic or nylon are flexible and durable, can reduce some vibrations, but make for unreliable production of thrust and can induce flutter. Carbon fibre meanwhile is very stiff, making thrust production reliable and more efficient, but remains very brittle, thus sensitive to impacts.

A definitive decision was made to use Carbon Fibre, as predictable thrust is highly important for autonomous controllers. Initially, with a 600mm frame, a propeller diameter of 11 inches was considered, however, after the frame change, it was reduced to 8 inches. After frame size was locked, a study into market, off-the-shelf options was performed, which revealed only one carbon-fibre dual propeller designed for multicopters, with a pitch of 5.5 inches. Thus, this propeller was selected for the final preliminary design.

Although the selected high-pitch propellers improve thrust generation and wind authority for the compact rotor diameter, they also increase the sensitivity to vortex ring state (VRS). VRS occurs when a rotor descends into its own induced wake, leading to recirculating flow through the rotor disk, reduced effective thrust, and potentially unstable and irrecoverable descent behaviour. This risk is most relevant during near-vertical descent at low horizontal velocity, especially when the descent rate becomes comparable to the rotor-induced velocity. To mitigate this, the control loop will avoid purely vertical descent profiles and instead command a small forward or corkscrew descent trajectory. By maintaining a horizontal component of motion, the rotors are kept in relatively undisturbed air, reducing the likelihood of VRS while preserving the benefits of the chosen high-pitch propeller configuration. For instances where purely vertical descents are demanded, the descent rate will be limited to $1m/s$.

8.3.7. Motor sizing and selection

The electric motors were selected together with the propeller and battery, since motor selection is strongly influenced by propeller size, battery configuration and thrust demands. During the sizing process, different battery voltages, propeller and frame sizes were considered, thus, different candidate motors were evaluated for these configurations, rather than one single motor selection at the beginning of the design process. Once battery and propeller configuration had converged, motor selection was iterated for the final propulsion-system architecture.

The minimum, maximum thrust that a motor should provide is calculated as follows:

$$T_{\text{max,motor}} = \left(\frac{T}{W}\right)\frac{mg}{N_{\text{rotors}}}, \tag{8.13}$$

Where $\left(\frac{T}{W}\right)$ is the value assumed in AS-PRE-10, Table 8.3, m is the mass of the drone, g is the gravitational acceleration and N_{rotors} is the number of rotors, which in the case of a hexacopter is 6. The motor must therefore provide this required thrust while remaining within its current, power, rotational speed and thermal limits.

One of the main characteristics of motors is the K_V rating which approximately relates the no-load motor speed to applied voltage:

$$n_0 \approx K_V V, \tag{8.14}$$

Where n_0 is the no-load rotational speed and V is the applied voltage. A higher K_V motor is generally able to reach higher rotational speeds for a given voltage while sacrificing torque, while a lower K_V motor typically produces more torque and is more suitable for larger or more heavily loaded propellers. While the K_V rating is important for motor sizing, it cannot be independently used to determine whether a motor is suitable, and must instead be combined with other parameters, such as the allowable battery voltage, maximum continuous current, peak current, motor mass, stator dimensions, thermal capacity, and expected efficiency.

Candidate motors were selected based on eCalc estimates and available manufacturer data. Principal screening criteria were:

- Sufficient maximum thrust;
- Acceptable current consumption during hover;
- Operation within the current, power, and thermal limits during Beaufort 6 hover and manoeuvring conditions;
- Compatibility with the selected propeller diameter and pitch; and
- Acceptable motor mass.

For the final configuration, the EMAX PRO Series 2808 motor with a K_V rating of 1350 was selected. This motor represents a good compromise between available thrust, current demand and voltage compatibility. One important thing to note is that, while the motor is rated for 6S batteries, this rating considers only LiPo batteries, for which a 6S configuration would entail a total voltage of 25.2V. However, the current battery configuration has a total, fully charged voltage of 25.8V. This is only 2.4% over the 25.2V tested margin, which is considered acceptable, as the motors would run at over their rated voltage for only a few minutes at a time. As the motor casing has very good thermal rejection capabilities, this is considered an acceptable compromise.

8.3.8. ESC selection

The Electronic Speed Controller was selected based on voltage compatibility, current capacity, mass and control reliability. The ESC must support the 6S battery and safely supply the maximum motor current under high-thrust manoeuvres.

The TBS Lucid 12S 90A ESC was selected as it provides a very lightweight package. Its 90A current rating also provides a substantial current margin relative to the approximate 45A maximum motor current. It is important to note that each of the motors requires its own ESC unit, bringing the total required number of ESCs to 6 per drone.

8.3.9. Flight controller

The Pixhawk 6C was selected as the flight controller due to its processing capability, sensor redundancy, and compatibility with the required propulsion and navigation hardware. Its STM32H7 processor provides sufficient computational capacity for flight-control algorithms, autonomous operation, sensor integration, and communication with the companion computer.

The controller incorporates redundant inertial measurement units and an integrated vibration-isolation system, improving the reliability and quality of attitude measurements. It also provides the interfaces required to connect the ESCs, RTK-GNSS receiver, telemetry system, and additional onboard sensors. Furthermore, compatibility with the PX4 flight-control software enables the implementation and modification of the control logic required for the mission.

8.3.10. Companion computer

The Raspberry Pi 4 Model B with 4 GB of RAM was selected as the companion computer for high-level onboard processing. It provides sufficient computational capability for sensor-data handling, communication, mission management, and the execution of autonomy-related software without placing these tasks on the flight controller.

Its USB, Ethernet, and serial interfaces allow integration with the Pixhawk 6C, RTK-GNSS system, Wi-Fi communication hardware, and additional sensors. The platform is also lightweight, widely supported, and compatible with commonly used robotics software, making it suitable for development and system integration.

8.3.11. Other parts

The drones also requires other, smaller supporting components. An RM3100 magnetometer must be included to ensure accurate orientation detection of the drone, complementing the RTK system. A DC/DC Voltage converter is needed to ensure the flight controller, companion computer, RTK rover, WiFi module and sensors can be powered by the required 5V. The Voltage converter selected is the Mateksys BEC12S PRO, as it has a low mass and supports a throughput of up to 12A, giving it a substantial margin over the maximum of 6A that the entire system would consume.

8.4. Preliminary mass and endurance estimation

8.4.1. Mass estimation

After selecting all components and creating a drone frame and sensor boom design, the final mass of the drone can be estimated. The bill of materials for the hardware of the drone can be found in **??**.

The resulting design weighs almost 3300g, meaning the weight of the drone will not be a problem when it comes to certification, as drones under $4kg$ are considered much safer than anything above, according to EASA standards.

8.4.2. Endurance estimation

Once the mass has been estimated, an estimation for the hover time at $0km/h$, $50km/h$ and $70km/h$ is required to ensure the drone fulfils requirements. However, in order to estimate the hover in windspeeds, the drag of the drone must be estimated first. Quan Quan [36] provides a method for estimating the drag coefficient of a drone as a function of its pitch angle. For this method, two different CFD simulations are run to calculate the drag force on the drone in 0° pitch and 90° pitch. Then, the drag coefficient of the drone can be estimated as a function of the pitch angle with the following formula:

$$C_D(\theta) = C_{D,0^\circ}\left(1 - \sin^3\theta\right) + C_{D,90^\circ}\left(1 - \cos^3\theta\right) \tag{8.15}$$

After running two different CFD simulations, it was identified that $C_{D,0^\circ} = 0.496$ and $C_{D,90^\circ} = 0.78$. By having these two coefficients and, thus, having a formula for the drag coefficient at any pitch angle, the pitch angle required for stable hover at $50km/h$ and $70km/h$ can be calculated to be $\approx 7^\circ$ and $\approx 9^\circ$, respectively. An upper margin of 2° will be taken on these results to ensure the approximated hover time is not over-approximated.

Hover endurance can be estimated by calculating the required thrust, which can be computed through the simple force equilibrium $\vec{T} + \vec{W} + \vec{D} = 0$. This can be translated into:

$$T = \frac{mg}{cos(\theta)}, \tag{8.16}$$

Which leads to $T_{50km/h} = 32.776N$ and $T_{70km/h} = 32.979N$. This results in a thrust per motor of $T_{motor,50km/h} = 5.463N$ and $T_{motor,70km/h} = 5.496N$. However, in $50km/h$ and $70km/h$ hover, most of the propellers would be located in crossflow conditions, which mainly redistributes the thrust over the propellers, but can also slightly reduce the total thrust produced by propellers. Since literature quantifying this effect is very sparse, after consulting with expert Dr. Daniele Ragni, it was decided that taking a 10% margin for required thrust is enough to account for this effect.

From actuator disk theory:

$$P = T\sqrt{\frac{T}{2\rho A}}, \tag{8.17}$$

Where P is the power required for hover, T is the thrust, ρ is the air density and A is the area of the actuator disk. Assuming $P = \eta IV$, where η is the propulsion efficiency, then thrust ideally scales with current:

$$T \propto I^{2/3}. \tag{8.18}$$

Using eCalc's estimation for hover current, which is $I_{motor,hover} = 5.38A$and taking the 10% margin for the required thrust, the final current values for hover are $I_{motor,50km/h} = 6.317A$ and $I_{motor,70km/h} = 6.38A$. Taking an average consumption of 3A for the auxilliary components (companion computer, flight controller, sensors, RTK rover and WiFi), the hover flight times are estimated to be $t_{50km/h} = 37.4mins$ and $t_{70km/h} = 37mins$. Of these, only 27 minutes are usable as the rest of 10 minutes are reserved for formation take-off and landing.

8.5. Block diagram

Figure 8.1 presents the block diagram of the complete system, combining the previously described drone components with the supporting ground-station infrastructure. It provides a system-level overview of the mechanical, power, and data interfaces between the main subsystems, illustrating how the individual drone components interact to enable propulsion, sensing, navigation, communication, and control. In addition, the diagram introduces the ground station and its connections to the drone, including the infrastructure required for battery charging, drone storage, RTK positioning, wireless communication, and on-site operation. The sizing of the ground station, together with an explanation and justification of its individual components, is presented in Section 9.1.

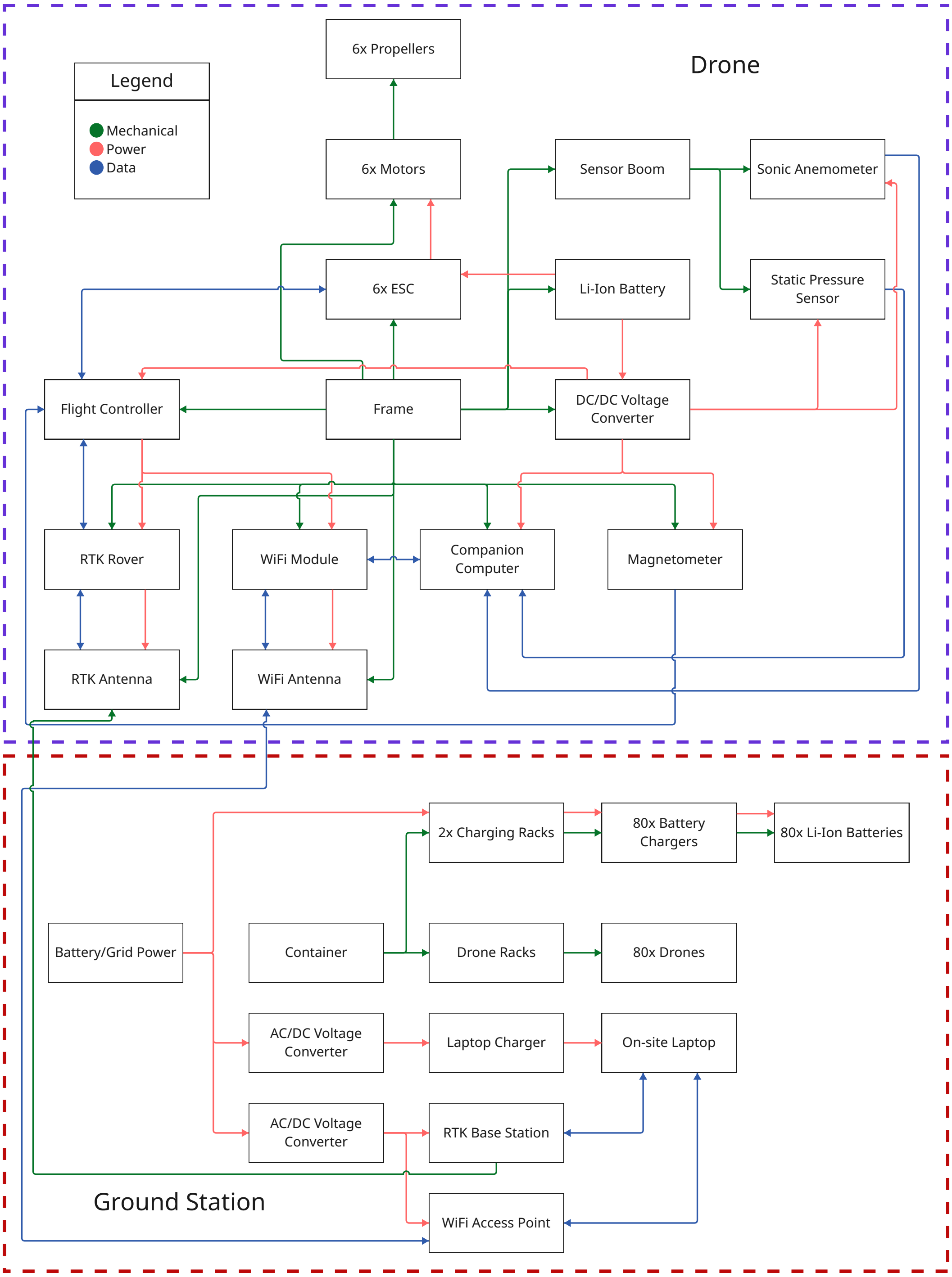


Figure 8.1: Hardware Block Diagram

RTK Antenna — RTK Base Station arrow should be blue

8.6. Data handling and electronics

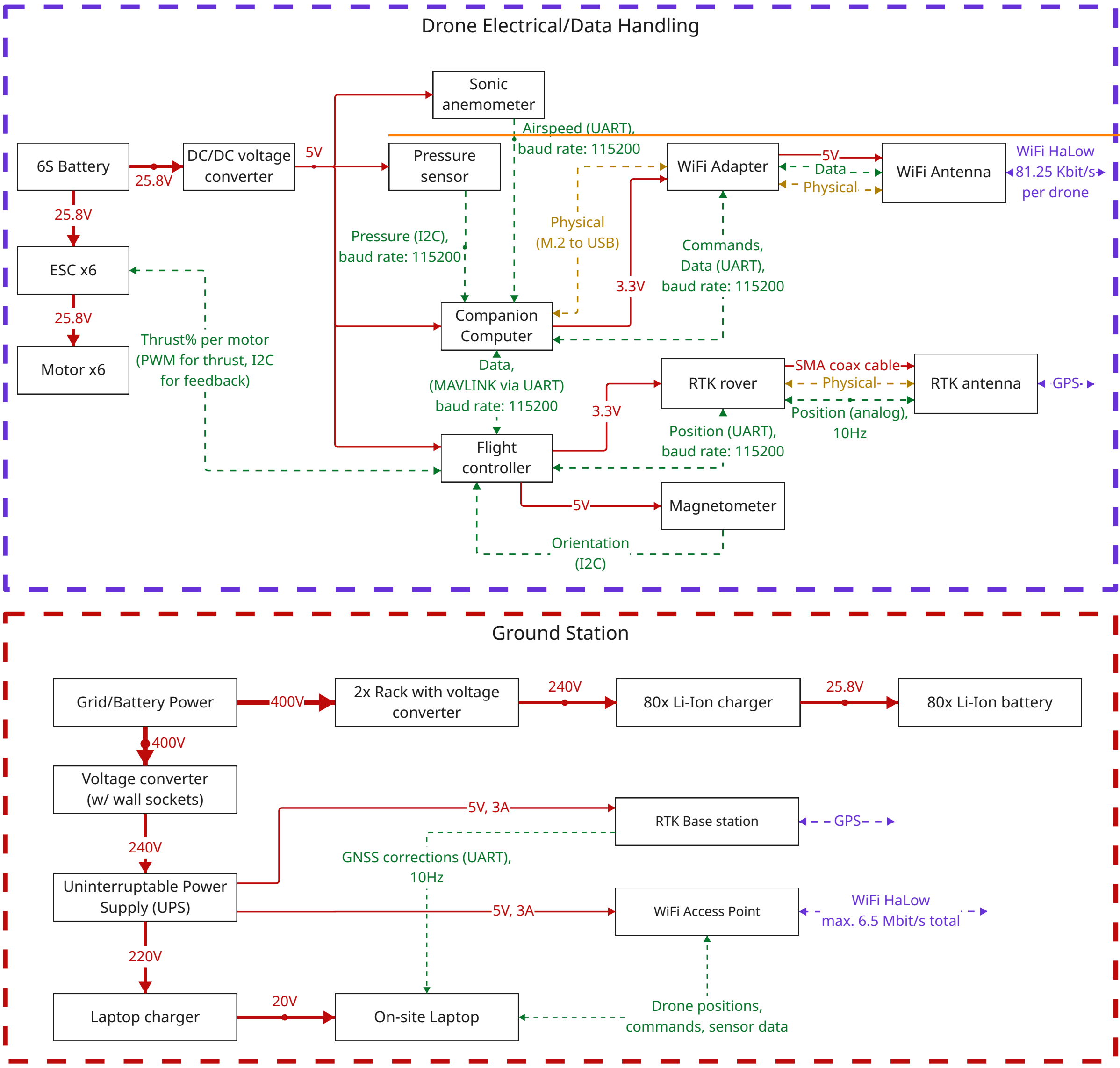


Figure 8.2: Data handling/Electronics diagram

Figure 8.2 shows the architecture of the electrical system as well as the data handling, for each drone in the swarm and the ground station. Both systems are discussed in detail in the following subsections.

8.6.1. Drone data electronics architecture and data handling

Per drone, the system is powered by 6S (25.8V) standard drone batteries (specs), which power the motors as well as the onboard flight controller, computer and payload. Standard DC/DC voltage converter provides power to the flight controller, companion computer and the sensors, all of which operate within 5V. Both the flight controller (Pixhawk 6C Mini[1]) and the companion computer (Raspberry Pi 4 model B[2]) offer 3.3V output, so they power their respective dependent electronics. Data handling follows a standard scheme for drones of comparable size.

The flight controller connects all related instruments to fly the drone autonomously, including the GPS-RTK unit and an external magnetometer for reliable readings unaffected by electromagnetic interference of power electronics near the flight controller. The communication protocols were selected according to the capabilities of the respective sensors. The flight controller supports UART connectivity

[1]https://docs.px4.io/main/en/flight_controller/pixhawk6c_mini
[2]https://www.raspberrypi.com/products/raspberry-pi-4-model-b/

specifically for the GPS and an extra I2C channel to support an external magnetometer. The flight controller is connected to a companion computer, which will control the measurement sensors. The companion computer will run on Ubuntu 20.04[3] on a standard 32GB SD card. The 4GB RAM on-board is necessary to run ROS2 nodes, which will handle velocity and pressure measurement filtering scripts, as well as to relay commands from the WiFi adapter to the flight controller (via mavsdk[4] or similar APIs). Here, the baud rate is set to 115200 for all sensors, which is equivalent to approximately 115.2 Kbits/sec per device. With a sampling rate of 10Hz per device in 32-bit floats, each sensor will collect:

$$10[\text{values/sec}] * 32[\text{bits/value}] * 1/8[\text{bytes/bit}] = 40[\text{bytes/sec}].$$

For serial digital communication protocols such as UART, the baud rate converts one-to-one to bits, therefore:

$$115200 * 1/1000 * 1/8 = 14.4[\text{Kbytes/sec}].$$

Together with communication overhead, the baud rate of 115200 will therefore support each individual sensor. With an identical estimation, the WiFi antenna data rate will also be supported by this baud rate.

The hardware interfacing with the flight controller was selected specifically with compatibility in mind. The GPS-RTK and Magnetometer units will work out-of-the-box with the selected flight controller.

is this right? diagram is wrong

9 Ground Control, Operations, and Logistics

9.1. Ground station

9.1.1. Container

As the system must be deployable in urban, rural, but also offshore environments, it was decided that the entire system must fit inside of a standard 20ft ISO container. This container can then be transported on land by a truck, or taken offshore on a boat, satisfying the requirements for the system's modular usage. To facilitate loading and offloading drones and batteries, the container must also be side-opening, as that allows operators to easily access all elements inside the container. The container will be modified to contain an extendable side ramp when opened, to allow side access for operators during on-land operation.

9.1.2. Transport truck

The truck must be capable of transporting a 20ft container, while being manoeuvrable in urban environments. For this reasons, a box truck was considered. To ensure sustainability for the system, only new, fully electric trucks were researched. The Mercedes-Benz eACTROS 400 was selected, with a 6x2 rigid configuration. This box truck is fully electric, supports the attachment of a 20ft ISO container and has a range of 390km on a full charge, for a 12.5 ton payload.

9.1.3. Drone storage

For the storage of drones, it has been decided that a shelf system is the best solution. Thus, an aluminium frame will be installed in the container, which can hold 6 removable shelves from bottom to top. Each shelf can hold three different drones, with each drone having a comfortable 700mm x 700mm area to be placed into (the drone is 625mm from tip of propeller to tip of opposite propeller), with the height of a shelf being 30cm. The aluminium frame would include 5 shelves next to one another, which are secured to both the top and bottom of the container for ideal stability.

[3] https://ubuntu.com/download/raspberry-pi
[4] https://mavsdk.mavlink.io/main/en/index.html

For each drone shelf, custom foam inserts would be manufactured for each drone, and the drones would be secured with a velcro strap around the top of the frame, to ensure no movement during transport.

The first drone shelf would start at just 5cm off the container floor, while the last one would be at a height of 185cm, making them just accessible for the operators. Unfortunately, this also sets a height constraint for the operators, as they must be tall enough to easily reach the last shelf.

This configuration for drone storage allows for the storage of 90 drones at a time.

9.1.4. Charging station

Charging of 85 batteries is a challenging endeavour, as 85 30Ah batteries hold a lot of energy, making the fire risk quite significant. Thus, it was decided that battery charging shall take place inside fire-controlled racks, designed specifically for lithium battery charging. It was decided that two ASECOS Lithium-ION ULTRA-PRO 90 are needed, which have smoke detection capabilities, fire suppression, active ventilation for charging heat rejection. Each rack has 5 shelves supporting 75 kg each. 9 batteries can be charged on a single shelf, meaning 45 batteries per shelf. However, to support the charging of 45 batteries, the cabinet's power system must be adapted from 400V 16A to 400V 32A.

The batteries must be specifically charged at 25.8V, 15A each, meaning that good, robust chargers are needed for the batteries. The chargers selected are SkyRC T1000 Maestro, which support charging with custom voltage and current, at a maximum of 480W. A total of 90 chargers are needed, one for each battery. As the batteries have XT90, and the chargers have XT60 plugs, 90 adapters are also needed.

Charging 90 batteries can generate substantial amounts of heat, which means small HVAC modifications must also be done to the container to safely dissipate this heat and ensure adequate airflow into and out of the charging racks.

As will be explained in Section 9.3, three different sets of batteries are needed to ensure measurement time is just around 4 hours. Consequentially, this means that, aside from the batteries on the drones, a further 170 batteries must be stored inside the container. The charging racks can be used for this purpose, as the space and weight limits allow for storage of this many batteries. Custom slots for the batteries can be manufactured, which would ensure the batteries would not tip over or move during transport.

9.1.5. Other Ground Station Elements

The Ground Station also necessitates a laptop used to receive real-time telemetry, readings and run real-time GPR on the measurements of the drones. Thus, a laptop with a good processor and high amounts of RAM is preferred. The laptop must be durable as well, since measurements will take place in different environments. For this reason, the GETAC X600 PRO was picked, as it is a fully rugged laptop with a state of the art Intel Xeon W-11865MRE and 128gb of RAM, making it capable of running GPR in real-time, while remaining very durable.

9.2. Setup and dismantling sequences

9.3. Operations Logistics

— setting up the coordinate system before measurements!!!! ...

As shown in Figure **??**, one measurement mission consists of four different phases (0-3).

1. **Phase 0: Initial Measurements** The freestream velocity needs to be known before sampling can start for the GPR. This is done by forming a cylinder around the building, populated with the 85 drones following a two dimensional (height and angular position) Latin Hypercube distribution. After six minutes, the freestream velocity is determined and the cylinder is tilted in the direction following the flow, with an angle of $\theta = 22°$. This angle is derived such that drones with a larger altitude don't influence the measurements below. From Table 8.5, the angle can be calculated to be $\theta = \arctan(2.8/7) \approx 22°$. The drones move so that they are more concentrated in the wake, as

this is poorly modelled by the potential flow.

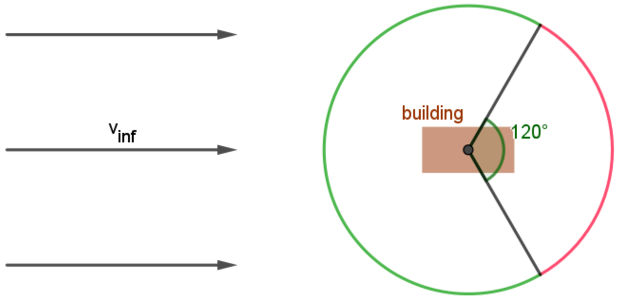


Figure 9.1: A diagram of the building from a bird's eyeview. 50% of the drones move to the red region on the figure, and 50 % stay in the green region. In both regions, the drones are spaced using a three-dimensional Latin Hypercube where height and angular position are the dimensions.

As can be seen in Figure 9.1, the red region, containing 50% of the drones, is more densely populated. Both red and green region contain 50 % of the drones each, and each is populated following the 2D Latin Hypercube again.

9.4. Software Sequence

9.5. Recharging logistics

10 Verification and Validation

This chapter defines the verification and validation plan, both during the design process and after the system is built. The aim of the plan is to set up a verification and validation framework within which not just the current design but also the final system should be tested before operational use. Verification is used to determine errors in the code and to ensure that the whole system is correctly built with respect to the requirements and design objectives. Validation is used to determine whether the system solves the intended problem, i.e. reconstructing the wake of a large structure from sparse measurements and estimating the resulting aerodynamic forces that act on the structure with acceptable accuracy.

The V&V strategy follows a staged approach based on the V diagram for Verification and Validation [8]. As such, we begin with the verification techniques for the tools used in the design (unit testing, Subsection 10.1.1) followed by the hardware level verification techniques (unit and functional testing, Subsection 10.1.2). Each of these verification stages is also followed by its own corresponding validation stage, mimicking the structure from the V diagram. Validation is performed using increasingly realistic data. Validation will start from synthetic high-fidelity CFD data (analysis, corresponding to software verification) and progressing towards controlled experiments and full-scale demonstrations (test, demonstration, corresponding to hardware verification). This is deemed necessary since, for a real large scale structure, the true aerodynamic force is generally not measurable, therefore, validation cannot rely on a single, perfect "ground truth" and must instead combine independent references and uncertainty bounds.

10.1. Verification Plan

The purpose of verification is to ensure that each component of LAWSS satisfies its requirements before the complete system is validated. Verification shall be performed at four levels: component level, subsystem level, system level, and ultimately operational level. Thus, the verification plan start with synthetic unit verification and then moves to functional testing.

10.1.1. Software Unit Testing

The software verification focuses on the wake reconstruction capability and force estimation pipeline. In order to verify it, a set of verification tests with criteria were laid out in the midterm report. Since then, the list has been augmented to include more tests, better reflecting activities needed to unit test the software. The complete list of verification tests, split between various used software tools is listed in Table 10.1. Importantly, each entry has its own unique ID, description, the criterion needed to consider it passed, level of priority (low, medium, high, critical), and general method used to verify it.

Tool Unit Tests Overview

Table 10.1: Tool unit tests planned for the detailed design.

Tool	Test ID	Description	Pass Criterion	Priority	Method
GPR	VER-GPR-01	Verify drag prediction accuracy	90% of cases below 10% error from CFD ground truth for drag in the measurement direction	Critical	Analysis
	VER-GPR-02	Verify uncertainty quantification calibration	2σ interval covers ≥ 0.90 of all test points	High	Analysis
	VER-GPR-03	Verify robustness against noise	Performance maintained when sensor errors within manufacturer-specified range are applied	High	Analysis
	VER-GPR-04	Verify hyperparameter optimisation convergence	Hyperparameter values converge as more samples are added	Medium	Analysis
	VER-GPR-05	Verify prior mean generation	Prior mean generated correctly and transferred to GPR; flow is solenoidal	Critical	Analysis
	VER-GPR-06	Verify kernel sufficient smoothness	Matérn 5/2 second derivative continuous on domain	Critical	Analysis
	VER-GPR-07	Verify reconstruction is divergence-free	Max divergence of reconstructed field $\leq 10^{-3}$ on analytical solenoidal test fields	Critical	Analysis
	VER-GPR-08	Verify far field GPR prediction approaches prior mean	Prior mean value reached at sufficient distance from measurement	Critical	Analysis
Adaptive Sampling	VER-AS-01	Verify adherence to geometric constraints	No samples requested inside the body or on the wall	High	Analysis
	VER-AS-02	Verify sample budget adherence	Total sampling points $\leq$ budget throughout runtime	Critical	Analysis
	VER-AS-03	Verify convergence of sampling results	RMSE decreases as sampling points increase over the budget range	High	Analysis
	VER-AS-04	Verify superiority over random sampling	Error drops faster than random sampling for N in the drone budget range	High	Analysis
	VER-AS-05	Verify non-interference	Samples never requested in propeller exclusion area of existing drones	High	Analysis
CFD	VER-CFD-01	Verify mesh convergence of integral quantities	GCI < 5%	Critical	Analysis
	VER-CFD-02	Verify mesh quality	Max skewness < 0.75	High	Analysis
	VER-CFD-03	Verify global mass conservation	Net mass imbalance < 0.1% of inlet flux	Critical	Analysis
	VER-CFD-04	Verify solver residual convergence	All residuals $< 10^{-5}$ within iteration limit	Critical	Analysis

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Table 10.1 – continued from previous page

Tool	Test ID	Description	Pass Criterion	Priority	Method
	VER-CFD-05	Verify turbulence model wall resolution	$y^+ < 350$ for k-ε; $y^+ < 5$ for k-ω	High	Inspection
	VER-CFD-06	Verify boundary condition implementation	Correct settings; result unaffected by boundary distance by more than 0.5%	High	Inspection
Trade-off Tool	VER-TOT-01	Verify percentages sum to 100%	Percentage win-rates of all options sum to 100%	Critical	Inspection
(relevant for midterm)	VER-TOT-02	Verify behaviour for extreme variance values	Correct operation when weights may exceed $[0, 100\%]$	High	Analysis
Momentum Integral	VER-MI-01	Verify CV integrator convergence	Recovered force converges as integration point count increases; error < 5% at converged count	Critical	Analysis

A selection of the most critical verification tests listed in Table 10.1 are documented below. Many other tests, although not addressed here directly for the lack of space, were also checked: for example, VER-AS-01, VER-AS-02, and VER-AS-03 clearly visible in the graphs in Section 7.4. The following are selected critical verification tests, visualized.

CV Integrator Convergence (VER-MI-01)

A convergence study was performed to verify the momentum integral computation from discrete testing points. To this end, a mesh of N points was constructed and the force value was computed by integrating over the created mesh surface. Each mesh point was fetched directly from the CFD data and linearly interpolated if it fell between mesh cells. As can be seen in Figure 10.1, as the integration points increase, the predicted F_y values converge to a constant value. This value was verified to be within 5% of the CFD value. The deviation is hypothesized to be due to our neglecting of the Reynolds stress contribution, as discussed in Section 5.6.

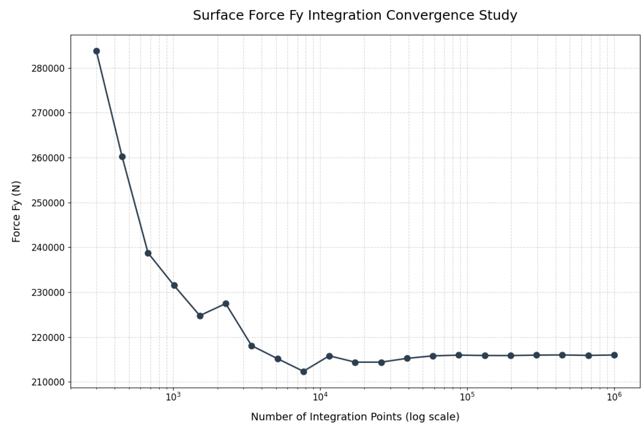

Figure 10.1: Convergence study on how the force in the y direction (drag) varies with number of points integrated

Sufficient Smoothness Test (VER-GPR-06)

The sufficient smoothness of the Matern 5/2 kernel was verified. As outlined in Chapter 5, the kernel to be used to build the divergence free GP needs to be twice continuously differentiable on the entire domain. To test this, the second derivative of the kernel was plotted and inspected for smoothness. The results are found in Figure 10.2 with Matern 3/2 plotted alongside for comparison.

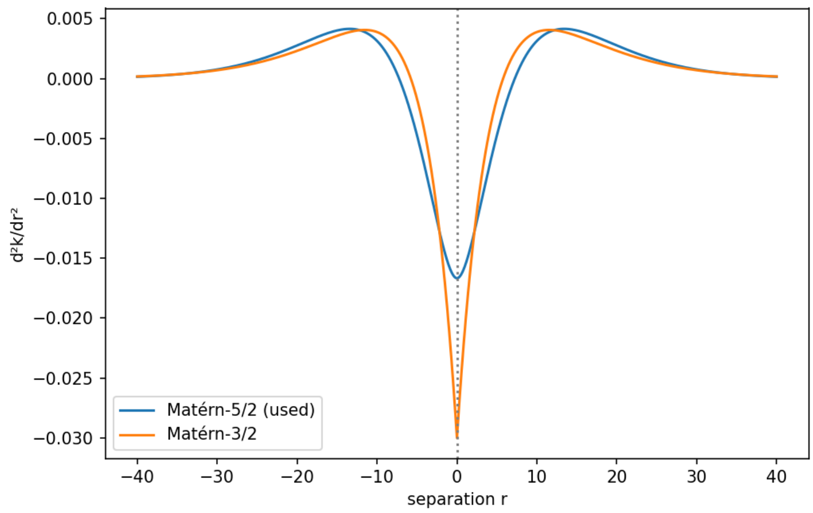

Figure 10.2: Second derivatives of the Matern 5/2 and Matern 3/2 kernels centred at zero. Matern 3/2 is continuous but not smooth at the origin, Matern 5/2 is continuously differentiable on the domain.

Divergence Free Field Verification (VER-GPR-07)

It was verified that the three dimensional field reproduced by the velocity Gaussian Process is indeed divergence free. How this is supposed to work was already visualized and demonstrated on a trivial case in Figure **??** . In order to verify this more rigorously, three divergence free analytical ground truth functions: $u_1 = [\sin(z), \sin(x), \sin(y)]^T$, $u_2 = [-y, x, 15]^T$, $u_3 = [-\frac{x}{2}, -\frac{y}{2}, z]^T$ were sampled at an arbitrary number of training points and the field was reconstructed using the divergence free GP pipeline. The streamlines of the analytical functions along with point locations are plotted in Figure 10.3; above every plot, the maximum divergence observed within the domain is stated. ref

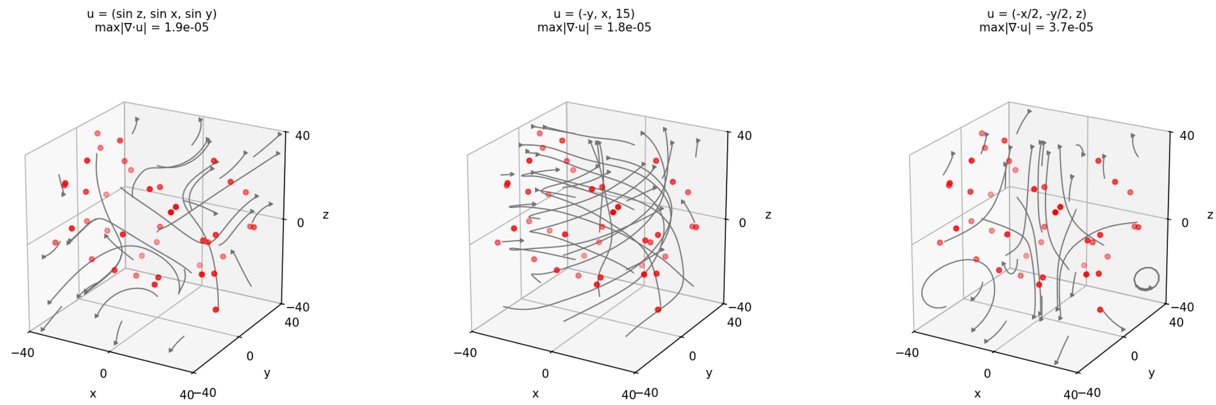

Figure 10.3: Reconstructed streamlines of the three divergence-free test fields u_1, u_2, u_3 with training points shown. The maximum divergence within the domain (above each subplot) remains on the order of 10^{-5} in all cases.

As can be seen in the plot, all three reconstructions resulted in a max divergence in the order of magnitude of 10^{-5}, easily meeting the 10^{-3} threshold. This verifies to a sufficient (for the design stage) extent that the fields produced by the GPR with this kernel can be considered divergence free. Addressing why the maximum divergence is close to but not exactly zero, it most likely has to do with numerical errors, particularly during the calculation of the derivatives.

10.1.2. Component Level Verification

At component level, each individual sensor, drone hardware, positioning system, communication link, and software modules shall be tested independently.

The velocity sensors (sonic anemometer) shall be calibrated against a reference anemometer or controlled flow facility. The miniaturised 3D sonic anemometers shall be tested over the expected velocity range, including both low speed wake regions and higher speed accelerated regions. The test shall verify velocity magnitude accuracy, flow direction accuracy, sampling frequency, sensor drift and sensitivity to mounting orientation w.r.t flow direction. A second test shall be performed with the sensor mounted on the drone in order to quantify the effect of rotor downwash (in addition to the literature findings on this topic), structural blockage, vibration sensitivity. Any systematic bias introduced by the drone shall either be shown to be negligible or corrected using a calibration model.

The pressure sensors shall be verified using a calibrated pressure reference. Since the momentum integral depends on pressure differences, the relative calibration between sensors is more important than the absolute value alone. The sensors shall be tested for offset, drift, temperature sensitivity, vibration sensitivity and sensor to sensor consistency. The pressure uncertainty obtained shall be included in the later force uncertainty budget.

The RTK-GNSS positioning system shall be verified by comparing drone position estimates against surveyed reference location or a higher-accuracy tracking system. Both static and dynamic tests shall be performed. The static test verifies absolute position accuracy, while the dynamic test verifies update rate, time synchronisation, and tracking performance during motion. The test can be considered passed if the positioning error for both the static and dynamic tests are below a certain threshold.

The communication system shall be verified under representative swarm load. This shall be done first with simulated drone nodes and later with real drones. The test shall verify that velocity, pressure, position, timestamp, and health data can be transmitted at the required rate from all drone to the ground station. It shall also verify command uplink reliability and packet loss. If communication degradation occurs, the system shall enter a safe state, rather than transmitting corrupted data.

Each drone shall be verified for payload capacity, endurance, vibrations and fail-safe behaviour. Since the drone carries the measurement payload, its flight stability directly affects measurement quality. The drone shall therefore be tested with the full sensor package installed. The test shall then verify that the drone can maintain its assigned position, correctly move to the required position for adaptive sampling, operate within the specified endurance time and safely return after low battery, communication loss or propulsion failure.

10.1.3. Functional Verification

Following the software and hardware unit testing, functional testing will be performed, as per the V model for Verification and Validation. The plan for the verification activities associated with this are listed in Table 10.2. It must be noted that these activities are intended for after a prototype of the system is built as they are physical tests testing interfaces of several components. In Table 10.2 each test has a unique identifier, a description, associated system requirement(s), facilities necessary to perform it, general method of performing the test (analysis, inspection, demonstration, test), passing criterion, and an associated financial cost expressed on a scale from 1 to 5.

Table 10.2: Post-build Verification Matrix.

Test ID	Description	Requirement(s)	Facilities	Method	Passing Criterion	Cost
PBVer-01	Single wind-tunnel session: flow-speed range + u/v/w component resolution	SYS-FUN-01.01.01; SYS-FUN-01.01.02	Wind tunnel with yaw/pitch rig	Test	13.6 m/s, no dropout; u/v/w within budget	5
PBVer-02	Temporal resolution on logged samples	SYS-FUN-04.01.01	Bench	Test	>10 Hz, no gaps >0.1 s	1
PBVer-03	Pre-session sensor readiness: field calibration + reference cross-check	SYS-PER-04.01.05; SYS-PER-04.01.06	Bench	Demo	Calibration stored; reference within tolerance	1
PBVer-04	Ambient T/P/RH sensor verification	SYS-FUN-02.01.03	Climate chamber	Test	T/P/RH channels present and traceable	4

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Table 10.2 – continued from previous page

Test ID	Description	Requirement(s)	Facilities	Method	Passing Criterion	Cost
PBVer-05	Single GNSS outdoor session: time sync + coordinate-frame resolution	SYS-FUN-01.01.03; SYS-FUN-01.01.04	Outdoor surveyed site	Test	Time within tolerance; coordinates within budget	3
PBVer-06	Single positioning range session: station-keeping + X/Y/Z tracking accuracy	SYS-PER-01.01.06; SYS-PER-01.01.07	Indoor positioning range	Test	$\leq \pm 1.5$ m; X/Y/Z within ASM-POS-02 (± 0.1m)	4
PBVer-07	Single open-field comms range test: 200 m link, latency and data-link presence	SYS-FUN-05.01.01; SYS-FUN-04.01.03; SYS-FUN-04.01.02	Open-field test range	Test	200 m link; latency $\leq$ ASM-COM-01 (1s); live data	3
PBVer-08	Buffer capacity with link-drop emulator	SYS-FUN-04.01.04	Bench	Test	Buffer $\geq$ ASM-COM-02 (30s); no data loss	1
PBVer-09	Single command/control demo with live data-link confirmation	SYS-FUN-05.01.02; SYS-FUN-04.01.02	Open-field test range	Demo	Commands acknowledged; live data visible	3
PBVer-10	On-board redundant local storage of measurement data	SYS-PER-04.01.03	Bench	Demo	Redundant data recovered, no corruption	1
PBVer-11	Single IPX4 spray session for platform, ground station, connectors and cables	SYS-CON-01.01.01; SYS-CON-01.01.02; SYS-CON-01.01.03	Environmental lab	Test	IPX4 passed; functions and continuity retained	4
PBVer-12	Sound power LWA measurement per ISO 3744 vs EU 2019/945	SYS-CON-05.01.01	Hemi-anechoic chamber	Test	LWA $\leq$ EU 2019/945 limit	5
PBVer-13	Declaration of Conformity package inspection	SYS-CON-05.01.02	Office	Inspect.	DoC complete and traceable	1
PBVer-14	Energy-storage recharge rate meets break window	SYS-OPS-01.01.01	Bench	Test	Recharge/swap within break window	2
PBVer-15	Ground-station charging infrastructure capacity for full swarm	SYS-OPS-01.01.02	Bench / ground-station shed	Inspect.	Full-system charging capacity verified	2
PBVer-16	As-built single-item mass	SYS-CON-08.02.01	Hangar	Inspect.	$\leq$25 kg per item	1
PBVer-17	Access-control: data inaccessible without credentials	SYS-OPS-04.02.05	Office workstation	Demo	Unauthenticated access denied	2
PBVer-19	Two trained operators run full system end-to-end	SYS-OPS-02.01.03	Field site	Demo	No supervision; no critical deviations	3
PBVer-20	Pre-deployment checklist procedure walk-through	SYS-OPS-02.01.04	Field site	Demo	Checklist complete; go/no-go recorded	3
PBVer-21	Deployment procedure executable up to 13.6 m/s wind	SYS-CON-02.02.01	Outdoor site or large wind tunnel	Demo	Safe deployment at 13.6 m/s	5
PBVer-22	All components remain secured during windy deployment	SYS-CON-02.02.02	Outdoor site	Demo	No displacement beyond tolerance	4

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Table 10.2 – continued from previous page

Test ID	Description	Requirement(s)	Facilities	Method	Passing Criterion	Cost
PBVer-23	Two-person ergonomic load/unload	SYS-CON-08.02.02	Vehicle loading area	Demo	Handled by 2 operators within limits	1
PBVer-24	Urban field session: footprint, signage and cable/tether layout	SYS-CON-03.01.01; SYS-CON-03.02.02; SYS-CON-03.02.03	Urban field site	Demo	Within parking space; signage visible; no public crossings	3
PBVer-25	Field session: 100 m × 100 m coverage + four time periods	SYS-FUN-01.02.01; SYS-PER-01.01.08	Field test site	Test	100×100 m covered; 4 valid periods	3
PBVer-26	Flow-feature length-scale resolution	SYS-FUN-03.01.01	Field test site	Test	Features ≥20% body length resolved	4
PBVer-27	Identification of wake deficit, pressure zones, shear, separation and vortices	SYS-FUN-03.01.02	Field site or large wind tunnel	Demo	Required flow features identified	5
PBVer-28	Real-time coverage-progress display	SYS-OPS-04.02.06; SYS-OPS-04.02.09	Field site	Demo	Progress display matches logged positions	3
PBVer-31	Outlier detection and flagging in raw data	SYS-PER-01.01.05	Office (replay)	Demo	Outliers flagged before force computation	1
PBVer-32	Reconstruction pipeline acceptance: force computation, freestream inputs and uncertainty output	SYS-FUN-02.01.01; SYS-FUN-02.01.02; SYS-FUN-02.01.04	Office	Test	Forces, inputs and uncertainty reported	1
PBVer-33	Wind direction relative to body recorded each session	SYS-FUN-02.01.05	Field site	Demo	Wind direction stored per session	3
PBVer-34	Operable only up to KNMI code yellow	SYS-CON-02.01.01	Field site	Demo	Go/no-go blocks unsafe weather	3
PBVer-36	System pauses and enters secure state above 13.6 m/s	SYS-CON-02.02.05	Field site or large wind tunnel	Demo	>13.6 m/s triggers secure state	5
PBVer-37	Safe radius from objects	SYS-CON-02.03.04	Field site	Demo	Radius ≥ 2×ASM-POS-01 (1.5m)	3
PBVer-38	Real-time alerts when parameters exceed safe operating bounds	SYS-OPS-04.02.07	Field site	Demo	Alerts triggered beyond bounds	3
PBVer-39	4 h operational endurance with breaks between measurements	SYS-PER-02.01.01	Field site	Demo	≥4 h operational endurance	3
PBVer-40	GPS position and timestamp stored for every measurement point	SYS-OPS-04.02.04	Field site	Demo	No missing GPS/timestamps	3
PBVer-41	Line-of-sight maintained for all units during operation	SYS-CON-02.03.01	Field site	Demo	Line-of-sight maintained	3
PBVer-42	Kill-switch / return-to-home triggered on link loss	SYS-CON-02.03.02	Field site	Demo	Link loss triggers safe state	3
PBVer-44	Scheduled/unscheduled maintenance without retiring system	SYS-PER-03.01.02	Hangar	Demo	Maintenance without retirement	2
PBVer-45	Calibration-drift flagging	SYS-PER-04.01.07	Field site (replay)	Demo	Drift above threshold flagged	2

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Table 10.2 – continued from previous page

Test ID	Description	Requirement(s)	Facilities	Method	Passing Criterion	Cost
PBVer-46	Field-replaceable maintenance components with standard tools	SYS-CON-07.01.03	Hangar	Demo	Replacement with standard tools	1
PBVer-47	Persistent calibration record traceable to national standards	SYS-OPS-04.02.03	Office (records)	Inspect.	Records persistent and traceable	1
PBVer-48	Disassembly of all material categories using standard hand tools	SYS-CON-07.01.06	Hangar	Demo	All materials separated with hand tools	1
PBVer-49	Operable from vessel-based or fixed-offshore platform	SYS-FUN-01.03.02	Vessel or offshore platform	Demo	Offshore mock-up operation completed	5

10.2. Validation Plan

The purpose of validation is to determine whether LAWSS provides an accurate and useful estimate of the wake field and aerodynamic force for the intended real-world application. Unlike verification, validation must compare the system output against physical reality in mission analogous scenarios. Because full scale aerodynamic force "ground truth" is difficult to obtain, validation shall be performed in stages of increasing realism.

10.2.1. Design Tools Validation

The validation starts with design tool validation, the aim of which is to validate the analytical tools used to justify the design before hardware is produced. Table 10.3 lists several the validation items planned for this stage, accompanied by a 1-5 scale financial cost estimation.

Table 10.3: Design Tool Validation Plan.

ID	Design Tool	Validation Method	Reference	Primary Metric	Acceptance Criterion	Cost	Required Facilities
DTV-CFD-01	CFD solver	Comparison against bluff-body literature wake studies and wind tunnel measurements	Wind tunnel data and literature data	Drag force error	< 5%	4	HPC cluster, CFD software licence
DTV-GPR-01	GPR-based reconstruction	Compare reconstructed forces from GPR and Momentum Deficit methods to the CFD "ground truth" output	High-fidelity CFD simulations based on various input conditions and analyzed structures	Force magnitude error	Target < 5%, max acceptable 10%	3	HPC cluster, CFD software licence

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Table 10.3 – continued from previous page

ID	Design Tool	Validation Method	Reference	Primary Metric	Acceptance Criterion	Cost	Required Facilities
DTV-D-01	Endurance model	Compare against hover power tests or manufacturer data for similar drone/payload configurations	Hover power tests and manufacturer data	Flight time in comparable conditions	< 5%	2	Workstation, manufacturer datasheets
DTV-D-02	Mass estimation model	Compare summed component masses with accurate CAD representations	CAD model	Mass prediction error	< 5%	1	Workstation
DTV-D-03	Control simulation	Compare virtual model response and position tracking	PX4 Software-in-the-Loop and flight-test logs	RMS position tracking error under gust disturbances	< 5%	3	Engineering workstation, PX4 SITL environment
DTV-D-04	Cost model	Validation against supplier quotes and material price quotes	COTS price quotes and material price quotes	Relative cost prediction error	< 10%	1	Office workstation
DTV-D-05	Power consumption model	Compare power draw during hover, maneuvers, and wind gust conditions	eCalc, motor datasheets, and bench tests of used components	Relative power prediction error	< 5%	2	Workstation, motor datasheets
DTV-D-06	Thrust estimation model	Compare thrust produced by the whole propulsion ensemble	Comparison with similar drones, eCalc, and manufacturer datasheets	Relative thrust prediction error	< 10%	2	Workstation, manufacturer datasheets
DTV-TO-01	Trade-off tool	Sensitivity analysis performed on all trade-offs using Monte Carlo simulations	Monte Carlo trade-off sensitivity study	Stability of trade-off rankings	Selected concept optimal in > 80% of cases	2	Workstation

10.2.2. Mission Validation

Consequently, Table 10.4 outlines the validation items to be implemented after the system is constructed, in real mission scenarios. Following this, Sections 10.2.3-10.2.8 rephrase the validation procedures in the context of synthetic-to-real hierarchy, which provides an additional perspective to how the real validation procedures should be carried out.

Table 10.4: Mission Validation Plan.

ID	Mission Element	Stakeholder Need	Validation Activity	Test Environment	Metric	Acceptance Criterion	Cost
MV-01	Drone hover stability	STK-CON-02	Outdoor hover test in gusty conditions (15 to 19 m/s gusts, as per Beaufort 6, revised)	Outdoor flight field	RMS position deviation from preplanned path	< 0.5 m RMS	3
MV-02	Swarm coordination	STK-CON-03, STK-CON-04	Multi-drone flight test	Outdoor test area	Minimum separation distance	> 2 m at all times	4
MV-03	Wake measurement mission	STK-FUN-01, STK-FUN-02, STK-FUN-03, STK-PER-01	Measure wake behind simplified obstacle	Wind tunnel / outdoor test	Relative aerodynamic force estimation error	< 5%	5
MV-04	Communication system	STK-FUN-04, STK-PER-04	Packet-loss test	Operational range test	Packet loss	< 1%	2
MV-05	Mission deployment	STK-CON-08, STK-OPS-01, STK-OPS-02	Full-scale deployment rehearsal	Simulated mission site	Setup and disassembly time	< 90 min	3
MV-07	Component endurance	STK-PER-03	Reliability and maintainability assessment using accelerated life testing	Component bench tests, supplier data, and operational usage profiles	Predicted mean time until critical failure	> 5 years	4

10.2.3. Synthetic and CFD-Based Validation

The first validation stage shall use synthetic CFD generated wake fields as reference data. A high resolution CFD solution shall be treated as the known reference field. Sparse measurements shall then be extracted from this field at locations representative of drone measurements. Realistic velocity noise, pressure noise, positioning uncertainty and drone dropout shall be added to the sampled data. LAWSS shall then reconstruct the wake field and estimate the force without access to the full solution.

This stage validates the reconstruction method in a controlled environment where the true field and reference force are known.

10.2.4. Controlled Experimental Validation

The second validation stage shall use flow measurements in a controlled environment, such as a wind tunnel. A scaled bluff body shall be placed in a known inflow. A dense reference measurement system, such as fixed sonic anemometers, pressure taps, PIV, or a force balance shall provide independent reference data.

The LAWSS measurement approach shall then be replicated using sparse sampling. If flying drones inside the facility is not practical, the sensor may be mounted on a fixed frame. This separates the validation of the reconstruction method from the complications caused by drone flight. The reconstructed wake and force estimate shall be compared with the dense reference measurements and direct force measurements. It also verifies that the method works on real turbulent flow data rather than only on CFD fields.

10.2.5. Drone-Mounted Sensor Validation

The third validation stage shall quantify how much the drone itself affects the measured flow. A drone carrying the full sensor package shall hover near a fixed reference anemometer or measurement mast in an open flow environment. The purpose is to determine whether rotor downwash, drone blockage, vibration or attitude changes introduce a significant measurement bias. If such bias exists, it shall be corrected using a calibration model. The corrected drone-mounted measurement shall be compared again against the fixed reference.This is a critical validation step, because even a perfect reconstruction algorithm would fail if the drones do not measure the undisturbed wake accurately.

10.2.6. Reduced-Scale Outdoor Validation

The fourth validation stage shall test the integrated LAWSS concept outdoors around a simple reference object, such as a container, wall or small building. The object should be instrumented where possible using pressure taps, reference anemometers or an independent force estimation method. A small swarm shall perform both the inlet condition measurement and downstream wake measurement phases.

The reconstructed wake shall be checked for physically expected features, including velocity deficit, shear layers, wake recovery and dependence on wind direction. The estimated aerodynamic force shall be compared against the independent reference estimate.

10.2.7. Matched Reynolds Number and Full-scale Validation

The final validation of LAWSS should not rely only on a full scale outdoor campaign, because the true aerodynamic force on a real large structure is generally not directly measurable. Instead, the final validation should be split into two complementary stages, a matched Reynolds number wind tunnel validation and a full scale outdoor operational validation.

The matched Reynolds number wind tunnel validation should be used to test the aerodynamic and reconstruction accuracy of LAWSS under conditions where a reliable reference is available. A scaled model of a representative structure, such as a rectangular building, bridge deck, tower section or a wind-turbine support structure, should be tested in a wind tunnel capable of approaching the relevant Reynolds number range. If a conventional atmospheric boundary layer wind tunnel cannot reach a sufficiently representative Reynolds number, a cryogenic or pressurised nitrogen tunnel could be considered. Such facilities increase Reynolds number by increasing density or reducing viscosity, allowing a smaller model to reproduce flow regimes closer to full scale.

In this test, the model should be instrumented with direct reference measurements. These may include balance, surface pressure taps, dense wake probes, PIV, or laser Doppler velocimetry. The dense reference measurements would define the best available “ground truth” for the wake field and aerodynamic force. The LAWSS reconstruction method would then be tested by artificially reducing the dense dataset to the sparse sampling pattern expected from the drone system. The reconstructed wake and estimated force would then be compared against the dense wind-tunnel reference.

The wind tunnel stage would therefore validate the aerodynamic reconstruction problem under controlled conditions, but not the complete LAWSS operation. It would show whether sparse sampling, GPR reconstruction and the momentum-integral force estimate can recover the correct wake field and aerodynamic force when a reliable reference is available.

A full-scale outdoor campaign would still be required to validate the operational system. This test would assess drone-mounted sensor interference, RTK-GNSS positioning, communication latency, swarm deployment, atmospheric variability, and repeatability between runs. LAWSS would be considered validated only if the wind-tunnel test shows accurate reconstruction against the reference data, and the outdoor campaign produces repeatable, physically consistent results within the reported uncertainty bounds.

10.2.8. Uncertainty Validation

Because LAWSS relies on sparse measurements, uncertainty quantification is part of the product rather than an optional output. The uncertainty model shall therefore be validated explicitly. Across CFD, controlled experimental, and field tests, the predicted confidence intervals shall be compared with the actual reconstruction and force errors. If the stated confidence interval contains the reference result at the expected frequency, the uncertainty model is considered calibrated. If not, the uncertainty model shall be corrected before operational use.

The final uncertainty budget shall include contributions from velocity sensor error, pressure sensor error, positioning error, timestamping error, drone-induced flow disturbance, GPR reconstruction error, control-volume placement, and numerical integration error.

11 Risk and Contingency

11.1. Risk Assessment Methodology

Risks are assessed using an expected value model, Risk = Probability × Severity, visualised on a 5 × 5 Risk Assessment Matrix. Probability is scored from 1 (Remote, <20%) to 5 (Certain, >80%), where a score of 5 applies to physics-driven degradation unavoidable over the system lifetime, and a score of 1 requires multiple simultaneous independent failures. Severity is scored from 1 (Negligible) to 5 (Catastrophic), where a score of 5 means the aerodynamic force error exceeds 5%, minimum spatial coverage cannot be maintained, *or* a safety incident involving the public occurs. The safety criterion is treated as equally catastrophic to mission failure, reflecting the urban operating environment of the dense swarm concept.

Risks are identified across four life-cycle phases: Manufacturing (TR-MAN), Deployment (TR-DEP), Operations (TR-OPS), and Maintenance (TR-MNT). All ground-based concept flags are removed; every risk applies to the selected architecture. Preventive mitigations target the likelihood of occurrence and drive design requirements; contingency plans target the consequence if the risk event happens and define response procedures. Post-mitigation scores are conditional on those mitigations being implemented.

11.2. Mitigation Strategy and Critical Risks

Several risks remain elevated after mitigation. **TR-OPS-01** (weather) is persistent as environmental conditions are beyond the team's control. **TR-OPS-04** (formation flow distortion) remains critical because mutual rotor-rotor interference across close-flying drones can only be corrected via CFD models that carry their own uncertainty. **TR-DEP-05** (battery pack ignition during transport) is treated as critical despite a low residual probability, since the consequence of a thermal runaway propagating through ~260 co-located Li-ion packs in an enclosed van remains severe even with certified cases and reduced state-of-charge.

The most significant new risks introduced by the swarm concept relate to position estimation and the communication architecture. **TR-OPS-13** (mid-air collision due to position estimation error) can have its probability bounded by separation and ORCA(Optimal Reciprocal Collision Avoidance) avoidance requirements, but not eliminated, as it depends on the reliability of the RTK link broadcast over the HaLow network. **TR-OPS-14** (common-cause swarm failure) remains severe even with diverse redundancy, since a shared fault, such as RTK ground station failure or a firmware bug, can simultaneously affect all units. **TR-OPS-21** Electromagnetic interference corrupting RTK/HaLow links) reflects the same underlying dependency in a denser form: a congested urban RF environment can degrade positioning and command links for the whole swarm at once, which is why pre-mission RF surveys and channel planning are treated as preventive requirements rather than optional checks.

Finally, **TR-OPS-17** (regulatory permit denial) remains an existential risk for the primary use case, as dense autonomous swarm operations in urban airspace remain an emerging and unsettled regulatory category. **TR-OPS-19** (GPR algorithm failing to reconstruct the flow field) and **TR-OPS-20** (operator error during swarm management) are retained as moderate residual risks specific to the swarm scale: the former because adaptive sampling depends on sufficient and well-distributed sensor coverage that cannot always be guaranteed in advance, and the latter because a two-person crew managing ~80 drones has a structurally higher chance of a missed abort or incorrect go/no-go call than a single-drone operation, regardless of checklist discipline.

Table 11.1: Risk Map Before Mitigation

	Severity				
Prob.	**1-Negl.**	**2-Minor**	**3-Mod.**	**4-Crit.**	**5-Catas.**
5-Certain		TR-DEP-01, TR-MNT-01	TR-OPS-05	TR-OPS-04	
4-Likely		TR-MAN-02	TR-MAN-01	TR-OPS-12, TR-OPS-13, TR-OPS-16, TR-OPS-17	TR-OPS-03, TR-OPS-15
3-Possible			TR-OPS-10, TR-OPS-18	TR-DEP-04, TR-OPS-01, TR-OPS-11, TR-OPS-19, TR-OPS-20, TR-OPS-21, TR-MNT-02	TR-DEP-05, TR-OPS-14
2-Unlikely	TR-DEP-03	TR-DEP-03	TR-OPS-22	TR-OPS-02, TR-OPS-08	
1-Remote					TR-DEP-02

Table 11.2: Risk Map After Mitigation

	Severity				
Prob.	**1-Negl.**	**2-Minor**	**3-Mod.**	**4-Crit.**	**5-Catas.**
5-Certain					
4-Likely	TR-DEP-01, TR-MNT-01				
3-Possible		TR-MAN-02	TR-OPS-18, TR-OPS-19, TR-OPS-20, TR-OPS-21		
2-Unlikely	TR-DEP-03	TR-DEP-04, TR-OPS-05	TR-MAN-01, TR-OPS-01, TR-OPS-02, TR-OPS-03, TR-OPS-11, TR-OPS-12, TR-OPS-13, TR-OPS-15, TR-OPS-16, TR-OPS-17, TR-MNT-02, TR-DEP-05, TR-OPS-22		
1-Remote	TR-OPS-10	TR-OPS-06, TR-OPS-08	TR-DEP-02, TR-OPS-04	TR-OPS-14	

Table 11.3: Technical Risk Register

ID	Risk Event	Impact	Pre- Prob.	Pre- Sev.	Preventive Mitigation	Contingency Plan	Post- Prob.	Post- Sev.	Responsible
Manufacturing									
TR-MAN-01	Unavailability of COTS parts required for sensor	Single sensor delay affects entire drone swarm. Schedule and cost overrun.	0.7	0.6	≥2 qualified vendor alternatives per part type; bulk orders placed early.	Deploy Minimum Viable System with lower-spec fallback sensors.	0.4	0.5	A. van Bastelaere
TR-MAN-02 **[N]**	Swarm integration effort underestimated	Calibrating hundreds of units far exceeds single-drone schedule; inconsistent calibration corrupts wake data.	0.7	0.3	Standardised per-drone built-in-test + sensor calibration + comms test <20 min; batch integration in project schedule.	If less than 80% of drones pass tests, fly with available units and increase spacing.	0.5	0.3	A. van Bastelaere
Deployment									
TR-DEP-01	Transport damage to drones	Damaged units reduce operational swarm; fewer drones degrade spatial coverage and force estimate accuracy.	0.8	0.3	Stackable padded batch crates; preload damage checklist per crate.	Carry ≥15% spare assembled drones on-site.	0.7	0.1	S. Beukeleirs
TR-DEP-02	Whole system destroyed in transport	Total hardware loss; mission cannot proceed.	0.1	0.9	Vetted transport company; split swarm across multiple vehicles where possible.	Full-fleet insurance; emergency procurement plan.	0.05	0.6	A. van Bastelaere
TR-DEP-03	Assembly exceeds 1.5 h	Missed weather window; mission delay.	0.4	0.3	Modular tool-free drone design; rehearsed role-assigned procedure with checklist.	Larger schedule buffer.	0.3	0.2	S. Beukeleirs
TR-DEP-04	Swarm exceeds van capacity	Extra vehicles required; violates transport requirement.	0.6	0.7	Max stowed dimensions from van interior as hard design constraint; folding arms required.	Roof rack or trailer fallback; split swarm across two vans.	0.3	0.4	B. Postrach
TR-DEP-05 **[N]**	Single battery pack ignition during transport	~260 Li-ion packs transported together; thermal runaway in one cell releases flammable gas before ignition, risking a chain reaction to adjacent packs; fire in an enclosed van is rapid and difficult to suppress; non-compliant transport may also violate ADR/IATA dangerous goods (UN3480/UN3481) regulations.	0.5	1.0	Certified packs in fire-resistant transport cases; packs stored at ≤50% SoC during transport; physical separation/spacing between battery crates to limit propagation; ADR-compliant labelling and shipping documentation; regular battery health checks; isolate and remove any pack showing swelling, heat, or voltage anomaly before departure; do not transport visibly damaged or swollen cells.	Class D / lithium-battery-rated fire extinguisher and fire blanket carried on board.	0.4	0.8	S. Vudutha
Operations									
TR-OPS-01	Weather corrupts measurements	Rain or high wind violates <5% force error requirement; mission delays.	0.6	0.7	Meteorological go/no-go thresholds derived from weather forecast.	Corrections in post-processing, based on sensor and weather data.	0.4	0.6	S. Beukeleirs
TR-OPS-02	Sensor inaccuracy exceeds 5%	Calibration drift or sensor limits violate force error requirement.	0.4	0.7	Error budget derived back from 5% requirement; frequent in-field calibration.	Post-processing drift detection; outlier removal; reference cross-validation.	0.3	0.6	M. Popescu

ID	Risk Event	Impact	Pre-		Preventive Mitigation	Contingency Plan	Post-		Responsible
			Prob.	Sev.			Prob.	Sev.	
TR-OPS-03	Drone malfunction and swarm collision	Uncontrolled descent in dense formation risks mid-air collision with adjacent units; debris hazard over public.	0.7	1.0	Redundant motor/ESC; automatic RTH on health threshold breach; onboard ORCA collision avoidance.	Geofenced exclusion zone; prop guards; swarm-wide hold on descent detection.	0.4	0.6	V. Velichkov
TR-OPS-04	Formation flow distortion	Mutual rotor-rotor interference across swarm corrupts velocity data beyond 5% budget; full-formation CFD needed.	0.8	0.8	Full-formation CFD/wind-tunnel validation of separation; sensor placement in quiet zone.	Formation-level aerodynamic correction model in post-processing.	0.2	0.6	B. Postrach
TR-OPS-05	Recharge cycle creates coverage gaps	20–30 min flight time requires repeated battery swaps or recharges; unplanned rotation gaps break GPR.	0.8	0.6	Pre-planned staggered swap/recharge rotation keeping coverage ≥80%; sufficient spare battery inventory for full-day ops.	On-site recharge station and pre-charged spare packs; coverage threshold as go/no-go per measurement phase.	0.4	0.4	S. Vudutha
TR-OPS-08	Propeller vibrations corrupt sensors	Blade-pass excitation propagates into sensor mounts; invalidates measurements.	0.3	0.7	Modal analysis; sensor mount frequency ≥20% from blade-pass; vibration-isolation mounts.	Vibration-correction algorithm in post-processing.	0.2	0.4	B. Postrach
TR-OPS-10	HaLow communication link failure or capacity overload	Loss of telemetry or command link to one or more drones; degraded situational awareness and inability to issue real-time corrections or emergency commands to affected units.	0.5	0.5	Pre-deployment link-budget verification at maximum operational range; TDMA scheduling ensures per-drone duty-cycle compliance within 868 MHz EU limits; redundant HaLow access point on standby with automatic failover.	Onboard local data storage ensures no measurement loss; post-mission upload on link restoration; 802.11ac fallback link maintains GCS command and control independently.	0.3	0.3	Y. Yovchev
TR-OPS-11	Noise exceeds EU/NL urban limits	Swarm with hundreds of drones exceeds noise limits thresholds far more acutely than a single unit; non-compliance blocks urban certification.	0.6	0.7	Fleet-level acoustic analysis against EU Directive 2002/49/EC and EASA SC-VTOL before propeller selection.	Higher operating altitude; temporary noise exemption; restrict to permitted hours.	0.3	0.6	M. Popescu
TR-OPS-12	Loss of flight stability at 8 Beaufort	Drone unable to maintain controlled hover; gust-induced position error causes final force estimation error to exceed 5%.	0.7	0.8	Active stabilisation validated at 8 Bft; position hold ≤ ±0.3 m demonstrated in simulated gust profiles.	Automatic safe-mode triggered on attitude deviation threshold; return-to-home on sustained instability; operations suspended above 8 Bft.	0.3	0.6	V. Velichkov
TR-OPS-13 **[N]**	Mid-air collision due to position estimation error	Corrupted or degraded position data (RTK dropout, GPS multipath, comms latency) displaces drone into adjacent unit or restricted volume; collision causes uncontrolled descent over public.	0.7	0.8	Minimum 3–4 m inter-drone separation enforced as hard constraint; RTK positions broadcast via HaLow AP to all drones with ORCA avoidance reaction ≤100 ms.	Prop guards; exclusion zone sized for worst-case debris field; swarm-wide emergency land triggered on imminent collision detection.	0.4	0.6	V. Velichkov
TR-OPS-14 **[N]**	Common-cause multi-drone failure	RTK ground station failure degrades all positions simultaneously; WiFi jamming disrupts swarm-wide command; firmware bug could affect all units; single-drone contingency overwhelmed.	0.5	1.0	Diverse redundant comm channels; firmware updates never swarm-wide simultaneously; RTK ground station redundancy (see TR-OPS-06).	Hardcoded lost-link failsafe: RTH or controlled descent within 5 s of heartbeat loss, independent of GCS.	0.1	0.8	Y. Yovchev
TR-OPS-15 **[N]**	Airspace conflict with third party	Another aircraft enters the operational area; collision risk in urban environment.	0.7	1.0	Obtain airspace permit before each mission; notify local authority; deploy ground-based visual observers.	ADS-B traffic monitor at GCS; automatic swarm descent on traffic alert; pre-defined abort corridor.	0.4	0.6	M. Popescu

ID	Risk Event	Impact	Pre-		Preventive Mitigation	Contingency Plan	Post-		Responsible
			Prob.	Sev.			Prob.	Sev.	
TR-OPS-16 **[N]**	GCS link failure during operations	Drones execute stale waypoints unsupervised; battery depletion risks uncontrolled descent; swarm conflicts unresolvable without ground intervention.	0.7	0.8	Redundant HaLow AP on standby with automatic failover; RSSI and packet-loss thresholds as hard go/no-go; hardcoded lost-link RTH within 5 s on MAVLink heartbeat timeout.	Pre-programmed RTH requiring no GCS confirmation; 802.11ac RPi4 fallback maintains command and control independently of HaLow link; backup command on secondary frequency band.	0.4	0.6	Y. Yovchev
TR-OPS-17 **[N]**	Regulatory permit denial	RVO/EASA denies urban swarm permit; primary commercial use case potentially unviable.	0.7	0.8	Early regulatory engagement; SORA analysis per EASA 2019/947 targeting lowest feasible ground risk class.	Peri-urban fallback test sites; adjust altitude and exclusion zone to meet lower risk category.	0.4	0.6	S. Beukeleirs
TR-OPS-18 **[N]**	Ground station compute failure during adaptive sampling	Gaussian Process Regression loop breaks; swarm reverts to static preplanned waypoints; adaptive coverage improvement lost for remainder of measurement phase.	0.5	0.5	Redundant compute node (hot standby); watchdog process auto-restarts ROS2 on crash; hardware sized with margin for GPR workload.	Pre-computed fallback waypoint grid used for remainder of phase; measurement still valid, just non-adaptive.	0.4	0.4	Y. Yovchev
TR-OPS-19 **[N]**	GPR algorithm fails to reconstruct flow field	Insufficient or poorly distributed sensor data prevents the model from converging on a valid flow solution; the 5% force error requirement cannot be met for that measurement phase.	0.5	0.7	Minimum sensor coverage threshold defined per phase; data distribution monitored in real time against GPR input requirements; simulation used to validate that planned drone positions yield sufficient coverage.	Flag phase as invalid; re-fly with a denser or more targeted waypoint grid before proceeding to the next phase.	0.4	0.6	M. Popescu
TR-OPS-20 **[N]**	Operator error during swarm management	With ~80 drones and a two-person crew, fatigue or miscommunication can lead to a missed abort, an incorrect go/no-go call, or a wrong parameter input.	0.5	0.7	Checklist-driven procedures with clearly assigned roles; any irreversible command requires confirmation from both operators; operator breaks scheduled on multi-day deployments.	GCS hard limits (geofence, minimum separation, max speed) enforce a safe operating envelope regardless of operator input; automated abort triggers on threshold breach.	0.4	0.6	S. Beukeleirs
TR-OPS-21 **[N]**	EM interference corrupts RTK/HaLow in urban environment	Urban RF environment (5G base stations, WiFi dense deployments, radar) degrades RTK fix quality or saturates HaLow band; position errors propagate into force reconstruction.	0.5	0.7	Pre-mission RF survey at operational site; RTK base station positioned to maximise line-of-sight and minimise interference; HaLow channel plan avoids identified interference sources.	Switch to SBAS/GNSS-only positioning with increased inter-drone spacing; log RF environment for post-processing correction.	0.4	0.6	Y. Yovchev
TR-OPS-22 **[N]**	RTK base station positioning or initialisation error	Systematic offset in base station coordinates propagates as a common bias to all drone positions; absolute position error does not trigger relative collision avoidance but corrupts georeferenced wake data.	0.4	0.6	Base station coordinates surveyed against known benchmark before each deployment; convergence time monitored; position fix quality logged.	Cross-check base position against GNSS reference network; re-initialise and re-survey before resuming measurements.	0.2	0.6	Y. Yovchev
Maintenance									
TR-MNT-01	Battery degradation over lifetime	Capacity fade reduces per-drone endurance; severely degraded cells risk thermal runaway.	0.8	0.3	Per-cell state-of-health monitoring; replacement threshold before capacity drops below endurance requirement.	Standard lifecycle and storage procedures; batch pack replacement at defined threshold.	0.8	0.1	S. Vudutha
TR-MNT-02	Moisture ingress damages electronics	UV-degraded seals allow moisture onto PCBs; corrosion and electrical failure in later deployments.	0.5	0.8	UV-resistant silicone seals; scheduled seal integrity check at each service interval.	Conformal coating on all internal PCBs as secondary barrier.	0.4	0.6	M. Popescu

12 Reliability, Availability, Maintainability and Safety

This chapter sets out the Reliability, Availability, Maintainability and Safety (RAMS) characteristics of the LAWSS design. Instead of basing it on similarity with existing products, it is built directly from the detailed design and reflects the architecture selected in the trade-offs. All four items of RAMS will be discussed: the redundancy philosophy, expected reliability and availability, the scheduled and non-scheduled maintenance activities. The RAMS analysis draws on the risks identified in Chapter 11 for the failure modes it must protect against, and on the verification plan in Chapter 10 for the tests.

12.1. Safety-Critical Functions

A function is identified as safety-critical if its failure can cause mid-air collision, a loss of separation from the surveyed structure or the public, or an uncontrolled descent. These are the ones scored as catastrophic in the risk assessment, because the dense swarm operates close to large structures and can be deployed above public space. Table 12.1 lists there functions with the dominant effect of their failure, the provision controlling that failure and the governing requirement.

Table 12.1: Safety-critical functions and their controlling design provisions

Function	Effect of failure	Design provision	Requirement
Collision avoidance	Drone-to-drone collision within the swarm	Onboard ORCA on every drone, resolving conflicts from neighbour states with a 3 to 4 m hard separation and a 50 ms reaction target	SYS-CON-02.03.01
Position estimation	Drift leading to loss of separation or collision	RTK-GNSS giving centimetre-level position, with corrections broadcast from the ground base to all drones	ASM-POS-01
Loss-of-link handling	Drone flying on with stale commands	Kill-switch and return-to-home triggered automatically on link loss	SYS-CON-02.03.02
Wind-limit protection	Loss of control in conditions beyond the airframe envelope	System pauses and enters a secure state above 20.7 m/s	SYS-CON-02.02.05
Separation from structure and public	Strike on the surveyed body or a bystander	Enforced safe radius of at least twice the position assumption, with line-of-sight maintained for all units	SYS-CON-02.03.04
Flight control and attitude hold	Uncontrolled descent on a propulsion fault	Hexa-rotor layout that remains controllable and lands safely after a single motor or propeller failure	AS-PRE-07
Battery thermal safety	Thermal runaway across co-located packs during transport	Certified transport cases and reduced state-of-charge for the packs in transit	TR-DEP-05 (mitigation)
Operator awareness	Operator unaware of an out-of-bounds state	Real-time alerts when parameters exceed safe operating bounds	SYS-OPS-04.02.07

The defining choice is that the collision avoidance runs onboard each drone and not the ground station. This results in a drone never having to wait for the ground to keep clear of its neighbours. A degraded or lost command link cannot disable avoidance. The link-loss response and the wind limit secure state can act as the next in line of defence. Namely, if a drone cannot be commanded, or the weather moves outside the velocity envelope, the system falls back to a defined safe state.

12.2. Redundancy Philosophy

Redundancy is applied on four levels, connected to the failures each is meant to survive.

On the airframe level, one of the advantages for the hexa-rotor is its motor-out behaviour. Unlike a quad-rotor, which enters a dive on a single motor failure, the hexa-rotor becomes controllable on five motors and can be brought to safe landing. This trades some endurance for survivability and was accepted on those grounds in the configuration trade-off.

On swarm level, redundancy is inherent to the dense-swatm concept. The reconstruction does not depend on any single drone, making a loss of one unit not ending up in a mission failure. The swarm is sized with a margin over the minimum training point count the convergence analysis requires. Because path planning assigns targets while each drone flies its own inner control and avoidance loops, a single unit can drop out without stalling the rest of the swarm.

On data level, each drone holds an onboard redundant local copy of its measurements, so a dropped packet or a brief link outage does not lose data. The record can be recovered after landing.

On communications level, the separate logical channels keep the safety-critical avoidance traffic distinct from the bulk data. The onboard placement of avoidance removes the ground link from the critical path.

The limit of this philosophy is that some elements cannot be made redundant by adding more drones. A single RTK ground base, or a firmware fault shared across multiple units can affect the whole swarm at once (TR-OPS-14). Diverse redundancy and pre-mission RF surveys reduce the likelihood, but these shared dependencies remain single points of failure at system level and are carried as critical risks in Chapter 11.

12.3. Reliability and Availability

The reliability target is a predicted mean time to critical failure greater than five years per unit (SYS-PER-03.01.01). Because no field history exists for this design, the figure needs to be substantiated by:

1. accelerated life testing of the critical components, and
2. supplier reliability data, and
3. expected operational usage profile

The corresponding validation activity is the reliability and maintainability assessment in the validation plan (MV-07).

It is useful to separate item reliability from mission reliability. Item reliability is the five year per-unit figure above. Mission reliability, the probability of completing a survey campaign, is higher than any single unit's reliability precisely because the swarm tolerates individual losses. As long as enough drones remain to meet the required training points, the campaign succeeds.

Availability is therefore best defined at campaign level rather than per unit. A drone that is down for a battery swap or a field repair does not make the system unavailable. The spares and swap batteries cover its absence within the operational break window (SYS-OPS-01.01.01). The dominant availability limiter is not hardware but the weather. Operations are cancelled above the 20.7 [m/s] wind limit and in unsuitable conditions, which is outside the team's control.

12.4. Maintainability

LAWSS is a maintainable system: it is built to be serviced and returned instead of being retired on a fault (SYS-PER-03.01.02). Maintenance is split into scheduled and non-scheduled activities.

Scheduled maintenance covers the activities done on a fixed interval regardless of the state of LAWSS. Before each deployment a checklist and go/no-go procedure is run (SYS-OPS-02.01.04). Batteries are tracked by cycle count and replaced on a schedule, well before their performance degrades. Sensor calibration is verified against a persistent record traceable to national standards (SYS-OPS-04.02.03). The carbon-fibre frame, sensor booms and propellers are inspected periodically for impact damage and fatigue.

Non-scheduled, corrective maintenance covers repairs triggered by a fault. The design supports this as the maintenance components are field-replacable using standard hand tools (SYS-CON-07.01.03). A faulty drone is repaired or swapped at the site rather than returned to a workshop. Faults are isolated per drone, and the redundant onboard data copy protects measurements taken before the fault. The dense swarm philosophy reinforces maintainability at system level: units are interchangeable and individually non-critical.

The verification of these provisions, including field-replacement with standard tools, maintenance without retirement and calibration-drift flagging, is covered by the corresponding PBVer items in the verification plan in Chapter 10.

13 Cost Analysis

13.1. Financial Budget

aaaaaaa The financial section will be split into 6 different tables, based on the subsystems in the project. The tables are the following:

1. Financial Cost overview
2. Frame Components Cost
3. Battery Components Cost
4. Propulsion Components Cost
5. Sensing and Computational Components Cost
6. Onboard Communication and Positioning Components Cost
7. Ground Station and Transportation Costs

13.2. Technical Budget

14 Sustainability Development Strategy

Throughout the development of this project, the LAWSS design team has been committed to internalising the external socio-economic and environmental costs by embedding sustainability into the manufacturing, operational, and end-of-life stages. This is to avoid simply offsetting external costs through increased pricing. Now that the detailed design of the wake survey system has been completed, this chapter expands on the sustainability analysis conducted during the preliminary design phase. The sustainability development strategy begins with specifying the functional unit to quantify the product's environmental impact followed by a recap of the life cycle assessment scope and life cycle phases. The manufacturing phase is detailed next with a deep dive into the updated bill of materials, embedded energy analysis, and greenhouse gas emissions, following changes in the drone design between the preliminary and detailed design phases. The operational phase's socio-economic impact, energy usage, community impact, and positive societal impacts are elaborated.

The chapter concludes with a look into the end-of-life phase of the wake survey system, how materials are recovered, and finishes with how the detailed design compares to the sustainability requirements established early in the project.

14.1. Sustainability Framework and Scope

14.1.1. Functional Unit

14.1.2. Life Cycle Assessment and Life Phases

14.2. Manufacture Phase - Environmental Impact

14.2.1. Bill of Materials

Table 14.1: Drone (Detailed Design) Bill of Materials (BoM)

Subsystem / Component	Component Material	Quantity per Drone	Component Mass / per Unit (g)	Total Mass (g)
Drone Structural Frame				
Frame Tubes	CFRP	6	13.5	81.0
Sensor boom	CFRP	1	32.2	32.2
Landing Gear Legs	CFRP	4	36.8	147.2
Landing Gear Skid	CFRP	2	32.9	65.8
Frame Screws / Fasteners	Aluminium	1	50.0	50.0
Propulsion System				
Propellers	CFRP	6	11.5	69.0
Battery Pack				
Battery Cells	Lithium Ion	18	280.0	5040.0
Battery Nacelle	Titanium	1	120.0	120.0
Compression Plate	CFRP	6	52.5	315.0
Payload Sensors				
Wiring	Copper	–	65.0	65.0

14.2.2. GWP and Energy Impact Analysis

Table 14.2: Drone Material Energy Consumption and Greenhouse Gas Emissions [$kgCO_2/kg$]

Material Category	Energy to Produce (primary) [MJ/kg]	Energy to Produce (recycled) [MJ/kg]	GWP (primary) [kgCO2/kg]	GWP (recycled) [kgCO2/kg]
CFRP	500.0	-	24.0	-
Aluminium	224.0	17.9	12.0	0.6
Lithium Ion	205.0	-	20.18	-
Titanium	553.0	258.0	29.4	13.70
Copper	70.6	11.65	2.42	0.4

Table 14.3: Environmental Impact Metrics Per Drone Breakdown

Material Category	Energy to Produce (primary) [MJ]	Energy to Produce (recycled) [MJ]	GWP (primary) [kgCO2]	GWP (recycled) [kgCO2]
CFRP	355.10	–	17.0	–
Aluminium	11.20	0.9	0.6	0.03
Lithium Ion	1,033.20	–	101.7	–
Titanium	66.36	31.0	3.53	1.64
Copper	4.59	0.757	0.157	0.0260
Total	**1,470.45**	**32.61**	**123.04**	**1.70**

14.3. Operational Phase - Socioeconomic Impact

14.3.1. Operational Energy Usage Analysis

14.3.2. Battery Life Cycle and Replacement

14.3.3. Noise and Community Impact

14.3.4. Positive Societal Values

14.4. End of Life Phase

14.4.1. Disassembly and Material Recovery

14.4.2. Battery Regulations

14.4.3. Sustainability Requirements and Changes

15 Project Design and Development

The project design and development logic refers to the actions regarding the project from the end of the final design (end of Design Synthesis Exercise) to the end-of-life of the system. This sequence includes activities that were deemed outside of the scope of the Design Synthesis Exercise but necessary to develop a compliant system.

The full Project Design and Development Diagram can be seen in Figure 15.1. The design synthesis exercise outputs a final design, which can be checked for completeness. Once it is checked, a first prototype can be built, using both off-the-shelf (OTS) parts and custom-built components. This prototype can go through a series of tests of increasing complexity and risk. However, prior to testing, the system must be certified to be able to fly in any airspace having been built. If one of these tests are to fail, the failure mode must be analysed. The design gaps identified post-analysis shall be resolved before starting the testing process again. Once all tests are successful, the final operational system can be assembled and deployed for the stakeholder-requested period of 5 years. Afterwards, end-of-life measures should be implemented, either recycling or refurbishing the system's components.

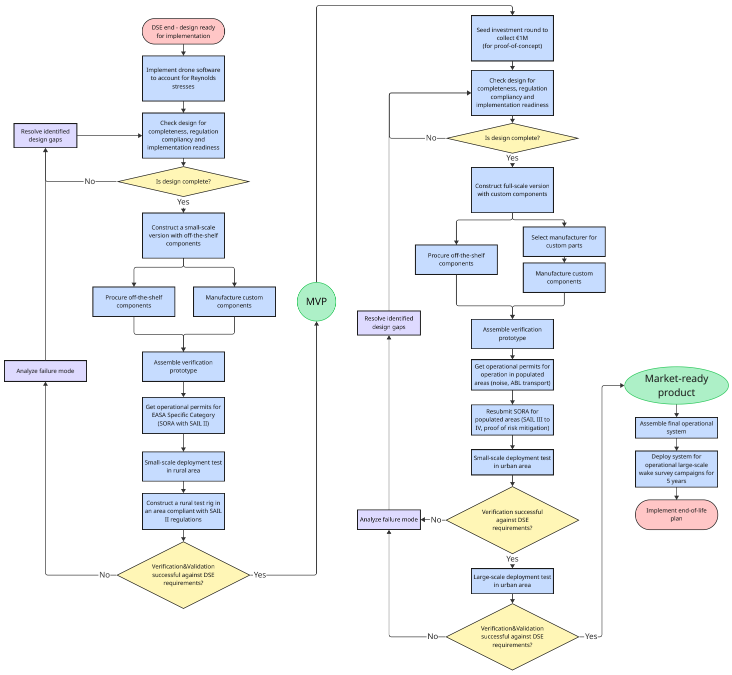

Figure 15.1: Project Design and Development Diagram

Subsequent sections discuss the need for system refinement and certification requirements mentioned in the timeline. 'Operational large-scale wake survey campaigns' assumed requested demand for the system as specified with occupancy days in financial analysis .

reference market analysis

15.1. Development logic and Investment Round

Although the DSE provided sufficient scope to review the product viability, before the system may be built, it must be financed. Firstly, this requires a Minimum Viable Product (MVP), the existence of which would warrant a seed round of investment for the project to come to life. Then, proper verification and validation of the design may proceed.

The MVP for LAWSS would be a down-scaled version of the product in terms of hover time and weather capabilities. Cheaper drone platforms may house lower-cost off-the-shelf sensors to provide a proof-of-concept real-world dataset, such that Software methodology may be verified and validated. Deploying this system in a rural area to conduct tests on a test-jig would offer insight into the viability of the product before incurring major risks during rounds of investment funding. The MVP could be funded via angel investors, university-level funds, creation of a Dream Team[1], crowdfunding, or other public methods.

Having created an MVP that passes all requirements set out my the DSE, verified and validated, a seed round of investment may begin. According to market analysis, there is potential for the system to fill a niche market need for analysis and optimization (see Chapter 2), and according to financial analysis, the system would require a seed investment round of €1 million to build a market-ready product (see Chapter 13). Then, with large-scale verification and validation complete, the system is deemed market-ready and the first deployment may be conducted for a client. After this, the system can serve for 5 years, its feasibility is evaluated, and the end-of-life plan is implemented accordingly.

[1]https://www.tudelft.nl/dreamhall/teams/tu-delft-dream-teams

15.2. Accounting for Reynolds Stress Contributions to Drag

In this section, preliminary guidelines are given to account for the effects of airspeed fluctuations around mean velocity and the resulting Reynolds stresses on aerodynamic force prediction. As explained in Section 5.6, the Reynolds stress contribution to the momentum integral has been neglected in the proof of concept steady state RANS CFD for lack of time-variable data to enable the breakdown of velocity into mean and fluctuation components. In this chapter, the methods to be applied to real flows sampled at 10 Hz are outlined to improve force estimation accuracy are outlined.

The Reynolds stress tensor (Equation 15.1) consists of 6 unique terms which need to be taken into account (in a bluff body flow measured close to the perturbing body both the diagonal and off-diagonal terms are non-zero [**<empty citation>**]). As the model assumes the z (ground-normal) force is not of interest, the number of components of interest reduces to 5 ($\overline{w'^2}$ omitted). empty citation

$$-\rho\,\overline{u_i'u_j'} = -\rho \begin{bmatrix} \overline{u'^2} & \overline{u'v'} & \overline{u'w'} \\ \overline{u'v'} & \overline{v'^2} & \overline{v'w'} \\ \overline{u'w'} & \overline{v'w'} & \overline{w'^2} \end{bmatrix} \tag{15.1}$$

Because every drone collects velocity component samples at the rate of 10 Hz, the time series of velocity fluctuations in each direction can be retroactively computed by subtracting the converged mean from each measured snapshot:

$$u_i'[t] = u_i[t] - \overline{u_i}.$$

Then, because the velocity components are sampled synchronously, we can compute the five time series $u'^2[t]$, $v'^2[t]$, $u'v'[t]$, $u'w'[t]$, and $v'w'[t]$ from individual velocity fluctuation components. The high frequency ensures the data is representative of actual Reynolds stresses. This results in pointwise time series approximating the Reynolds stresses at these specific points.

We propose that in the post-processing stage of the operations (after the final measurement), these can be averaged pointwise and five separate Gaussian Processes can be trained to approximate the mean Reynolds stresses at the testing points based on the data measured at the training points. Following this, Equation 5.34 can be applied to retrieve a more accurate prediction of the streamwise and lateral aerodynamic forces.

15.3. Regulations Adherence

Due to the nature of the system, consisting of about 80 large drones, each powered by a battery and also needing to transport several backup batteries, the project falls into several regulation areas that will need to be covered during system deployment. Below, these areas are detailed and discussed.

15.3.1. EASA Specific Operations Risk Assessment (SORA)

Given the system's maximum take-off mass (MTOM) of 3.308 kg per unit, the operation falls below the 25kg ceiling of the EASA (European Union Aviation Safety Agency) Open Category. However, autonomous beyond visual line of sight (BVLOS) operation of a fleet of 80 units over potentially populated areas excludes the system from the Open Category regardless of mass, since Open Category rules require continuous remote pilot oversight and impose strict limits on flight over uninvolved persons [13]. The system therefore falls under the EASA Specific Category, requiring an Operational Authorisation from the relevant national competent authority, in the Netherlands this is the Human Environment and Transport Inspectorate (Inspectie Leefomgeving en Transport, ILT) [19].

The required methodology to obtain this authorisation is the Specific Operations Risk Assessment (SORA), developed by JARUS (Joint Authorities for Rulemaking on Unmanned Systems) and adopted by EASA as the Acceptable Means of Compliance for the Specific Category. As of September 2025, EASA has formally incorporated SORA version 2.5 into the regulatory framework via ED Decision 2025/018/R [10]. The SORA process consists of the following key steps, each of which would need to be addressed in detail for the system prior to operational deployment:

1. **Concept of Operations (ConOps)**: A description of the operational volume, flight geography, population density of the overflown area, and whether the operation is conducted within visual line of sight (VLOS) or beyond visual line of sight (BVLOS) [22].
2. **Intrinsic Ground Risk Class (iGRC)**: Determined from the maximum characteristic dimension of the UA (Unmanned Aircraft), its maximum speed, and the population density of the area overflown [40]. For a multicopter, the characteristic dimension is the rotor tip to rotor tip span of the frame.
3. **Final GRC**: The intrinsic GRC reduced through mitigations such as M1 (operational restrictions or sheltering) and M2 (ground impact effect reduction, e.g. parachute or controlled descent systems). The final GRC must not exceed 7, operations exceeding this threshold fall outside the scope of SORA and require the Certified Category [22].
4. **Air Risk Class (ARC)**: Determined from the airspace class, altitude, and encounter rate with manned aircraft. In the Netherlands, Specific Category operations are restricted to 120 meters above ground or water unless otherwise specified (or specifically requested) in the operating permit [19].
5. **Specific Assurance and Integrity Level (SAIL)**: Derived from the combination of final GRC and residual ARC, on a scale from I to VI [11].
6. **Operational Safety Objectives (OSOs)**: A set of 24 safety objectives that must be satisfied at a robustness level (low, medium, or high) corresponding to the determined SAIL [14].

Based on a preliminary assessment using the system's MTOM, operating altitude (below 120 m), and the assumption of operation over populated areas with autonomous BVLOS flight of multiple simultaneous units (STS 02), an early estimation of the system is that it will fall into the **SAIL III to IV** range, requiring medium to high robustness compliance for most OSOs. It is possible that the system would fall under **SAIL II** in sparsely populated areas with little encounters with manned aircraft, under stricted operational restrictions (thereby reducing both GRC and ARC). This has direct implications for the design that should be addressed in future iterations, including:

- Mass and power budget allowance for ground impact mitigation systems (M2), such as a parachute or controlled flight termination system, required to reduce the final GRC.
- Integration of a Detect and Avoid (DAA) system if BVLOS operation is confirmed, with associated mass, power, and cost impact on the design presented in this report.
- Operational constraints arising from the ground risk buffer, which may restrict the minimum standoff distance between the drones and the building, directly affecting the flow estimation operations the system is designed to perform.
- Given the population density assumptions used in determining the final GRC, it is likely that an exclusion zone (ground risk buffer) around the operational volume will need to be enforced during flight operations, restricting uninvolved persons from the area directly beneath and surrounding the drones' flight paths near the building. The size of this buffer depends on the final GRC and the chosen ground impact mitigations (M2), and should be incorporated into the operational procedure and site planning for the system [22].

A full SORA submission to the ILT, including a detailed ConOps, GRC and ARC determination, and OSO compliance evidence, is recommended as a follow-up activity before operational deployment of the system.

15.3.2. ADR Battery Transport Regulations

The system requires 255 batteries to support 85 drones, with each drone carrying its operating battery (85 units) and the remaining 170 spare batteries transported by a support van, as per Subsection 8.3.5.

Lithium-ion batteries transported separately, as is the case for the spare batteries, are classified as dangerous goods under UN3480, while batteries installed in the drones fall under UN3481 [16].

Each 6S1P 30 Ah battery has an energy content of approximately 666 Wh and a mass of 1.845 kg, far exceeding the 100Wh threshold of ADR [1]. Above this threshold, the orange placards on the transportation vehicle must be displayed. With 170 spare batteries in the van, the total battery mass is 314 kg, which close to but below the 333 kg (1000-point) threshold; below this threshold, the orange placards can remain closed and the driver does not require an ADR certificate, though the transport documents must still contain all ADR data including the UN number, ADR name, and label number. Given that the calculated mass of 314 kg sits close to the 333 kg boundary, the van shipment is considered as requiring full ADR compliance (placarding, transport documentation, and a certified driver), as slight design changes are still possible. Therefore, post-DSE activities must account for this for the system to be transportable in any capacity.

15.3.3. Data, Electronics, and Noise Regulations

Beyond the airspace and transport regulations discussed above, two additional regulatory areas are relevant but impose limited concrete requirements on the current design. Regarding data and electronics, Commission Delegated Regulation (EU) 2019/945 mandates that all UAS above 250g broadcast a Direct Remote Identification (DRI) signal via Wi-Fi or Bluetooth throughout the flight; the Wi-Fi adapter already present in the architecture satisfies this hardware requirement, provided the broadcast conforms to the prEN 4709-002 encoding standard [13]. Regarding noise, EASA has published harmonised measurement guidelines for UAS below 600kg operating in the Specific Category, defining SEL-based hover and level-flight measurement procedures; however, the section on noise limits is explicitly marked as reserved, meaning no binding numerical limits are currently in force [12]. Noise characterisation may nonetheless be requested by the ILT as part of the operational authorisation process, particularly given the simultaneous operation of 80 units over populated areas. Therefore, a separate phase of certification in public areas must be complete before the system is fully operable.

16 Production Plan

The drone is produced as a custom hexarotor platform designed to carry the LAWSS measurement sensors far enough from the wake of the propellers. The vehicle consists of a central carbon fibre frame, six rotor arms, a large battery compartment, landing gear and a long sensor boom carrying the sonic anemometer and pressure sensor. The structure is built from available carbon fibre plates and tubes, combined with custom connectors pieces and mounting brackets.

The production process is divided into five main stages, component procurement, structural manufacturing, frame assembly, electronics integration and final testing.

16.1. Component Procurement

The first step is to procure all components required for one complete measurement drone. The onboard drone components are listed in Table 16.1, while the detailed frame components, battery pack components, field support components and charging station components are listed in Tables 16.2–16.5. The components are split into structural, propulsion, power, avionics, positioning, communication and measurement payload subsystems.

The structural procurement includes the carbon fibre sheets used for the central plates and battery cage plates, the carbon fibre tubes used for the six motor arms, the forward sensor boom, the landing gear legs and the landing skids, and the titanium threaded rods used in the battery cage. The custom connector pieces, such as the arm clamps, motor mounts, landing gear connectors and sensor mount, are produced from the CAD model. These parts should be manufactured before the final assembly starts, because they define the geometry and alignment of the drone.

The propulsion system consists of six EMAX PRO 2810–1350 motors , six 8 × 5.5 inch carbon fibre update motor name propellers, six ESCs and the required propeller adaptors. Each motor is mounted at the outer end of one carbon fibre arm. Each ESC is installed close to its corresponding motor, either on the arm or near the arm root, so that the motor cables remain short while still allowing cooling and inspection access.

The power system consists of three 6S1P 30 Ah battery packs per drone, one voltage converter and the required wiring. Only one battery pack is mounted on the drone during flight, but three packs are produced per drone to reduce downtime during field operation. The battery is placed inside the lower battery cage, below the central frame. The battery cage is built from carbon fibre plates, titanium threaded rods and carbon fibre compression spacers.

The avionics and data system consists of the Pixhawk 6C Mini flight controller, Raspberry Pi 4 companion computer, RTK rover, RTK GNSS antenna, magnetometer, WiFi HaLow adaptor and WiFi HaLow antenna. The flight controller is installed near the centre of the drone and as close as possible to the centre of gravity. The Raspberry Pi is mounted inside the central electronics bay, close enough to the flight controller for reliable data and power connections. The RTK GNSS antenna is mounted on the upper side of the drone with a clear sky view. The magnetometer is mounted externally and away from the battery, ESCs and high-current power wires. The WiFi HaLow adaptor is connected to the companion computer, while the WiFi HaLow antenna is mounted externally so that the carbon fibre frame does not block the signal.

The measurement payload consists of the TriSonica Mini sonic anemometer and the Adafruit BMP581 pressure sensor. These sensors are mounted at the end of the forward carbon fibre sensor boom. This position is selected to place the measurement volume away from the main body of the drone and away from the strongest rotor induced flow. The sensor cables are routed along the boom and fixed to the structure to prevent cable motion during flight.

Table 16.1: Main onboard components required for one LAWSS measurement drone.

Subsystem	Component	Qty.	Specification / note	Mass / cost
Frame	Custom frame assembly	1	Modified carbon fibre frame with smaller motor booms and smaller landing gear. Detailed structural components are given in Table 16.2.	0.450 kg / 200 EUR
Power system	Battery pack	3	6S1P 30 Ah battery pack. Three battery packs are produced per drone to reduce on field downtime, while only one battery is mounted during flight.	1.845 kg each / 2535 EUR total
Propulsion	Motors	6	EMAX PRO 2810–1350 motors.	0.306 kg / 124.20 EUR
Propulsion	Propellers	6	8×5.5 carbon fibre propellers, 203.2 mm diameter.	0.069 kg / 28.50 EUR
Propulsion	Propeller adaptors	6	Motor to propeller adaptors. These may not be required depending on the final motor and propeller interface.	0.005 kg / TBD
Propulsion electronics	ESCs	6	TBS LUCID 12S 30 A ESCs, 57.75 mm $\times$ 29 mm.	0.150 kg / 260 EUR
Power electronics	Voltage converter	1	Mateksys BEC12S PRO, 35 mm $\times$ 24 mm $\times$ 5.5 mm.	0.005 kg / 20 EUR
Wiring	Wiring set	1	Power wiring, signal wiring, connectors, heat shrink, cable ties and routing hardware.	0.200 kg / 20 EUR

Subsystem	Component	Qty.	Specification / note	Mass / cost
Measurement payload	Sonic anemometer	1	TriSonica Mini. Used as the main velocity measurement sensor.	0.060 kg / 2211.80 EUR
Measurement payload	Pressure sensor	1	Adafruit BMP581 pressure sensor. Requires I2C extension and shielded cable.	0.010 kg / 27.04 EUR
Flight control	Flight controller	1	Pixhawk 6C Mini, 55 mm × 39 mm × 18 mm.	0.043 kg / 113 EUR
Onboard computing	Companion computer	1	Raspberry Pi 4 Model B, 4 GB, 85 mm × 56 mm × 17 mm. The cheaper 2 GB option may be sufficient.	0.046 kg / 110 EUR
Positioning	RTK rover	1	Unicore UM980 RTK GNSS receiver, 26 mm × 39 mm. Provides 20 Hz update rate and connects to the flight controller.	0.010 kg / 170 EUR
Positioning	RTK GNSS antenna	1	Ardusimple compact helical triple-band GNSS antenna, 43 mm × 43 mm × 41 mm. L1/L2/L5 compatible.	0.026 kg / 128 EUR
Positioning	Magnetometer	1	RM3100 magnetometer, 33 mm × 26 mm × 10 mm. Should be mounted away from high current wiring and magnetic interference.	0.007 kg / 38.61 EUR
Communication	WiFi HaLow adaptor	1	Gateworks GW16167 WiFi HaLow M.2 2230 E-Key card. Interfaces with the companion computer.	0.006 kg / 38.16 EUR
Communication	WiFi HaLow antenna	1	Taoglas FXP890 860–928 MHz flexible PCB antenna.	0.002 kg / 3.36 EUR

Table 16.2: Structural components required for one drone frame.

Part	Component	Qty.	Specification / note	Mass / cost
Central structure	Central plates	set	Double sided high strength carbon fibre sheet, 3 mm thick. Approximate sheet area used: 0.024 m^2.	75.419 g / 16.11 EUR
Motor support	Frame arms	6	Woven roll wrapped carbon-fibre tube, 18 mm OD, 16 mm ID, 160 mm length.	81.216 g / 16.90 EUR
Sensor support	Sensor boom	1	Woven roll wrapped carbon-fibre tube, 12 mm OD, 10 mm ID, 600 mm length.	32.160 g / 7.05 EUR
Landing gear	Vertical landing gear legs	4	Woven roll-wrapped carbon fibre tube, 14 mm OD, 12 mm ID, 140 mm length. The legs are angled approximately 15 degrees outwards.	36.792 g / 7.18 EUR
Landing gear	Landing gear connectors	set	Custom connector pieces for joining the landing gear legs to the frame and skids.	TBD / TBD

Part	Component	Qty.	Specification / note	Mass / cost
Landing gear	Landing gear skids	2	Woven roll wrapped carbon-fibre tube, 14 mm OD, 12 mm ID, 250 mm length.	32.850 g / 6.17 EUR
Battery cage	Battery cage rods	4	Titanium threaded rods, M4, 140 mm length. These clamp the two battery cage plates together.	28.000 g / 11.24 EUR
Battery cage	Battery cage spacers	4	Carbon-fibre tube spacers, 6 mm OD, 4.8 mm ID, 140 mm length. These prevent the plates from being crushed when the rods are tightened.	7.504 g / 3.78 EUR
Battery cage	Battery cage plate	set	Double sided high strength carbon fibre sheet, 180 mm × 95 mm × 3 mm.	75.419 g / 16.11 EUR

Table 16.3: Components required to manufacture one 6S1P 30 Ah battery pack.

Part	Specification	Mass [g]	Cost [EUR]
Cells	6S1P 30 Ah lithium-ion cell assembly	1680	825
Main connector	XT90 connector	15	1.50
Balance connector	JST-XH 7-pin balance connector	10	1.50
Fish paper	0.38 mm thick insulation paper	7	5.00
Kapton tape	25 mm wide tape	2	0.30
Compression plate	Carbon fibre compression plate	105	9.00
Compression tape	Fibre reinforced packing tape	5	0.20
Wires	Estimated power and balance wiring	5	1.00
Heat shrink tube	Battery pack heat shrink	15	0.80
Foam padding	2 mm thick foam padding	1	1.00
Total		**1845**	**845.30**

Table 16.4: Field support infrastructure required for operation of the drone measurement system.

Subsystem	Component	Qty.	Specification / note	Cost [EUR]
Positioning infrastructure	RTK base station	1	Emlid Reach RS3. Provides RTK corrections for the drones.	2500
Communication infrastructure	WiFi HaLow access point	1	Morse Micro HaLowLink 2, 88 mm × 68 mm × 24 mm, 10 W. Used as the ground side communication access point.	111.80
Power infrastructure	Mobile battery storage	1	Mobile battery storage trailer, approximately 210 kWh, with two charging points.	100000
Charging infrastructure	Charging station	1	Custom charging cabinet/rack system. The detailed breakdown is given in Table 16.5.	65740

Subsystem	Component	Qty.	Specification / note	Cost [EUR]
Ground control	On-field laptop	1	GETAC X600 Pro, Xeon processor, 64 GB RAM, GTX1650.	12000
Transport	Electric truck	1	Mercedes-Benz eACTROS 400.	140000
Ground control power	UPS	1	Back-UPS BX950MI IEC C13, 520 W.	128.16
Storage	Drone shelves	set	Storage shelves for transporting and storing multiple drones.	15000
Storage	Container	1	Field container for storing the drone system and supporting equipment.	10000

Table 16.5: Charging station component breakdown.

Component	Qty.	Unit cost [EUR]	Total cost [EUR]
Charging rack	2	11480	22960
Power upgrade, 400 V 16 A to 400 V 32 A	2	10000	20000
Li-ion charger	80	230	18400
XT60 to XT90 adaptor	80	4.75	380
HVAC supplies	2	2000	4000
Total			**65740**

16.2. Manufacturing of custom parts

After procurement, the custom structural parts are manufactured. The carbon fibre tubes are cut to the required lengths given in Table 16.2. The six motor arms are cut from 18 mm OD carbon fibre tube, the sensor boom is cut from 12 mm OD carbon fibre tube, and the landing gear legs and skids are cut from 14 mm OD carbon fibre tube. The tube ends are sanded flat and cleaned before assembly. The central plates and battery cage plates are cut from 3 mm carbon fibre sheet according to the CAD model.

The central plates contain the mounting holes for the arm connectors, battery cage, flight controller stack and sensor boom mount. The battery cage plates hold the battery pack and are connected using the titanium threaded rods and carbon fibre spacers.

The custom connector parts are then produced. These include the motor arm clamps, motor mounts, landing gear connectors, battery cage supports and sensor boom mount.

16.3. Frame Assembly

The mechanical assembly starts with the central frame. The lower central plate is placed on a flat surface and the battery cage is attached underneath it. The battery cage rods pass through the carbon fibre cage plates and are tightened using nuts and washers. Carbon fibre spacers are placed around the threaded rods between the plates. The titanium rods provide the clamping force, while the carbon fibre spacers prevent the plates from being crushed. Threadlocker should be applied to the metal nuts to prevent loosening under vibration.

Once the battery cage is installed, the large battery pack is test-fitted. The battery must be removable without disassembling the frame, and the connector must remain accessible.

The six motor arms are then installed around the central frame. Each arm is held by two custom connector pieces at the root. The arms are inserted into the connectors and aligned so that all motor locations lie in the same horizontal plane. The arm clamps are then tightened gradually and symmetrically. At this stage, the frame should be checked against the CAD model to verify the arm angles and rotor spacing.

After the motor arms are installed, the motor mounts are attached at the outer end of each arm. The motor axes must be vertical and parallel to each other. The motors can then be bolted to the motor mounts.

The landing gear is assembled after the main frame and motor arms. The vertical landing gear legs are connected to the lower frame and angled 15 degrees outwards. The horizontal skid tubes are mounted at the bottom of the legs. The landing gear must provide enough ground clearance for the battery cage and must support the full drone mass during landing.

16.4. Propulsion Integration

The propulsion system is installed after the structural frame is complete. Each motor is connected to one ESC. The ESCs are mounted close to the corresponding motor. The three motor phase wires are kept as short as practical and are fixed to the arm using cable ties or clips.

The ESC signal wires are routed from the arms back to the central frame and connected to the flight controller. The power wires from all ESCs are routed to the main power distribution point connected to the battery. All power cables should be fixed to the frame so that they cannot move into the propeller disks. The propellers are only installed after the electrical tests and motor direction checks have been completed.

16.5. Battery Pack and Power System Integration

The battery packs are manufactured separately before final drone assembly. Each battery pack consists of the 6S1P 30 Ah cell assembly, XT90 main connector, JST-XH 7-pin balance connector, fish paper insulation, Kapton tape, carbon fibre compression plate, reinforced compression tape, wiring, heat shrink tube and foam padding. The complete battery pack has a mass of approximately 1.845 kg.

During drone assembly, one battery pack is placed inside the battery cage below the central frame. The battery is restrained using foam padding and the cage structure so that it cannot move during flight. The voltage converter is installed inside the central electronics bay and connected to the main battery line. It supplies the regulated voltage required by the flight controller, companion computer and payload electronics.

16.6. Avionics, positioning and communication integration

The avionics are installed after the propulsion and power systems are in place. The Pixhawk 6C Mini flight controller is mounted close to the centre of gravity and aligned with the drone body axes. The Raspberry Pi 4 companion computer is placed in the central electronics bay, where it connects to the flight controller, sensors and communication hardware.

The positioning system consists of the RTK rover, GNSS antenna and magnetometer. The RTK rover is placed in the electronics bay, while the GNSS antenna is mounted on top of the drone to provide a clear sky view. The magnetometer is mounted away from the battery, ESCs, motors and high current wiring to reduce magnetic interference.

The WiFi HaLow adaptor is connected to the Raspberry Pi, and its antenna is mounted externally on the frame to maintain a clear communication path to the ground station. All avionics wiring is routed neatly through the central frame and fixed to prevent movement during flight.

16.7. Sensor boom and measurement payload installation

The sensor boom is installed at the front of the drone after the main frame and avionics have been assembled. The 12 mm OD carbon fibre boom is fixed to the central frame using a reinforced root mount, since this connection carries the bending load from the boom and sensors.

The TriSonica Mini sonic anemometer and BMP581 pressure sensor are mounted at the end of the boom to reduce interference from the drone body and rotors. Their cables are routed along the boom and fixed to the structure to prevent vibration or cable motion during flight.

16.8. Ground support and charging infrastructure

The field support equipment is prepared separately from the drone assembly. This includes the RTK base station, WiFi HaLow ground access point, charging station, mobile power supply, field laptop, UPS, storage shelves and transport container.

The RTK base station and WiFi HaLow access point support positioning and communication during operation. The charging station and mobile battery storage provide battery turnaround between flights, while the laptop and UPS are used for ground control and monitoring.

16.9. Assembly sequence

The assembly sequence is summarised in Table 16.6. The order is chosen so that structural parts are completed first, followed by propulsion and power wiring, then avionics, sensors and testing.

Table 16.6: Assembly sequence and component placement for one measurement drone.

Step	Assembly stage	Components installed	Location on drone
1	Prepare custom parts	Carbon fibre plates, tubes, custom connectors	Cut and prepared according to CAD before assembly.
2	Build battery cage	Battery cage plates, titanium rods, carbon fibre spacers, nuts and washers	Below the central frame, forming the lower battery compartment.
3	Assemble central frame	Upper/lower central plates and cage interface	Main body of the drone, around the centre of gravity.
4	Install motor arms	Six carbon fibre motor arms and twelve arm clamps	Radially around the central frame; two clamps per arm.
5	Install landing gear	Four angled landing gear legs and two horizontal skids	Under the central frame and battery cage.
6	Install motor mounts and motors	Six motor mounts and six motors	At the outer end of each motor arm.
7	Install ESCs and power wiring	Six ESCs, power wires, signal wires and power distribution wiring	ESCs close to motors; wires routed along arms to the central frame.
8	Install battery and voltage converter	One flight battery, voltage converter and main power connector	Battery inside lower cage; voltage converter in central electronics bay.
9	Install flight electronics	Pixhawk 6C Mini and Raspberry Pi 4	Near the centre of the drone, inside the central electronics bay.
10	Install positioning hardware	RTK rover, GNSS antenna and magnetometer	RTK rover in central bay, GNSS antenna on top, magnetometer externally away from power wiring.
11	Install communication hardware	WiFi HaLow adaptor and antenna	Adaptor near Raspberry Pi; antenna externally mounted with clear signal path.
12	Install sensor boom	Carbon-fibre sensor boom and boom root mount	Forward facing boom attached to the central frame.
13	Install measurement sensors	TriSonica Mini and BMP581 pressure sensor	At the end of the sensor boom, away from rotor/body disturbance.
14	Final mechanical inspection	Fasteners, clamps, battery cage, boom, landing gear	Full drone inspection before powering the system.
15	Electrical and software checks	Flight controller, ESCs, Raspberry Pi, sensors, RTK and communication	Tested without propellers installed.
16	Install propellers and flight test	Six propellers	Installed only after motor direction and failsafe checks are complete.

17 Limitations and Recommendations

V&V has to be done

Velocity in the z-axis measured by sensor is limited

Drone potentially oversized

Weatherproof? unknown

Effects of positional uncertainty on GPR reconstruction

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